Assessing pre-restoration fine sediment dynamics and freshwater macronutrients at an ambitious restoration site on the River Breamish, Northumberland.

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Abstract:

Over 97% of the river network of the UK has been modified, with large-scale evidence of river straightening and embankment installation. Channelisation fundamentally alters the natural flow regime, river-floodplain connectivity has longstanding, detrimental impacts on fine sediment dynamics and macronutrient cycling. Hydrological disconnection between rivers and floodplains as a result of channelisation reduces opportunities for temporary storage of sediment and pollutants, catalysing sediment transport and increasing downstream accumulation. Restoration is planned to reverse the historical pattern of floodplain disconnection on the River Breamish, Northumberland as part of Natural England's LIFE WADER project. The restoration aims to reinstate the natural function of the river system, restore biodiversity and improve water quality in relation to fine sediment and nutrient loadings. This research applies a high resolution, hydrological monitoring programme from February 2024 to May 2025 alongside time integrated sediment sample composition analysis and diffuse pollutant critical source area modelling to assess the pre-restoration fine sediment and macronutrient dynamics at this ambitious restoration site. This methodology addresses the prevalent underrepresentation of diffuse pollutants in contemporary monitoring programmes. Results indicate that high nutrient rich sediment loads (<312.04 tonnes average) are transferred through the reach during high flows. A dominance of clockwise hysteresis indicates that fine sediment is delivered predominantly via rapidly responding pathways from sources proximal to the channel. Evidence of complex figure of eight hysteresis during the highest flow periods is attributed to potential re-suspension and bank incision mechanisms. Modelled critical source areas include headwaters and areas of agricultural land activity where transport pathways facilitate delivery of nutrient rich sediment directly into the channel. This study builds on existing theory to develop an enhanced understanding of fine sediment and macronutrient dynamics, providing a crucial baseline which will be used to inform post-restoration monitoring following ~8 weeks of works commencing in June 2025.

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List of Abbreviations:

Abbreviation	Meaning
ANOVA	Analysis of variance
DEFRA	Department for Environment, Food and Rural Affairs
DTM	Digital terrain model
EU	European Union
F8	Figure of eight
HI_{mid}	Hysteresis Index
IQR	Inter quartile range
N	Nitrogen
NH_4	Ammonium
NO_2	Nitrite
NO_3	Nitrate
P	Phosphorus
PO_4	Phosphate
SAC	Special area of conservation
SCIMAP	Sensitive Catchment Integrated Modelling and Planning
SSSI	Site of special scientific interest
SSY	Specific sediment yield
TIMS	Time integrated mass-flux sampler
TU_{FL}	Turbidity on the falling limb of the hydrograph
TU_{RL}	Turbidity on the rising limb of the hydrograph
UKCEH	United Kingdom Centre for Ecology and Hydrology

List of Abbreviated Units:

Abbreviation	Unit
ha	Hectares
km	Kilometers
m	Meters
mm	Millimeters
NTU	Nephelometric turbidity units
t	Metric tonnes
t/km^2	Metric tonnes per kilometer squared
µm	Micrometer
σ	Standard Deviation

Chapter 1 – Introduction:

1.1 Background and Rationale:

The function of freshwater ecosystems is declining at an unprecedented rate and is identified as one of the most impacted ecosystems globally as a result of anthropogenic activities (Sala *et al.,* 2000). Fine sediment dynamics and associated nutrient fluxes have a critical impact on ecological functioning, but remain poorly understood due to an inherent complexity, consequently the relevance of research contributing to environmental management regimes has only recently been acknowledged (Fielding, 1993; Zhang, 2014).

The legacy of agricultural expansion and increased fertiliser use represents a longstanding pressure on freshwater systems, elevating fine sediment and macronutrient fluxes such that current levels exceed pre-industrial conditions (Jones *et al.,* 2011). Agricultural activity is the dominant source of nitrate (NO_3) within freshwater systems (Bell *et al.,* 2021); however, macronutrient losses are spatially variable depending on soil characteristics, land use practice and hydrological connectivity (Ekholm *et al.*, 2000). The impacts on water quality and ecology are poorly characterised on a catchment scale, where multiple overlapping pressures and process based modifications create complex pathways of sediment and macronutrient delivery (Jarvie *et al.*, 2008). Critical source areas are defined as areas where high pollutant availability coincides with strong hydrological connectivity, consequently delivering disproportionate amounts of material to the channel (Pionke, Gburek and Sharpley 2000). The delivery of excess nutrient rich fine sediment accelerates eutrophication, consequently deteriorating water quality and detrimentally impacting aquatic ecosystem health (Ekholm *et al.*, 2000). Thus, identification and management of critical source areas is essential for effectively targeting restoration to improve water quality and ecosystem functioning within agricultural catchments.

Sediment and macronutrient dynamics exhibit additional complexity within anthropogenically modified catchments, particularly in channelised systems where an increased water surface slope and velocity during high flow events accelerates bed erosion, generating surplus sedimentary input through bank incision and undercutting (Todd-Burley *et al.,* 2021). There is a longstanding history of channelisation in the UK (Brookes, 1988), such that just 1% of UK rivers are classified as unimpacted by anthropogenic modifications (Jones *et al.,* 2019). Channelisation simplifies the

natural flow regime (Zhou and Endreny., 2020) and resultant floodplain disconnection critically impacts key sedimentary deposition and nutrient cycling processes (Hohensinner *et al.,* 2004). A functional floodplain must be laterally connected to the river channel to facilitate sediment and macronutrient exchange (Opperman *et al*., 2010; Thoms, 2003). This is closely related to longitudinal flow, depending on lateral hydrological connectivity (Zeiringer *et al.,* 2018), thus floodplain disconnection (as a result of channelisation) reduces the potential for sediment retention, consequently catalysing sediment transport and increasing accumulation risk in downstream areas with inefficient filtration (Noe and Hupp, 2005). Floodplain reconnection has gained recognition as an effective strategy for reducing sediment and nutrient fluxes through restoring the depositional functions and enhancing sediment sink retention, consequently reducing pollutant flux (Sear *et al.,* 2010; McMillan and Noe, 2017). However, there is limited integration between using critical source area modelling outputs to identify targeted locations for intervention, consequently floodplain restoration schemes are often implemented based on geomorphic and ecological criteria rather than hydrological delivery risk (Forbes, Ball and McLay, 2015). Given the increasing degraded state of freshwater environments, implementing effective restoration to improve water quality by reducing downstream sediment transport and macronutrient loading is imperative. It is critical to establish the direction of connection being discussed within this research, where in the context of SCIMAP modelling the term connection refers to the transfer of materials from the catchment slope into the river, but in the context of floodplain reconnection it refers to the transfer of materials out of the river onto the floodplain.

1.2 Study Area:

The River Breamish is a main tributary of the River Till, Northumberland, draining a catchment area of 90km^2 (DEFRA, 2023). Situated within the Till Catchment Site of Scientific Interest (SSSI) and The River Tweed Special Area of Conservation (SAC) (Tweed Forum, 2024), the Breamish flows North-West from its source in the Cheviot Hills to its confluence with the River Till at Bewick Bridge (Figure 1) (Chessell, 2014). The catchment is a designated SSSI and SAC due to the diverse landscape, habitats and biodiversity; many species of conservation value inhabit the Breamish including *Salmo Salar* (Atlantic Salmon), *Lutra Lutra* (Eurasian Otter) and *Petromyzontidae* (Lamprey) alongside vulnerable aquatic macrophytes such as *Ranunculus Aquatilis* (Water Crowfoot) (Chessell, 2014). Under natural conditions the Breamish would exhibit riffle-pool,

wandering channel morphology, migrating across its floodplains (Environment Agency, 2022). However, due to historic channel straightening and lack of contemporary management, the reach exhibits limited floodplain connectivity (Figure 2) and is classified as unfavourable and declining (LIFE WADER, 2024). This study focuses on the 1km stretch of previously straightened channel (Figure 1B) due to undergo works to restore ~50ha of floodplain wetland habitat (Figure 2) in aim of restoring natural system functioning and improving water quality (Williams *et al.*, 2023).

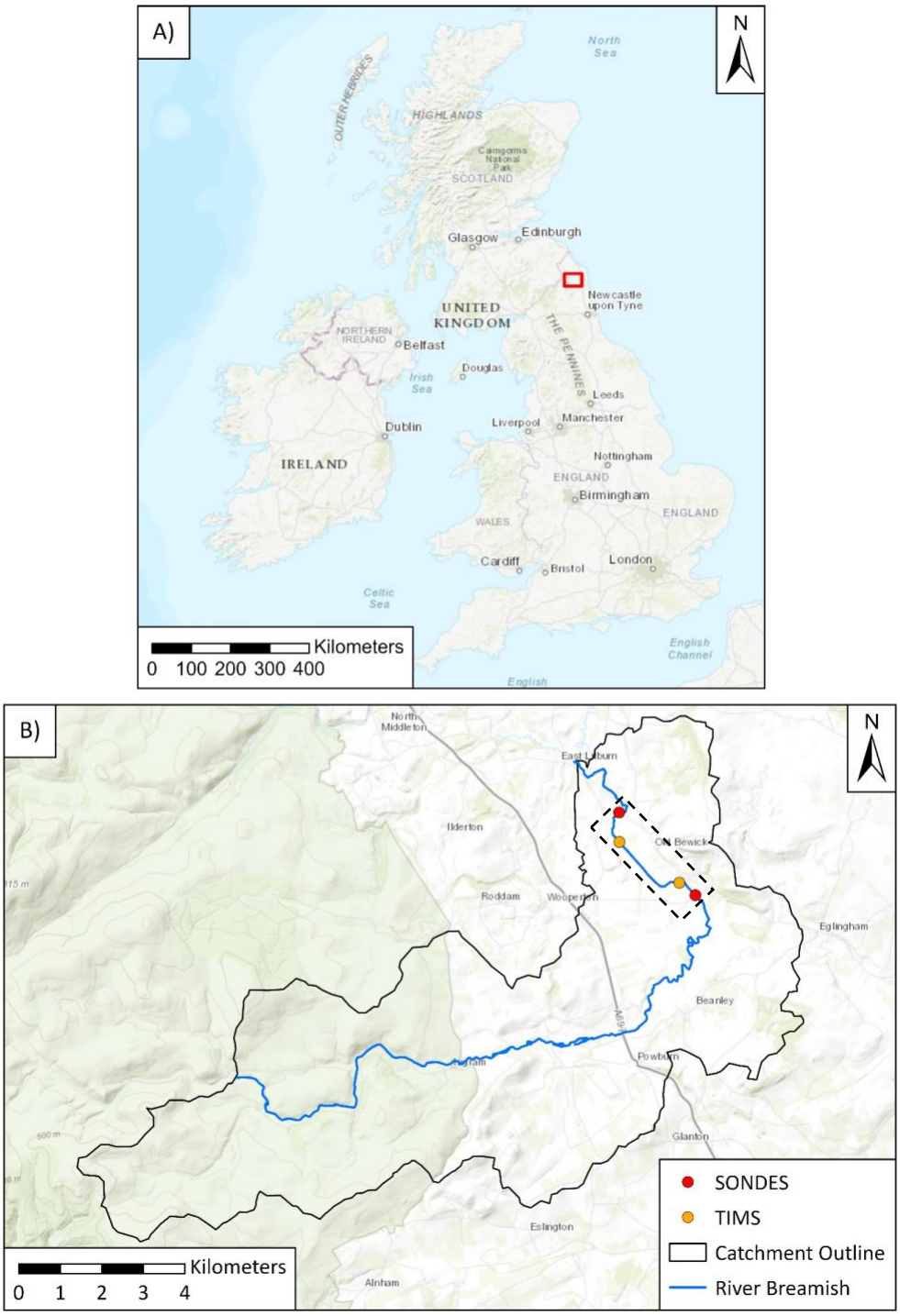


Figure 1. Study site map showing the River Breamish catchment at A) National scale and B) Drainage basin scale. The red rectangle indicates the location of map B within map A, and the black dashed rectangle indicates the River Breamish restoration area. Basemap: World Topographic Map, Esri OS Open Map, Available at: www.arcgis.com/home/webmap/viewer.html?useExisting=1&layers=30e5fe3149c34df1ba922e6f5bbf808f



Figure 2. Drone imagery showing the existing River Breamish channelised reach (left) and an interpretation of the proposed post-restoration landscape (right)
(Images Source: Williams et al., 2023, pp.4-5)

1.3 Research Problem + Study Significance:

Current channelised conditions within the Breamish reduce lateral hydrological connectivity and access to temporary storage within the adjacent floodplains. Elevated macronutrient concentrations in flux within the Breamish contribute to opportunistic macroalgae growth downstream at the North Northumberland coast, consequently smothering seagrass and marine infaunal species (LIFE WADER, 2024). There are outstanding scientific questions surrounding the effectiveness of floodplain reconnection for mitigating such impacts of elevated fine sediment and macronutrient loads within fluvial systems. This research forms an integral part of Natural England's LIFE WADER project, delivering a comprehensive pre-restoration baseline audit of fine sediment and macronutrient dynamics which will act as a reference for post-restoration monitoring. This will enable future evaluation of the effectiveness of floodplain reconnection, ensuring societal water quality and ecosystem management goals are met. Moreover, there is a prevalent underrepresentation of diffuse source pollutants in contemporary monitoring programmes, addressing this through the methodologies employed within this research is critical for future environmental monitoring and management as well-informed strategies are imperative for effectively targeting restoration.

1.4 Research Questions:

1. What are the critical source areas of fine sediment, nitrate and phosphate contributing to poor water quality pre-restoration within the River Breamish?

2. What is the baseline nutrient composition of sediment flux and pre-restoration pressures impacting delivery and transport within the River Breamish catchment?
3. Is floodplain reconnection a practical and effective solution for reducing downstream sediment transport and improving water quality within agricultural catchments like the River Breamish?

1.5 Scope of Research:

1.5.1 Aim:

To quantify the fundamental behaviour of fine sediment and macronutrient dynamics within the River Breamish catchment, prior to restoration in June 2025.

5.1.2 Objectives:

1. To acquire and quality control high-resolution hydrometeorological (river level) and water quality (turbidity) datasets between February 2024 to May 2025 from two Environment Agency monitoring stations located upstream and downstream of the River Breamish Restoration area.
2. To assess sediment source proximity through river level-turbidity hysteresis analysis and implement a hysteresis index (Lawler *et al*., 2006) to determine the hysteric strength.
3. To determine the particle size and nutrient composition of sediment flux at regular intervals through analysis of material captured by time integrated sediment samplers during pre-restoration sampling (February 2024 to October 2024) (Phillips, Russell and Walling, 2000).
4. To identify the critical source areas delivering fine sediment and macronutrient pollutants within the River Breamish catchment using the SCIMAP model (Reaney *et al.,* 2011).
5. To ground truth SCIMAP outputs during fieldwork, identifying potential pollutant sources and transport pathways where land connectivity facilitates delivery into the channel.
6. To integrate data from objectives 1-5, to provide a pre-restoration baseline of fine sediment and macronutrient dynamics, against which post-restoration conditions can be evaluated.

Chapter 2 – Literature Review:

This chapter is a contextual literary review of the academia and policy underpinning this research. First, a conceptual model is utilised to introduce the complexity of fine sediment dynamics from source, mobilisation and delivery including potential lags / delays. Then, the primary sources and mobilisation pathways of sediment and macronutrient pollutants are subsequently outlined. Next, the factors impacting pollutant delivery are discussed including the biogeochemical transformations controlling macronutrient cycling, retention and delivery. Building on this, the critical source area approach is introduced and model development is outlined. Then, the complex biogeochemical impacts of historic channelisation and floodplain disconnection are discussed and the success of floodplain reconnection within Europe and North America reducing macronutrient loadings and downstream transport is highlighted. Finally, the issues with UK regulatory management and monitoring will be summarised and the planned River Breamish restoration will be introduced, justifying this study as enhancing knowledge, addressing policy omissions and progressing towards developing guidelines to ensure societal water quality / ecosystem management goals are met.

2.1 Controls on Fine Sediment dynamics:

Fine sediment (<63µm), composed of organic and inorganic matter (Chang, 1988), is a natural element of freshwater systems, critical for ecosystem functioning (Owens et al., 2005). Due to an inherent complexity, fine sediment is one of the least understood aspects of physical sciences, consequently the relevance of research contributing to environmental management regimes has only recently been acknowledged (Fielding, 1993; Zhang, 2014). The quantity and quality of fine sediment available for delivery within freshwater systems is controlled by various interconnecting factors (White, 2008), impacted by natural and anthropogenically enhanced processes (Figure 3). Once fine sediment is delivered into the channel particles are either transported through the system, temporarily stored as surface deposition, or infiltrated into the bed for longer term storage (Wilkes et al., 2019). The impact of fine sediment within fluvial systems is subject to potential lags due to temporary storage / resuspension and delayed ecological response (Figure 3) (Wilkes et al., 2019).

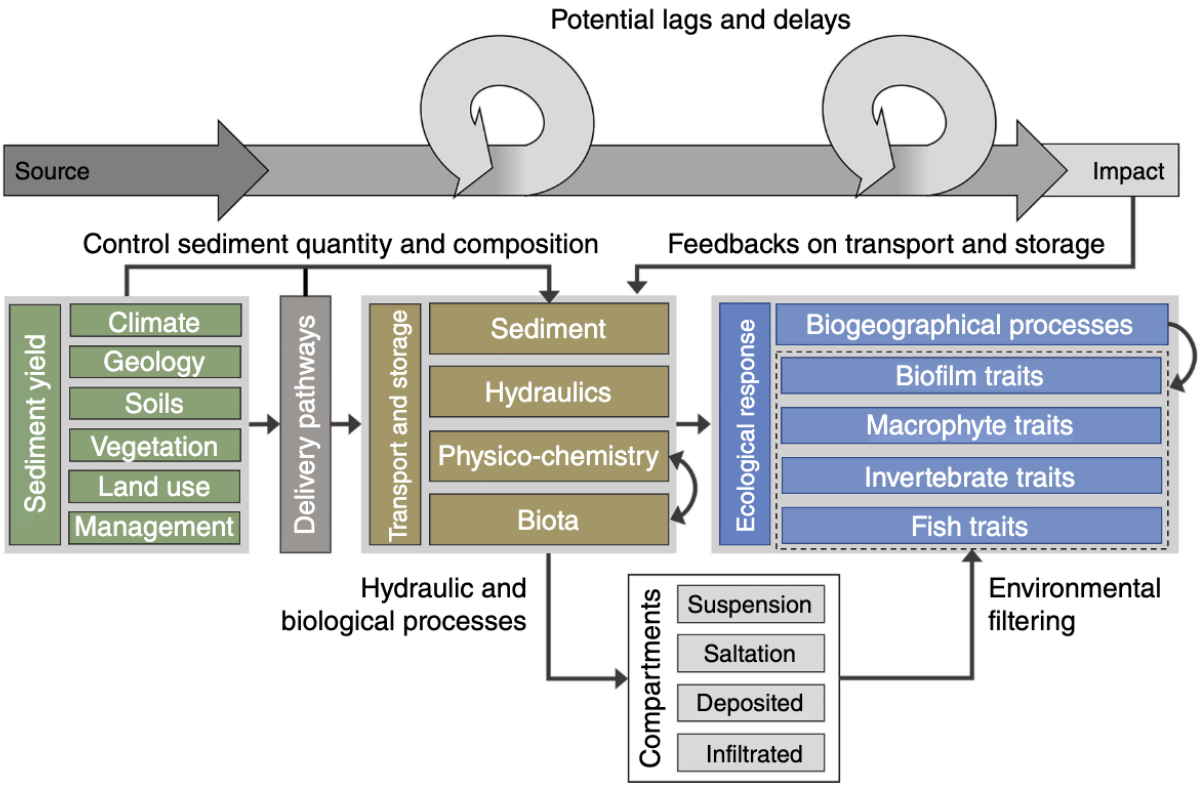


Figure 3. Natural and anthropogenic controls of the source, delivery, transportation / storage of fine sediment and related ecological response within fluvial environments. (Source: Wilkes et al., 2019, p.2, Figure 1)

2.1.1 Fine Sediment Sources + Availability:

Complex natural and anthropogenic factors combine to influence a catchment's erosional processes that control fine sediment supply (Reid *et al.,* 1997). Sedimentary and nutrient sources within fluvial environments are often split into point and diffuse sources to enable development of management practice. Point sources originate from singular sites and discharge directly into the watercourse, whereas diffuse sources originate from on the adjacent land within the wider catchment (Novotny, 2003). Changing land use during the Anthropocene continues to negatively impact the environment through progressive increase of sediment and macronutrient flux (Walling and Fang, 2003) such that current levels exceed pre-industrial conditions (Jones *et al.,* 2011). Agricultural intensification has increased fine sediment delivery by ~20%, thus it is recognised as a major sedimentary source within fluvial environments (Wilkinson and McElroy, 2007). Within agricultural catchments fine sediment predominantly arises from widespread diffuse sources. Vegetation cover exerts critical control on a catchment's sedimentary supply and hydrological regime (Knighton, 1998), protecting the riverbank from soil erosion, preventing

compaction and reducing overland flow (Aber, Marzolff and Ries, 2010). Thus, arable headwaters with poor ground cover represent a critical source of fine sediment, especially where steep sloping fields are situated adjacent to the channel (Hudson and Soar, 2023). Vegetation in areas proximal to the channel is highly susceptible to anthropogenic disturbance, whereby the natural vegetation is stripped to allow agricultural activities, consequently increasing fine sediment availability and delivery into fluvial systems (Knighton, 1998; Aber, Marzolff and Ries, 2010). Excess sediment supplied from riverbanks can be attributed to discharge conditions and damage resulting from access (White, 2008). Trampling of riparian zones from livestock and humans removes vegetation, compacts soil and destabilises the structure of the riverbank, consequently increasing surface runoff and elevating fine sediment availability near the channel (Heathwaite, Burt and Trudgill, 1990). Thus, promoting catchment wide sustainable land practice is critical for reducing fine sediment inputs within agricultural catchments.

2.1.2 Fine Sediment Mobilisation Pathways + Transport Mechanisms:

The mobilisation of fine sediment into the channel represents a critical process impacting the delivery of pollutants, subsequently deteriorating downstream water quality. The main mechanisms within agricultural catchments include overland flow, subsurface flow and bank erosion (Palmer, 2012). Diffuse sedimentary sources are strongly event driven, whereby intense rainfall triggers fine sediment mobilisation. For example, within headwater areas enhanced sediment erosion via the mechanisms of rainsplash and sheet wash occurs as a result of intense rainfall events (Kinnell, 2005; Stella et al., 2013), consequently, saturation excess overland flow transports fine particles into the channel (Hudson and Soar, 2023). Where rainfall events exceed soil infiltration capacity, surface runoff rapidly transports loose particles downslope towards the channel as a dominant pathway within arable headwaters and areas of compacted soil surfaces (Williams-Mounsey et al., 2024). Contemporary research argues that tractor wheel tracks create channels of concentrated surface runoff which further enhance the erosion and delivery of fine sediment into the river channel (Williams-Mounsey et al., 2024).

Within the river channel, fine-grained sediment is transported via the mechanisms of suspension / saltation as flocculated rather than discrete particles (Droppo, 2001). The inherent cohesion of fine grains causes the formation of bonds between particles, leading to aggregation (Zhang, 2014). Previous research suggests that suspended sediment load is supply limited, determined by

the quantity of fine sediment supplied from riverbanks and the adjacent landscape with external processes influencing the highly episodic nature of suspended sediment load (Knighton, 1998). However, recent studies found that fine sediment transport via suspension is positively correlated with flow intensity in rivers close to a natural flow regimes (Wohl *et al.,* 2015), suggesting transport function as a further control (Reid *et al.,* 1997). Fine sediment transport is strongly event driven, with concentrated flux occurring during high flow conditions. This theory is evidenced within European catchments where ~70% of suspended sediment transport occurs during the top 20% of flows (White, 2008). In historically channelised systems, such high-flow conditions increase the transport capacity as incised reaches increase stream power and reduce natural attenuation (Wohl *et al.*, 2015). The scientific robustness of contemporary understanding of fine sediment transport is limited by a lack of standardised methodology for sampling and measuring fine particles, thus leading to assumptions often being made within research (Droppo, 2001). The supply and transport of fine sediment within a drainage basin is overall influenced by a range of processes related to anthropogenic disturbance on various spatio-temporal scales (Reid *et al.,* 1997).

2.2 Controls on Macronutrient Dynamics:

2.2.1 Macronutrient Pollutant Sources + Availability:

There are four basic forms of phosphorus (P) within freshwater environments: dissolved inorganic / soluble reactive phosphate (PO_4), dissolved organic, particulate organic and particulate inorganic (Allan, Castillo and Capps, 2021). Dissolved inorganic P is immediately bioavailable for aquatic produces and micro-organisms, whereas dissolved organic and particulate P particles require chemical transformations to become soluble reactive and bioavailable within the system (Weigelhofer, Hein and Bondar-Kunze, 2018). Nitrogen (N) occurs in three forms: dissolved inorganic N (including ammonium (NH_4), nitrate (NO_3) and nitrite (NO_2), dissolved organic N and particulate organic N (Allan, Castillo and Capps, 2021). Similarly to P, dissolved inorganic N is readily available. Within fluvial environments, NO_3 and PO_4 are the primary macronutrient types impacting the systems productivity and nutrient cycling, with elevated concentrations impeding water quality, leading to eutrophication (Carpenter et al., 1998).

The natural functioning of fluvial systems faces significant challenges due to the negative legacy of agricultural expansion, increased use of artificial fertiliser and enhanced erosional processes. Due to its mobility in soils and susceptibility to leaching following application of artificial fertilisers, NO_3 is the dominant species of N in agricultural runoff (Withers and Jarvie, 2008). PO_4 is less mobile than NO_3, with sources including soil erosion, fertiliser inputs and wastewater point source discharge (Withers and Jarvie, 2008). Environmental reconstruction models indicate that the rise in macronutrient flux between 1800-2010 corresponds with a rapid increase in fertiliser use on arable land following World War 2 (Bell *et al.,* 2021). Delivery of material enriched with macronutrients from losses during crop production principally accelerates eutrophication within aquatic systems (Ekholm *et al.*, 2000). Early 21st century research identifies that agricultural land use elevates the risk of macronutrient losses at a field scale (Withers and Lord, 2002); this is evidenced where NO_3 and P inputs continuously exceeds production requirements and farming practice increases runoff and erosional processes (Withers and Lord, 2002). Contemporary research suggests agricultural activities are the largest source of NO_3 within freshwater systems (Bell *et al.,* 2021), however, underlying theory indicates NO_3 and P losses are spatially variable due to differing soil characteristics and land use practice (Ekholm *et al.*, 2000). A key research gap is presented as the impacts on water quality and ecology are poorly characterised on a catchment scale due to the complex process-based modifications (Jarvie *et al*., 2008). Overall, macronutrient sources vary depending on pollutant composition including concentration, speciation, bioavailability and mode / timing of delivery, therefore diffuse nutrient sources challenge the management of pollutants within freshwater environments as a combination of mitigation measures on both a catchment and reach scale are necessary (Withers and Jarvie, 2008). Identification of fine sediment / nutrient sources and delivery pathways is critical for developing strategies to restore the ecosystem's natural functioning (Jarvie *et al*., 2008).

2.2.2 Mobilisation and Transport of Macronutrient Pollutants:

Macronutrient mobilisation within agricultural catchments is governed by hydrological processes and land use / management practice (Heathwaite and Dils, 2000). NO_3 is predominantly transported through subsurface pathways due to its high solubility (Heathwaite, Sharpley and Gburek, 2000). These include groundwater flow and interflow which rapidly mobilise NO_3 within well drained soils and leaches following periods of high rainfall (Heathwaite, Sharpley and Gburek,

2000). Storm events often lead to overland flow within artificially fertilised fields which further elevates in-channel NO_3 concentrations (Jordan et al., 2007). Comparatively, PO_4 is strongly particulate-bound and is mobilised alongside sediment via erosion and surface runoff pathways (Ballantine, Walling and Leeks, 2009). During rainfall storms, sheet wash detaches sediment bound PO_4 from the landscape, subsequently transporting it into the channel (Ballantine, Walling and Leeks, 2009). Cultivated land with high amounts of fertiliser use proximal to the channel contributes more PO_4 enriched runoff than un-cultivated areas (Sharpley and Smith, 1990). Additionally, sediment associated PO_4 transport during storm events increases with the amount and intensity of rainfall (Fraser, Harrod and Haygarth, 1999). Studies indicate that once in the channel >90% of macronutrient flux occurs in association with suspended sediment transport (Walling et al., 2003). Therefore, the transfer and dispersal of sediment bound nutrients is strongly governed by the transport, deposition and remobilisation of suspended sediments (Walling et al., 2003).

2.3 Application of Connectivity for Understanding Delivery:

The delivery of fine sediment, NO_3 and PO_4 into a river channel is fundamentally governed by hydrological connectivity and the surrounding landscape characteristics. The connectivity concept is central to understanding delivery mechanisms in fluvial catchments, with emphasis shifting from traditional runoff models which prioritise infiltration excess overland flow as the dominant delivery mechanism (Bracken and Cooke, 2007). Instead, connectivity theory emphasises the spatio-temporal variability of material delivery occurring due to heterogeneous catchment landscapes (Najafi et al., 2021). For example, landscape features including tractor wheel tracks, ditches and field drainage channels create preferential flow pathways which bypass retention zones, consequently enhancing delivery (Withers et al. 2006). Conversely, land disconnection occurs due to vegetation buffers constraining sediment delivery, barriers preventing longitudinal flow and surfaces inhibiting fine sediment entrainment, consequently decreasing connectivity (Wainwright et al., 2011). The connectivity framework considers the interconnectivity and interactions between catchment characteristics such as landscape, climate, flow pathway and lateral hydrological connectivity to understand the transfer of materials from source areas into the river channel (Bracken et al., 2015). However, there is a lack of standard definition, consequently limiting the application of connectivity within empirical research (Bracken and

Cooke, 2007). Quantification of connectivity remains a significant challenge as current metrics do not effectively capture the dynamic and episodic nature of material transfer from the landscape into the channel (Preston and Schmidt, 2003). Consequently, the application of connectivity within research is predominantly informed by conceptual understanding of delivery pathways (Bracken et al., 2013). Critical source areas represent areas where high pollutant availability coincides with strong hydrological connectivity, consequently delivering disproportionate pollutant loads into the channel (Pionke, Gburek and Sharpley 2000). Identification and management of critical source areas is essential for targeting management strategies for improving water quality and ecology within agricultural catchments.

Moreover, multiple processes lead to the ubiquitous deposition of fine sediment within fluvial systems, for example discharge modifications, changing bed geomorphology and changes to adjacent land use (Leitner et al., 2015). Deposition predominantly occurs in low energy environments including channel margins, near woody debris and on connected floodplains during overbank flow (Noe and Hupp, 2005). The frequency, duration and depth of inundation controls sediment accumulation on floodplains with high frequency and prolonged inundation enhancing deposition rates and nutrient retention (Noe and Hupp, 2005). Fine sediment particles are also deposited in the riverbed and subsequently resuspended (Preziosi-Ribero et al., 2020) due to variations in surface flow conditions, driven by natural storm events (Drummond et al., 2017).

2.4 Biogeochemical Transformations Facilitating Macronutrient Cycling:

Once within the river system, P and N macronutrients undergo multiple biogeochemical transformations (Figure 4) including heterotrophic and autotrophic uptake of nutrients and release of nutrients through excretion and microbial decomposition (Weigelhofer, Hein and Bondar-Kunze, 2018). Biogeochemical transformations of macronutrients from dissolved to particulate / particulate to dissolved are critical for understanding macronutrient delivery and deposition within fluvial systems. High flows not only enhance macronutrient transport but also drive these key biogeochemical transformations. Nutrient release from organic matter occurs due to the drying and rewetting of floodplain sediments (Schönbrunner, Preiner and Hein, 2012). Within inundated floodplain environments, under anaerobic conditions denitrifying bacteria uses NO_3 as an electron acceptor to oxidise organic matter and convert NO_3 into inert N gas, thus permanently removing it from the system (Weigelhofer, Hein and Bondar-Kunze, 2018).

In addition to biological processes, P is also influenced by multiple physio-chemical transformations. P adsorption to sediment particles is critical for P cycling within agricultural catchments, where P pollutants are predominantly bound to particles (Weigelhofer, Hein and Bondar-Kunze, 2018). Sediment bound P retains adsorbed nutrients, however these can be released into the water depending on sediment adsorption capacity (highest in clay and sand textured sediments) (Weigelhofer, Hein and Bondar-Kunze, 2018). Particulate-bound PO_4 is desorbed under low redox potential and anoxic conditions, resulting in the transformation into bioavailable dissolved reactive P (Withers and Jarvie, 2008). This contrasts with the oxic conditions associated with high flows where enhanced erosion and mobilisation of P rich fine sediment contributes to elevated macronutrient loads (Sharpley et al., 2011). Generally, adsorption occurs where there are high concentrations of soluble reactive P, whereas desorption occurs where there are low concentrations. The zero equilibrium phosphorus concentration (EPC_0) refers to the concentration of soluble reactive P where no net exchange of P occurs. The EPC_0 value is high within agricultural catchments as a result of P overloading, consequently sediment functions as an internal P source (Sharpley et al., 2013). Overall, both biotic and abiotic processes combine to influence macronutrient retention in river systems, with fine sediment accumulations acting as a potential sink and source for P.

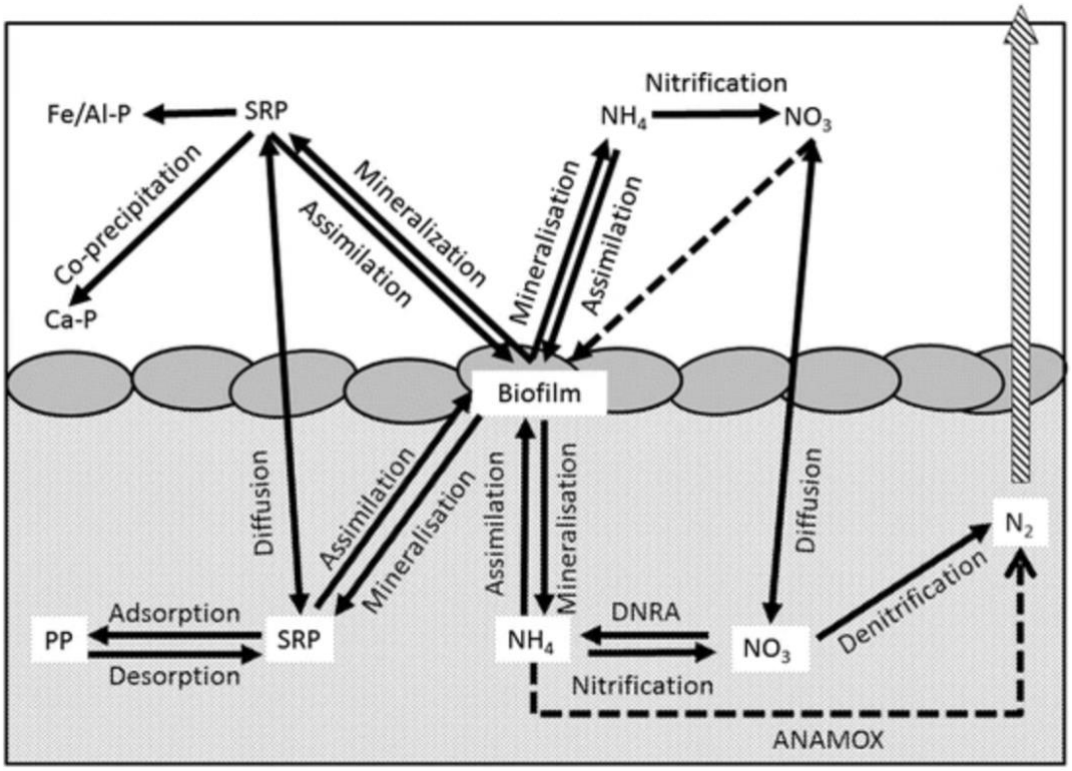


Figure 4. Dominant transformation processes for macronutrients in fluvial systems. (Source: Weigelhofer, Hein and Bondar-Kunze, 2018, p.191, Fig.10.2)

2.5 Critical Source Area Approach + Model Development:

The water quality of European waterbodies has significantly improved since the 1990s, largely due to implementation of contemporary policies for tackling point source pollutants (Reaney *et al.*, 2019). However, diffuse sources remain a prominent threat to water quality, acting as a barrier to achieving 'good status' under the Water Framework Directive (Council of the European Communities, 2000). The spatial and temporal heterogeneity of diffuse pollution poses additional challenges to source identification, thus impeding management and regulation (Reaney *et al.*, 2019; Patterson, Smith and Bellamy, 2013).

Understanding the delivery of material within drainage basins represents a longstanding concern within environmental science; historically, research followed a predominantly descriptive framework, largely based on empirically derived formulas and field observations (Yang, 1996). Conceptual development of critical source area models began during the 1980s where Walling (1983) identified continuous development of spatially distributed models as essential for understanding the complex interrelationships between the environmental factors and processes

impacting material delivery (Walling, 1983). However, models during this period could not include all relevant processes to accurately predict sediment yield due to the high-resolution data requirements (De Vente *et al.*, 2007).

Technological advances during the early 21st century facilitated a quantitative shift in research and wider application of critical source area models. The Universal Soil Loss Equation, proposed in 1978 (Wischmeier and Smith, 1978) and revised in 1997 (Renard et al., 1997), was pivotal in the evolution of quantitative modelling through application of an empirical, field derived index. The equation calculates mean annual soil loss through considering catchment factors including rainfall, runoff, soil erodibility, crop management and slope characteristics (Pham, Degener and Kappas, 2018). However, this approach is limited as it does not account for the spatial variability / disproportionate contribution of material into the channel between areas where the same land use is observed and can only be applied for long periods of time (Charlton, 2007).

Over the last few decades, the critical source area approach has evolved such that at present, there is a plethora of sophisticated models available to accurately simulate and predict soil erosion on various spatio-temporal scales. Inclusion of GIS and remote sensing technologies supported incorporation of spatially variable inputs including hydrological connectivity, vegetation cover and sediment transport capacity, thus expanding the capabilities of spatially distributed models for identifying critical source areas (Reaney *et al.*, 2011). The SCIMAP model captures the complexity of localised delivery processes within small scale systems (Reaney *et al.*, 2019), allowing rigorous hypotheses testing and providing a framework to consider the source of diffuse pollutants and identify areas where mitigation would be most effective (SCIMAP, 2025). Determination of diffuse pollutant critical source areas is achieved through application of models whereby high-risk sources and strong pathway connections are combined (Reaney *et al.*, 2019; Lane, Reaney and Heathwaite, 2009). Research suggests that agricultural activities account for approximately a third of the total diffuse pollutant sources within UK river systems (Parliamentary Office of Science and Technology, 2014); therefore, application of the critical source area modelling within agricultural catchments is critical for facilitating targeted management practice to improve water quality standards (Reaney *et al.*, 2019).

2.6 Floodplain Dynamics:

2.6.1 Biogeochemical Impacts of Floodplain Disconnection:

Sediment / macronutrient dynamics exhibit additional complexity within anthropogenically modified catchments, particularly in channelised systems where increased water surface slope and velocity during high flow events accelerate bed erosion, generating surplus sedimentary input through bank incision and undercutting (Todd-Burley *et al.,* 2021). Due to the cohesive properties, fine-grained particles are easily mobilised during peak flows (Zhang, 2014), consequently downstream sedimentation occurs. This shows how historic river management inhibits the natural functioning of freshwater systems (Hohensinner, Hauer and Muhar, 2018), with floodplain disconnection following channelisation fundamentally altering the channel morphology and natural flow regime (Arnaud *et al*., 2019). The longstanding, detrimental impacts of channelisation can be managed and reduced by reconnecting the channel with the adjacent floodplains (Walling, Owens and Leeks, 1998).

A functional floodplain must be laterally connected to the river channel, facilitating sediment and macronutrient exchange (Figure 5) (Opperman *et al*., 2010; Thoms, 2003). This is closely related to longitudinal flow, depending on lateral hydrological connectivity (Zeiringer *et al.,* 2018), thus, floodplain disconnection reduces the potential for sediment retention, catalysing sediment transport and increasing accumulation risk in downstream areas with inefficient filtration (Noe and Hupp, 2005). Between 70-90% of European floodplains are classified as environmentally / ecologically degraded due to anthropogenic activities and poor land practice altering vital sediment deposition and macronutrient cycling processes (European Environment Agency, 2016; Hohensinner *et al.,* 2004). Consequently, habitat degradation occurs and biodiversity declines (Kennedy and Turner, 2011). Floodplain reconnection restores the depositional functions and enhances sediment retention by creating a sediment sink, consequently reducing suspended sediment concentrations in the channel (Sear et al., 2010). Additionally, a connected and functioning floodplain enhances nutrient retention through increasing residence time and creating conditions for microbial processing, thus it is effective in reducing nutrient export during high flow events (McMillan and Noe, 2017). Overall, channelised and disconnected reaches experience accelerated downstream pollutant transport (Sear et al., 2010), while connected, low-

energy, vegetated floodplains act as natural sediment traps, promoting deposition during overbank flow events and reducing pollutant fluxes (McMillan and Noe, 2017).

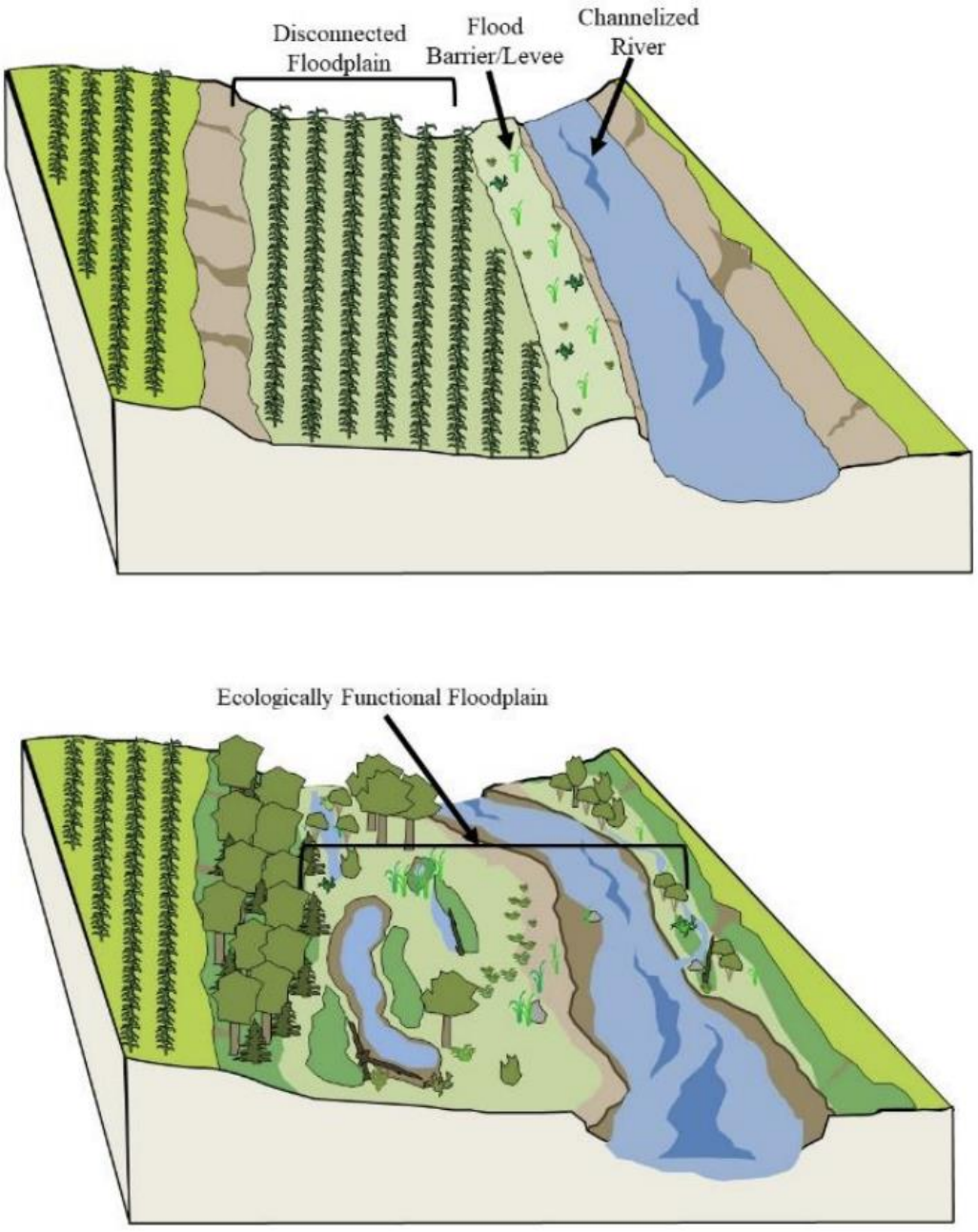


Figure 5. Conceptual diagram showing the difference between a disconnected non-functional floodplain with an ecologically functional floodplain that exhibits lateral hydrological connection with the river channel.
(Source: Gordon, Dorothy and Lenhart, 2020, p.2, Figure 1)

2.6.2 Floodplain Reconnection / Restoration Schemes:

Given the increasing degraded state of European floodplains, implementing effective restoration to improve water quality by reducing downstream sediment transport and macronutrient loading is imperative. Following restoration, key water quality improvements will occur by routing nutrient rich water / sediment through floodplains to enable removal of pollutants via plants and microbial activity (Robotham and England, 2025). Localised estimates of sediment / macronutrient retention are critical for identifying target areas for restoration (Hopkins *et al*., 2023).

Research in Europe and North America shows that floodplain reconnection is viable and effective in restoring biogeochemical functioning and reducing agriculturally elevated loads of NO_3 by ~200kg N ha^{-1} $year^{-1}$ and total P by ~21kg-P ha^{-1} $year^{-1}$ (Gordon, Dorothy and Lenhart, 2020). There are multiple examples of catchments where floodplain reconnection has successfully increased macronutrient retention, including the River Danube and River Rhine (Schneider, 2010). Notably, within the Upper Sandy Creek catchment, North Carolina, a previously channelised reach was reconnected to the floodplain and adjacent wetlands through a multi-phased restoration scheme (Richardson et al., 2011). Resultantly, the riparian system's hydrological functioning increased, reducing downstream sedimentary and macronutrient pulses (Richardson et al., 2011). NO_3 and PO_4 loads were reduced by 64% and 28% respectively and increased sediment retention totaling ~500 tonnes/$year^{-1}$ was evidenced by riparian wetland area accretion rates of ~1.1cm/$year^{-1}$ (Richardson et al., 2011). This shows the success of floodplain restoration measures similar to those proposed in the River Breamish, increasing lateral connectivity and reducing fluxes of sediment and macronutrients to overall improve water quality.

However, the scale of restoration success is dependent on multiple interconnecting factors. For example, shear stress assessments within floodplain environments indicate that fine-grained sediment deposition and consequential nutrient deposition are both variable and limited depending on the frequency and duration of floodplain inundation (Remo and Krienert, 2022). Further studies have identified restoration age as an additional control on the extent of macronutrient retention, as the development of stable vegetation increases soil carbon and accelerates microbial activity, subsequently driving macronutrient removal via denitrification and N / P mineralisation processes (Gordon, Dorothy and Lenhart, 2020). Overall, floodplain

restoration schemes offer multiple benefits for fluvial environments, including increasing sediment stores, improving water quality and increasing biodiversity (Robotham and England, 2025). While floodplain reconnection has gained recognition as an effective strategy for reducing sediment and nutrient fluxes, there is limited integration and use of critical source area modelling outputs to identify target intervention locations. Thus, floodplain restoration schemes are often implemented based on geomorphic and ecological criteria rather than hydrological delivery risk (Forbes, Ball and McLay, 2015).

2.7 Issues with UK / EU Regulatory Management + Monitoring:

Despite the ecological importance of sediment and macronutrient dynamics, suspended sediment was historically excluded from environmental quality standards and compliance targets under the EU Water Framework Directive (Perks *et al.,* 2017). However, in the last four years UK environmental policies have developed to ensure fine sediment has a higher management priority (UK Environment Act, 2021). Resultantly, despite the known role in degrading ecology, transporting macronutrient pollutants and altering nutrient cycling, fine sediment related pressures have historically been poorly represented in river basin management plans. Effective management was further impeded by monitoring programmes prioritising point source pollution and underestimating the detrimental impacts of the agriculturally elevated loads and diffuse sediment inputs that are prominent within UK catchments (McGonigle et al., 2012).

Moreover, while floodplain management is not directly required by European policy, restoration of natural conditions supports the achievement of multiple legally binding objectives (European Environment Agency, 2016). For example, under the EU Nitrates Directive, promotion of good farming practice to prevent agricultural sources of NO_3 from polluting watercourses is legally required (European Union, 1991). Floodplain reconnection has only recently been acknowledged as critical for reducing macronutrient pollution to achieve 'good' water quality status (European Union, 2015). However, as this is non-legally binding, scientifically justifiable and robust targets cannot be set (Kemp *et al.,* 2011).

2.8 Concluding Remarks + Research Justification:

Key scientific questions remain about how floodplain restoration can mitigate the impacts of elevated sediment and macronutrient loads within fluvial systems. This study builds upon existing

theory and research to develop an enhanced understanding of fine sediment / macronutrient dynamics and provide a pre-reconnection baseline audit within the River Breamish, Northumberland. A 1km stretch of previously straightened channel (currently classified as unfavorable and declining) is undergoing works to restore ~50ha of floodplain wetland habitat in aim of restoring natural system functioning and improving water quality (Williams *et al.*, 2023). Current channelisation reduces lateral hydrological connectivity and potential access to temporary storage within the floodplains, consequently contributing to downstream opportunistic macroalgae growth which smothers seagrass and marine infaunal species (LIFE WADER, 2024). This research forms an integral part of Natural England's LIFE WADER project, delivering a comprehensive pre-restoration baseline audit which will inform the design of post-restoration monitoring strategies to ensure the societal water quality / ecosystem management goals are met. The study also addresses the prevalent underrepresentation of diffuse source pollutants in contemporary monitoring programmes, this is critical for future environmental monitoring and management, as well-informed strategies are imperative for effectively targeting restoration.

Chapter 3 – Methodology:

3.1 Study Site:

The River Breamish is a main tributary of the River Till, Northumberland, draining a catchment area of 90km^2 (DEFRA, 2023). Situated within the Till Catchment Site of Scientific Interest (SSSI) and The River Tweed Special Area of Conservation (SAC) (Tweed Forum, 2024), the Breamish flows north west from its source in the Cheviot Hills to confluence with the River Till at Bewick Bridge (Chessell, 2014). The catchment is designated a SSSI and SAC due to the diverse landscape, habitats and biodiversity; many vulnerable aquatic macrophytes and species of conservation value inhabit the Breamish (Chessell, 2014). Under natural conditions this section of the Breamish would be characterised by riffle-pool / wandering channel morphology, meandering through the alluvial valley floor, fringed by fluvioglacial deposits (Chessell, 2014). However, embankments were implemented to confine the channel between the late 18th and early 19th century to enable agricultural access to fertile loamy floodplain soils (Sykes, 1933). Due to the historic channel straightening and lack of contemporary management, the reach currently exhibits limited floodplain connectivity and is classified as unfavorable and declining (LIFE WADER, 2024). The surrounding land use includes agricultural / arable farms, pastureland, woodlands and rural estates (Northumberland County Council, 2010).

The Breamish was selected as the study site as it represents a key example of historic management of river systems inhibiting natural functioning. This research focuses on a 1km stretch of previously straightened channel undergoing works to restore ~50ha of floodplain wetland habitat in aim of restoring the natural system functioning and improving water quality (Williams *et al.*, 2023). The ambitious restoration project involves reconnecting the River Breamish with its western floodplain through excavating a new channel, embankment breaches and construction of woody dams in the existing channel (LIFE WADER, 2024). It is key to note that there are three additional tributary inputs present within the channelised reach, located between the upstream and downstream monitoring sites (Figure 6).

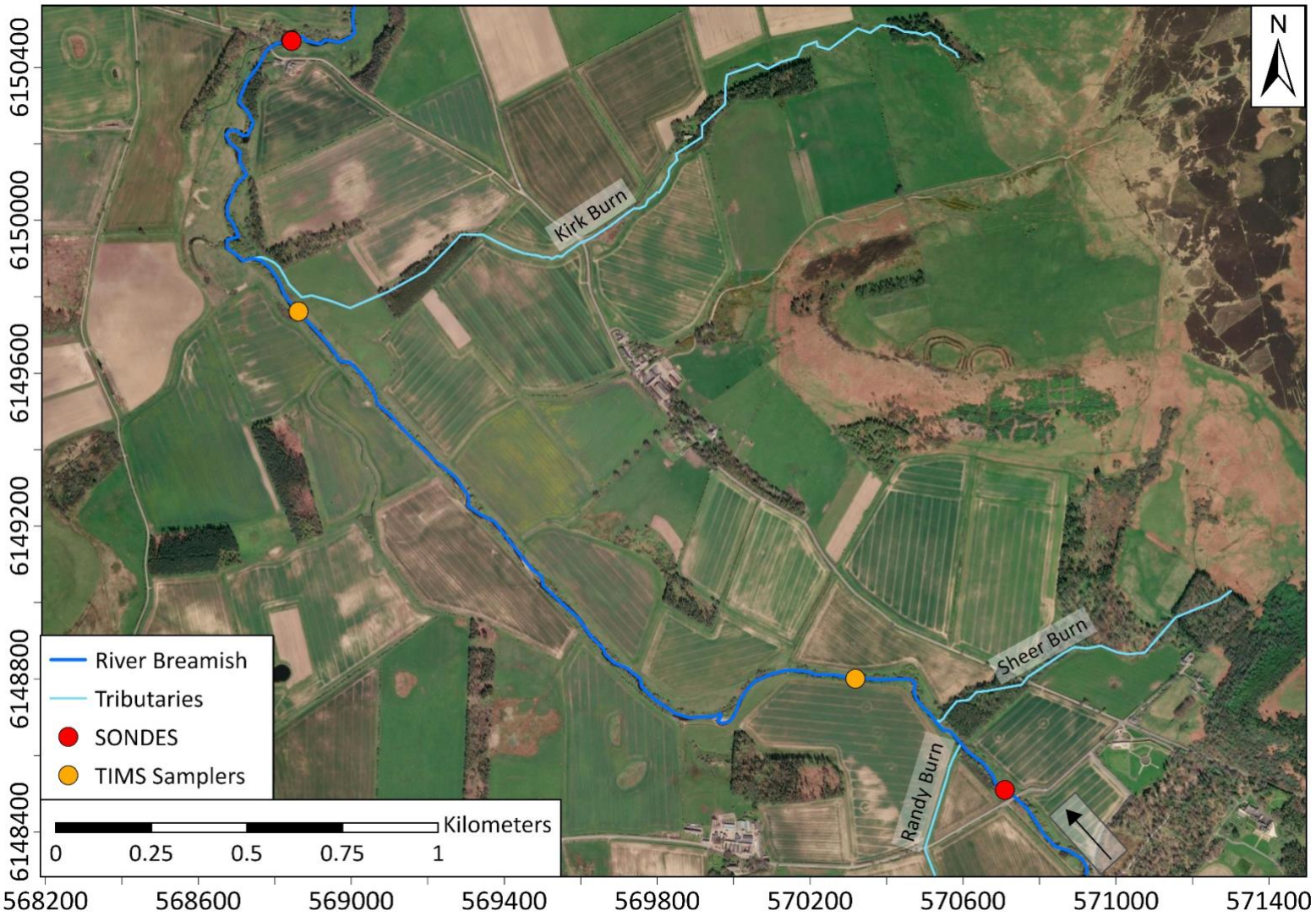


Figure 6. Map showing the River Breamish restoration area, illustrating the location of two sondes monitoring stations and three TIMS sediment samplers (downstream TIMS and sondes overlap). The direction of flow from south east to north west is indicated by the arrow.

3.2 Sample Design:

3.2.1 Data Sources:

The secondary data acquired to enable this research and analysis are detailed in Table 1.

Table 1. Summary of secondary data sources.

Data:	Spatial resolution:	Application:	Source:
River level data		Hysteresis analysis	Environment Agency sondes
Water quality data (multiparameter)		Hysteresis analysis	Environment Agency sondes
Digital elevation model	5m	SCIMAP input	(SCIMAP, 2025)
Landcover	10m	SCIMAP input	(Morton *et al.,* 2011)
Precipitation	25km	SCIMAP input	(Met Office, 2025)
Composite LIDAR digital terrain model	1m	Channel geometry / cross sectional area	(DEFRA, 2022)
Time integrated sediment samples		Sediment and macronutrient flux analysis	Environment Agency; Tweed Forum; UKCEH
Catchment outline		Producing maps and accurately clipping raster datasets	(DEFRA, 2023)

3.2.2 Monitoring Data Acquisition:

High frequency monitoring data (~30 minute intervals) was obtained from two Environment Agency EXO3 Multiparameter Sondes (YSI, 2024) located upstream [55.4775°, -1.8810°] and downstream [55.4955°, -1.9102°] of the River Breamish restoration area (Figure 6). Data collection spanned the duration of the pre-restoration study (February 2024-May 2025) with the aim of assessing fine sediment dynamics and indicating source proximity. These datasets are appropriate as point measurements correlate with mean cross sectional concentrations (Gray and Gartner, 2009) and Environment Agency data is widely utilised by government agencies for environmental monitoring / management (Davies, 2023). Turbidity and river level were the parameters selected for analysis of fine sediment dynamics, as turbidity is the most commonly utilised surrogate for suspended sediment within fluvial research (Gray and Landers, 2014).

3.3 Data Processing + Quality Control:

3.3.1 Testing For Interval Consistency:

Prior to analysis, the monitoring data was pre-processed and quality controlled to ensure interval consistency and remove spurious values through application of a pre-defined conditional interpolation rule. This is essential when analysing high frequency datasets from in-situ sensors as environmental conditions and technological failures may cause sensor malfunction and lead to inconsistent intervals between measurements (Horsburgh *et al.*, 2010; Wagner *et al*., 2000). Additionally, consistent data is a requirement for statistical tests utilised in fluvial research (Tijdeman, Stahl and Tallaksen, 2020). The raw data included four csv files, one for river level and one for water quality at each site. Exploration of this data revealed a shift in resolution from 1 hour to 30 minute intervals alongside additional periods of higher frequency measurements. To treat this issue measurements of <30 minute intervals were removed as the higher resolution was not necessary for analysis. Then measurements with >30 minute intervals were treated through application of a conditional rule, whereby gaps of <2 missing values were treated by linear interpolation, but >2 missing values were removed and replaced by NaN (Horsburgh *et al.,* 2011). Interpolation of gaps larger than 2 missing values would not produce representative surrogate values as turbidity and river level exhibit rapid change in response to environmental factors (Lawler *et al.,* 2006). This method successfully identified and treated missing observations following standardised protocols utilised within contemporary fluvial research (Horsburgh *et al.,* 2011; Lawler *et al.,* 2006) to ensure a continuous 30 minute interval dataset, maintaining the resolution required for effective analysis. It is important to address inherent methodological limitations such as the lack of pre-defined thresholds for the number of missing values where data should be interpolated up to, this leaves scope for variable thresholds being chosen by researchers during study replication.

3.3.2 Quality Control:

The standard of results is strongly dependent on the quality of data utilised for analysis. Thus, it is critical that raw data derived from in-situ sensors (Horsburgh *et al.,* 2011) pass a series of quality control procedures to detect and remove spurious values before analysis (Mourad and Bertrand-Krajewski, 2002). Standardised procedures utilised within contemporary research (Rodriguez-Perez *et al.,* 2020; Yousif *et al.,* 2022) to treat quality control errors within water quality datasets

were followed (Wagner *et al.,* 2006). Data was first checked to ensure measurement intervals were consistent before running gross range and multivariate tests within MATLAB (R2024a). The gross range test applied minimum and maximum values from the sensor specification to define thresholds for flagging out of range values caused by sensor malfunctions (Horsburgh *et al.,* 2011). Manufacturer specifications indicate the EXO3s sonde sensor threshold for river level is 10m and turbidity is 4000NTU (YSI, 2024). However, 2000NTU was utilised for quality control purposes as research suggests sensor accuracy decreases for readings >2000NTU and guidelines suggest maximum turbidity values recorded within freshwater systems typically range between 1000NTU to 2000NTU (Wagner *et al.,* 2006). No measurements were flagged by the gross range test at the downstream site, whereas 1580 turbidity and 1084 river level values recorded on the upstream sensor were flagged. The flagged measurements were removed / infilled utilising the aforementioned conditional rule, whereby a gap of <2 missing values was linearly interpolated, but >2 missing values were replaced by NaN (Horsburgh *et al.,* 2011). Additionally, the multivariate test was incorporated to analyse the covariance between river level and turbidity and identify periods where the relationship weakens. Reformatting was necessary to facilitate analysis, whereby MATLAB arrays were first converted into timetables to synchronise river level and turbidity data. The output of multivariate quality control indicated no flagged measurements at either site. The final, quality-controlled dataset used for analysis included 21,865 river level and turbidity values at each site. A prominent limitation of applying automated flagging methodology is the risk of error whereby a value is incorrectly identified as falling outside the expected range and conversely where a spurious value that should be flagged is not (Campbell *et al.*, 2013).

3.4 Hydrological Event Classification and Extraction:

Hydrologically significant events were extracted through manual hydrograph analysis. This was supported by classification of daily mean base / storm flow using the HYDSEP function within MATLAB (R2024a) (Figure 7) (Burkey, 2024). Resultantly a total of 44 events occurring at the upstream site and 54 events occurring at the downstream site between February 2024 and May 2025 were extracted. More events were analysed from the downstream site as frequent upstream sensor malfunction caused multiple gaps in data availability before the sensor was replaced.

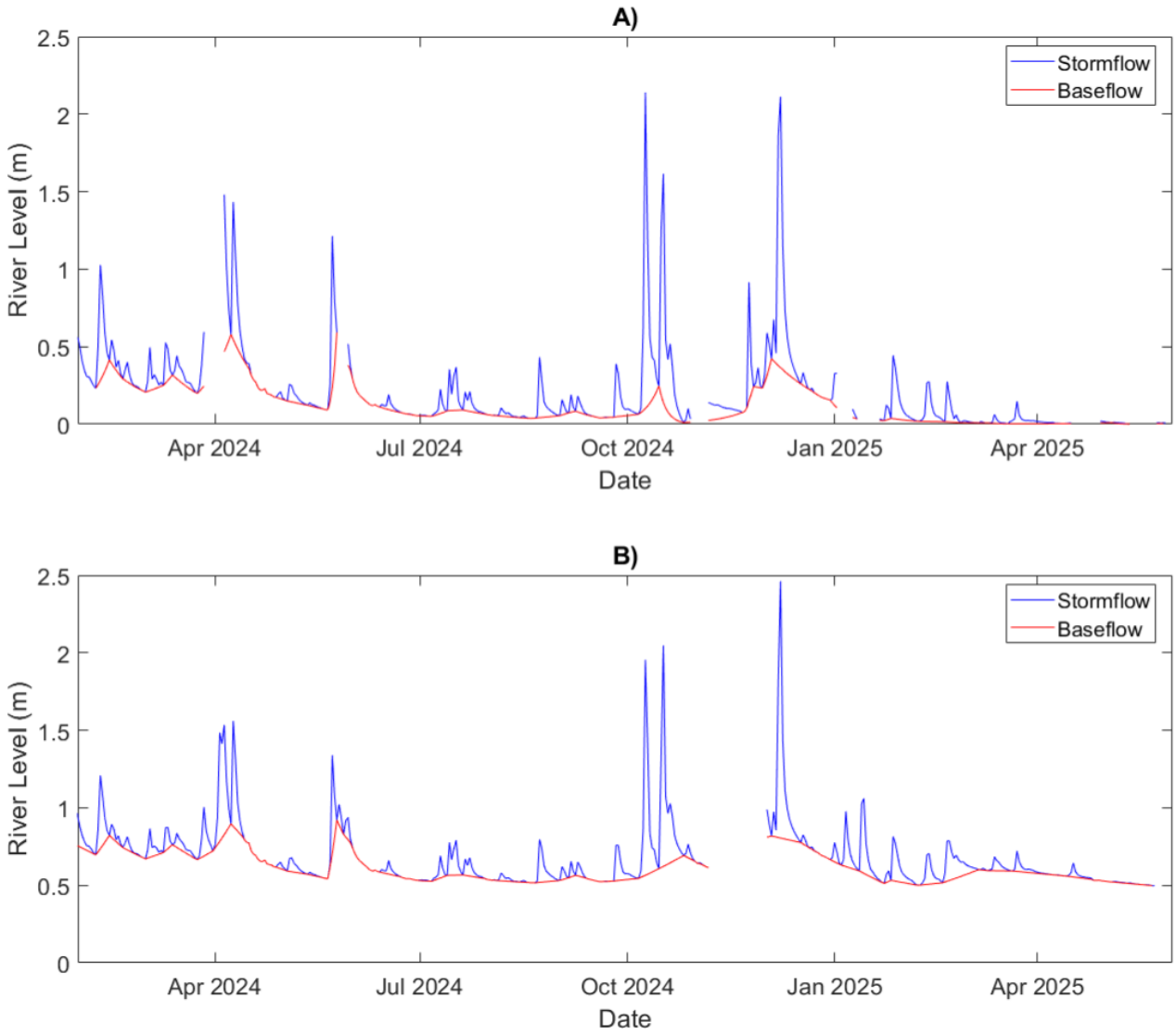


Figure 7. HYDSEP separation of daily mean river level baseflow and stormflow from A) Upstream and B) Downstream of the River Breamish restoration area.

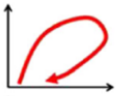
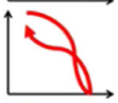
3.5 Hysteresis:

3.5.1 Hysteresis Loop Classification:

To investigate the processes underpinning sediment dynamics within the Breamish catchment, river level-turbidity hysteresis loops were plotted on scatter graphs through coding an array loop within MATLAB (R2024a). For each event the hysteric pattern and behaviour (direction and shape) were visually inspected and split into categories: clockwise, anticlockwise, clockwise figure of eight (F8), anticlockwise figure of eight (Table 2) (Zuecco *et al.,* 2016) and non-discernible pattern. On a broad scale, clockwise hysteresis indicates near channel sources and anticlockwise hysteresis indicates diffuse upper slope sources (Klein, 1984). This methodology follows standardised

analysis for determining sedimentary source areas (Lloyd *et al.,* 2016), whereby concentration-discharge relationships are used to identify underlying metrological, biological and geological processes driving sediment export patterns within a catchment (Speir *et al.,* 2024). Due to the limited data availability and constraints of this study, river level was used instead of discharge as river level is a prominent hydrometeorological factor impacting fine sediment transport (Vercruysse, Grabowski and Rickson, 2017) and previous research has demonstrated turbidity is an excellent surrogate for suspended sediment concentration (Perks *et al.,* 2015).

Table 2. Hysteresis classes used for analysis including: clockwise, anticlockwise, clockwise figure of 8 and anticlockwise figure of 8.
(Adapted from: Zuecco et al., 2016, p.1452, Table 1).

Hysteresis class	Loop	Dependent variable
Clockwise		Increases from the initial state
Clockwise figure of 8		Increases from the initial state
Anticlockwise figure of 8		Increases from the initial state
Anticlockwise		Increases from the initial state
Clockwise		Decreases from the initial state
Clockwise figure of 8		Decreases from the initial state
Anticlockwise figure of 8		Decreases from the initial state
Anticlockwise		Decreases from the initial state

3.5.2 Quantifying Hysteresis Magnitude:

The magnitude and direction of hysteric effect was quantified through calculation of a dimensionless hystersis index (HI_{mid}) (Figure 8), whereby the greater the hysteresis magnitude, the higher the HI_{mid} value (Lawler *et al.,* 2006). Only singular looped events were included in this analysis to ensure effective representation of HI_{mid} and minimise potential errors.

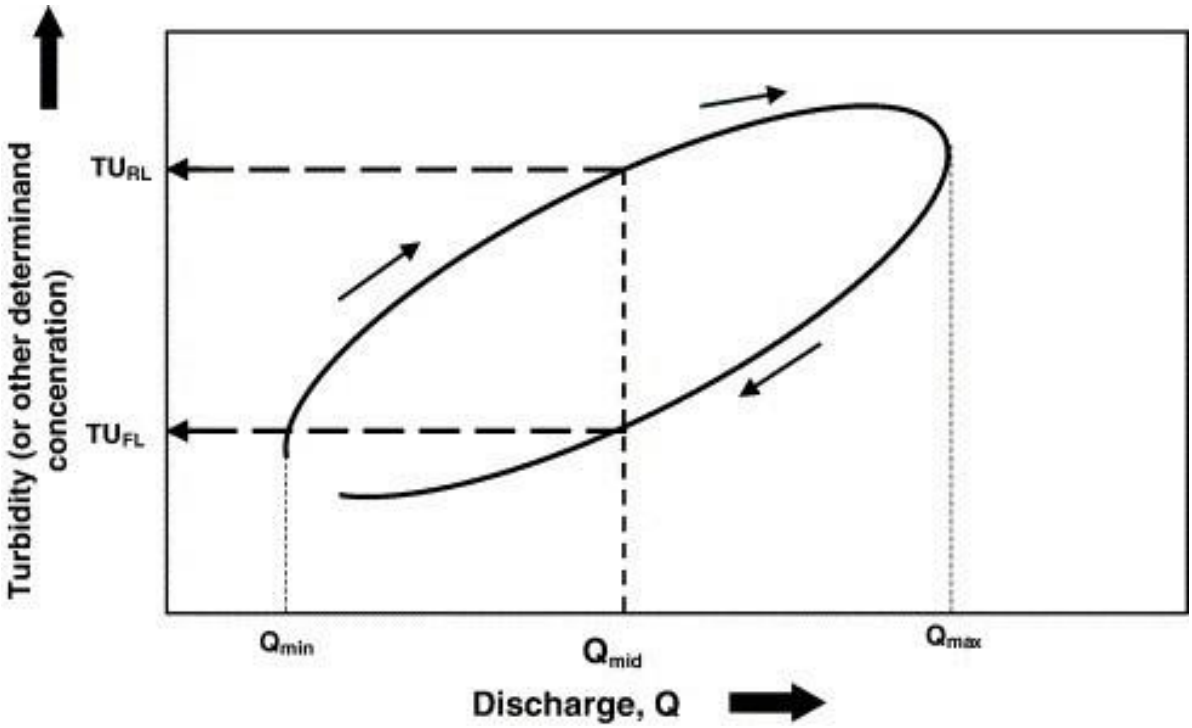


Figure 8. Schematic illustration showing the derivation of the hysteresis index (HI_{mid}). Where TU_{RL} is turbidity on the rising limb of the hydrograph and TU_{FL} is turbidity on the falling limb of the hydrograph.
(Source: Lawler et al., 2006, p.118, Fig.6)

River level data was extracted for each event, then the mid-point river level was determined utilising Equation 1. In this study, k was assigned a fixed value of 0.5, thus the hysteresis loop is measured at the midpoint of the rising limb. Previous studies have utilised the midpoint for analysis as it avoids potential small-scale conventions emerging at the top and bottom of hysteresis loops impacting analysis (Lawler *et al.,* 2006).

$$Q_{\mathrm{mid}} = k(Q_{\mathrm{max}} - Q_{\mathrm{min}}) + Q_{\mathrm{min}}$$

Equation 1. Where Q_{max} is peak river level of the event, Q_{min} is the river level at the start of the event and k represents the position at which the loop breadth is assessed relative to flow range (0.5).

Then, the data was linearly interpolated to allow extraction of turbidity values recorded at Q_{mid} on the rising limb (TU_{RL}) and falling limb (TU_{FL}) through array indexing in MATLAB (R2024a). This

enabled determination of HI_{mid} for clockwise hysteresis where $TU_{RL}>TU_{FL}$ (Equation 2) or anticlockwise hysteresis where $TU_{RL}<TU_{FL}$ (Equation 3).

$$\mathrm{HI_{mid}} = (\mathrm{TU_{RL}} / \mathrm{TU_{FL}}) - 1$$

Equation 2. Where TU_{RL} is turbidity on the rising limb of the hydrograph and TU_{FL} is turbidity on the falling limb of the hydrograph.

$$\mathrm{HI_{mid}} = (-1/(\mathrm{TU_{RL}} / \mathrm{TU_{FL}})) + 1$$

Equation 3. Where TU_{RL} is turbidity on the rising limb of the hydrograph and TU_{FL} is turbidity on the falling limb of the hydrograph.

The hysteresis direction was solved and calculations performed in Microsoft EXCEL using a conditional IF statement (Equation 4). The index assumes positive HI_{mid} values for clockwise hysteresis and negative HI_{mid} values for anticlockwise hysteresis (Lawler *et al.,* 2006).

$$= \mathrm{IF}\,(A7 < B7, -1/(A7/B7) + 1, (A7/B7) - 1)$$

Equation 4. Where A7=TU_{RL} and B7=TU_{FL}.

Overall, implementing a hysteresis index enables expression of both hysteresis magnitude and direction as a single value and normalises turbidity and river level data, thus it is critical for making direct comparisons between multiple sites (Aich, Zimmerman and Elsenbeer, 2014).

3.6 Measuring Fine Sediment + Nutrient Flux:

Material from three time integrated mass-flux sediment samplers (TIMS) (Figure 9) (Phillips, Russell and Walling, 2000) was analysed at approximately 2 month intervals between January 2024-October 2024. The samplers were located upstream [55.4803°, -1.8871°], mid-reach [55.4896°, -1.9108°] and downstream [55.4956°, -1.9103°] of the River Breamish restoration area (Figure 6). Although the sample design initially planned for monthly collections over a 12 month period, insufficient mass accumulated for analysis. Therefore, the interval was extended to 2 months to enable sufficient sediment accumulation while maintaining a reasonable temporal resolution (Perks *et al.*, 2017). The equipment was installed by the Tweed Forum into the river

channel facing upstream parallel to the direction of flow (Figure 9). As the stream water enters the inner chamber, reduced flow velocity encourages sedimentation of suspended fine particles within the chamber as the water flows to the outlet (Perks *et al.*, 2017). This allows collection of statistically representative samples for grain size distribution and nutrient composition analysis following the principles of sedimentation and flow continuity (Phillips, Russell and Walling, 2000). Sediment samples were collected by the Tweed Forum and were stored at -20°C prior to analysis (Olszewska *et al*., 2025).

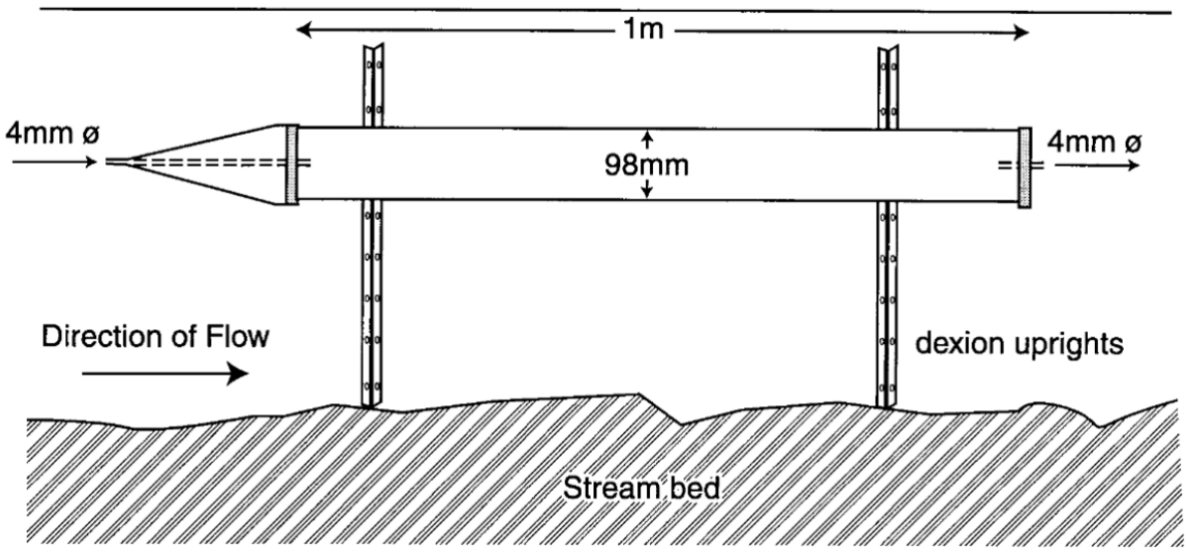


Figure 9. Cross section of a time-integrated sediment sampler.
(Source: Phillips, Russel and Wailing, 2000, p.2591, Figure 1)

3.6.1 Obtaining Channel Cross Sectional Geometry:

The channel cross section was surveyed at the upstream monitoring site using an EMLID RS2 Reach rover alongside a fixed base station (EMLID, 2024). The base station transmits Real Time Kinematic (RTK) corrections to the rover, with GPS observations recorded over a 60 second averaging period to improve positional stability. Data collection was impeded by dense riparian vegetation on the southern bank interrupting the connection between the rover and the base station, consequently resulting in 5 single and 25 fixed RTK points. To enhance the positional accuracy, post processing was conducted using the EMLID Studio Software to apply base corrections to the raw rover logs (EMLID, 2022). Utilisation of field equipment such as the EMLID RS Reach for global navigation satellite system positioning is widespread practice within environmental geoscientific research (Dabove and Pietra, 2024).

Due to bank access limitations, the surveyed cross section only extended ~2.5m above the riverbed, however sondes stage flow data indicated that river levels exceeded 2.5m on multiple occasions. To resolve this issue, a composite LIDAR digital surface model (DSM) obtained in July 2025 was integrated to further extend the cross section. The Warp Reproject function in QGIS was utilised to reproject the imagery from British National Grid to WSG1984 to ensure coordinates were in decimal degrees (QGIS, 2025). The QGIS profiler tool was then used to extract the cross sectional profile and the data was exported into a CSV file for plotting in MATLAB (R2024a). This profile was integrated with the field survey through linear interpolation, maintaining the central surveyed section and utilising the LIDAR profile to extend up the adjacent banks, leading to a complete representation of channel cross sectional geometry (Figure 10). Mid-reach and downstream channel geometries were also remotely estimated following the same process (Figure 11) but using a LIDAR digital terrain model (DTM) raster at 1m resolution (DEFRA, 2022). DTMs record bare surface elevation, therefore riparian vegetation does not disturb the results (Parrot and Núñez, 2016).

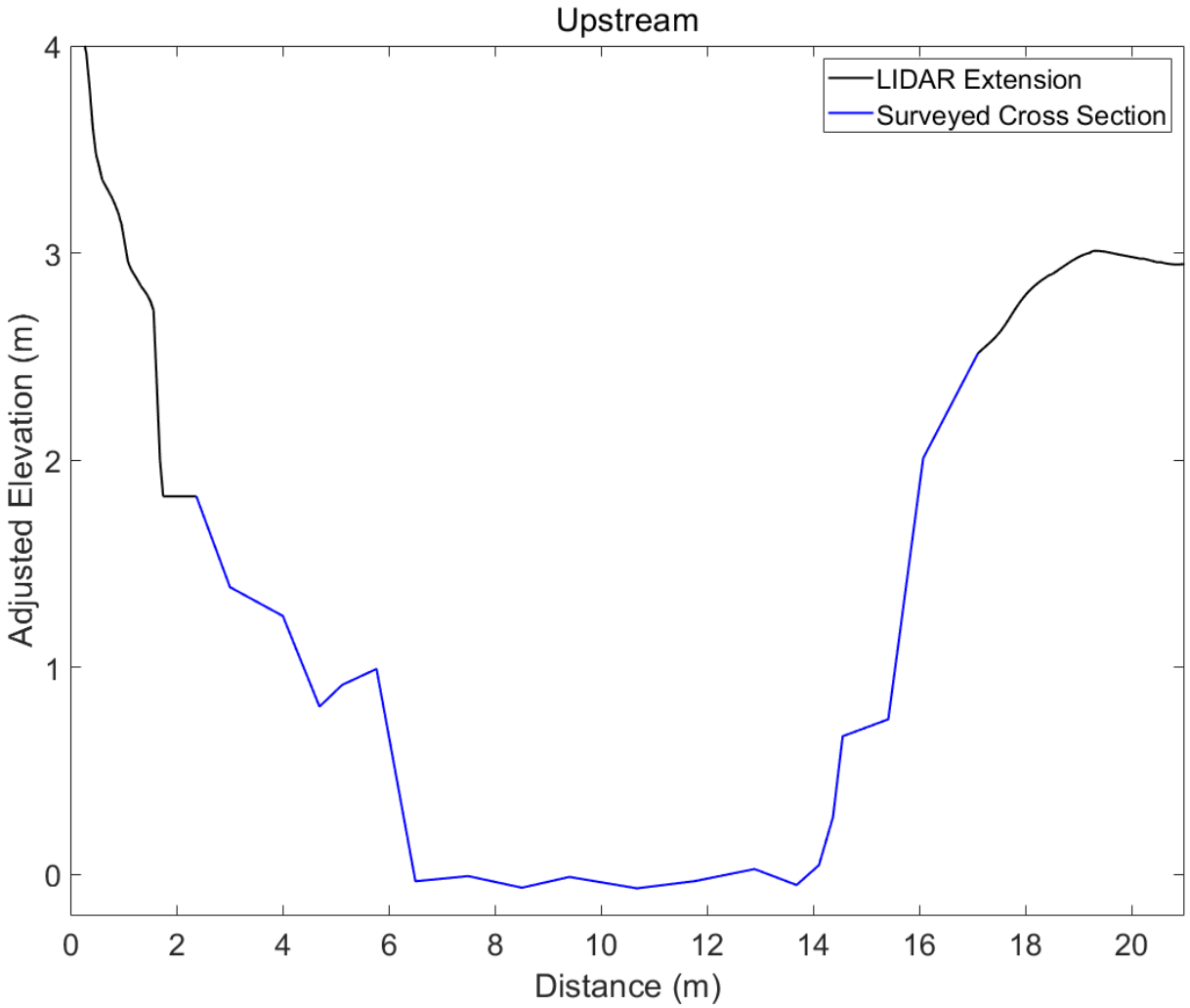


Figure 10. Channel cross section profile obtained from Upstream of the River Breamish restoration area. The blue line indicates the field surveyed cross section and the black line shows where LIDAR DSM imagery has been used to extend the profile. Elevations have been adjusted as outlined in section 3.6.2.

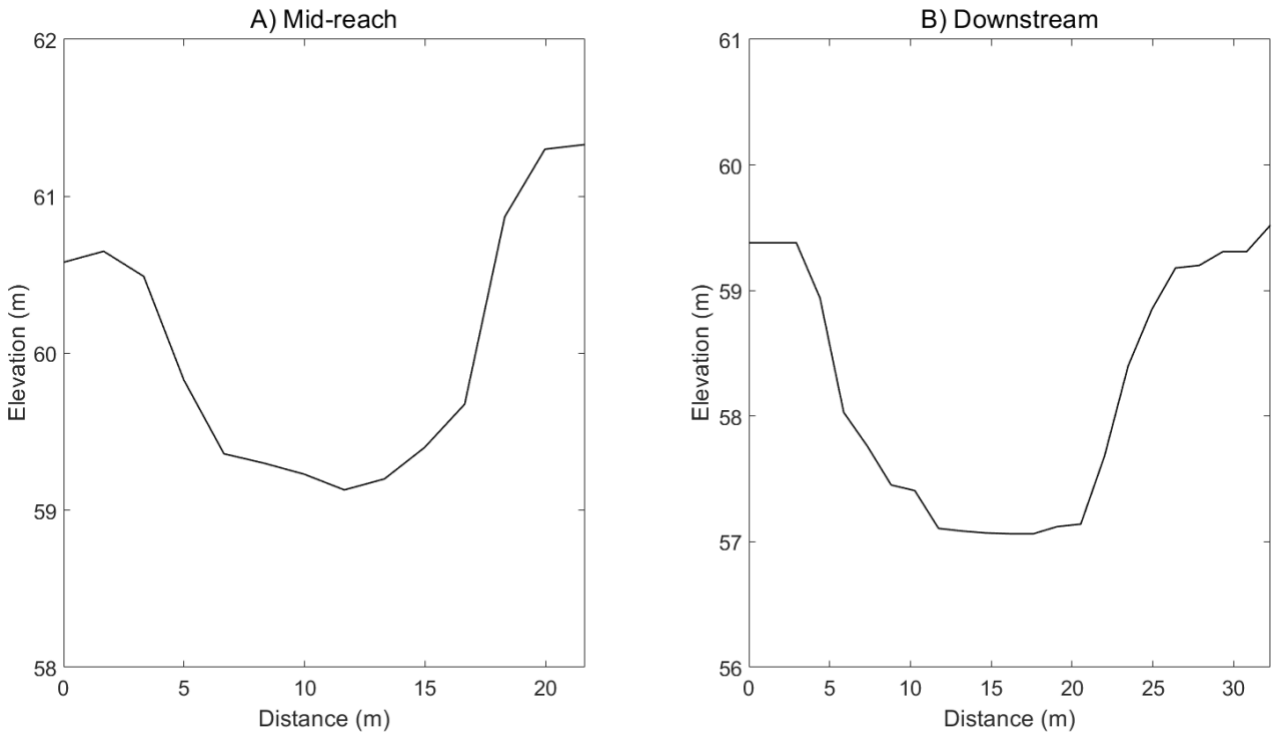


Figure 11. Channel cross section profiles obtained from A) Mid-reach and B) Downstream of the River Breamish restoration area.

3.6.2 Estimating Suspended Sediment Load + Specific Sediment Yield:

The surveyed upstream site was utilised to estimate the mass of sediment in flux. First, the data was normalised through vertical adjustment to determine the relationship between surveyed elevations (referenced to WGS84 datum) and real time levels recorded in-situ by the sonde sensor. The measured water level by the sensor at the time of surveying was 0.02m, while the corresponding WGS84 water level was 112.545m, thus a vertical offset of 112.525m was applied to all Z-elevations. This enabled direct comparison between the sensor measurements and surveyed cross sectional geometry. Using the adjusted data, an inverted trapezoidal numerical integration method was applied in MATLAB(R2024a) to calculate the cross sectional area of flow (m^2) for each river level measurement (depth) recorded during sampling periods. This created a time series array for each sample period containing continuous cross sectional areas. Manning's equation was then implemented to calculate velocity (Equation 5), where the average depth profile was calculated for each sampling period and the surveyed slope was 0.0062m/m. Manning's *n* roughness coefficient (*n*=0.046) was estimated using Cowan's (1956) equation method (Equation 6) and based on field observations (Table 3) as this is standard procedure recommended by the US Geological Survey (Coon, 1998). Finally, once velocity and cross-

sectional area were determined for each sampling period, discharge was calculated using the continuity equation (Equation 7).

$$v = \frac{\left(d^{\frac{2}{3}} \cdot S^{\frac{1}{2}}\right)}{n}$$

Equation 5. Where d is average flow depth, S is channel slope and n is Manning's roughness coefficient.

$$n = (n_0 + n_1 + n_2 + n_3 + n_4)\, m$$

Equation 6. Where n_0 is a base value for channel straightness / uniformity, n_1 accounts for the effect of cross section irregularity, n_2 accounts for variations in channel size and shape, n_3 accounts for the effects of obstructions, n_4 accounts for the type and density of vegetation and m is an adjustment factor for the degree of channel meandering.

Table 3. Values used in Cowan's equation (Equation 6) and the justification for choice.

	Value	Justification
n0	0.028	Gravel bed, straight channel.
n1	0.011	Severe irregular channel surfaces, eroded banks.
n2	0.000	Channelised reach so gradual changes.
n3	0.005	Some woody debris and boulders in channel.
n4	0.002	Not much live vegetation in channel, some algae.
m	1.000	Channelised reach, no meandering present.

$$Q = A \cdot v$$

Equation 7. Where A is cross sectional area and v is velocity.

The discharge values (m^3/s^{-1}) and high frequency turbidity data (mg/L^{-1}) were utilised to calculate the total mass of sediment transported during each TIMS sampling period. Literature indicated that $1m^3$=1000L (Nashima *et al.*, 2013), thus the process exemplified in Figure 12 was implemented within a MATLAB for loop to calculate the total sediment load at 30 minute intervals. These values were added up relative to each TIMS sampling period to calculate the total mass of sediment mobilised within the Breamish during each TIMS sampling period.

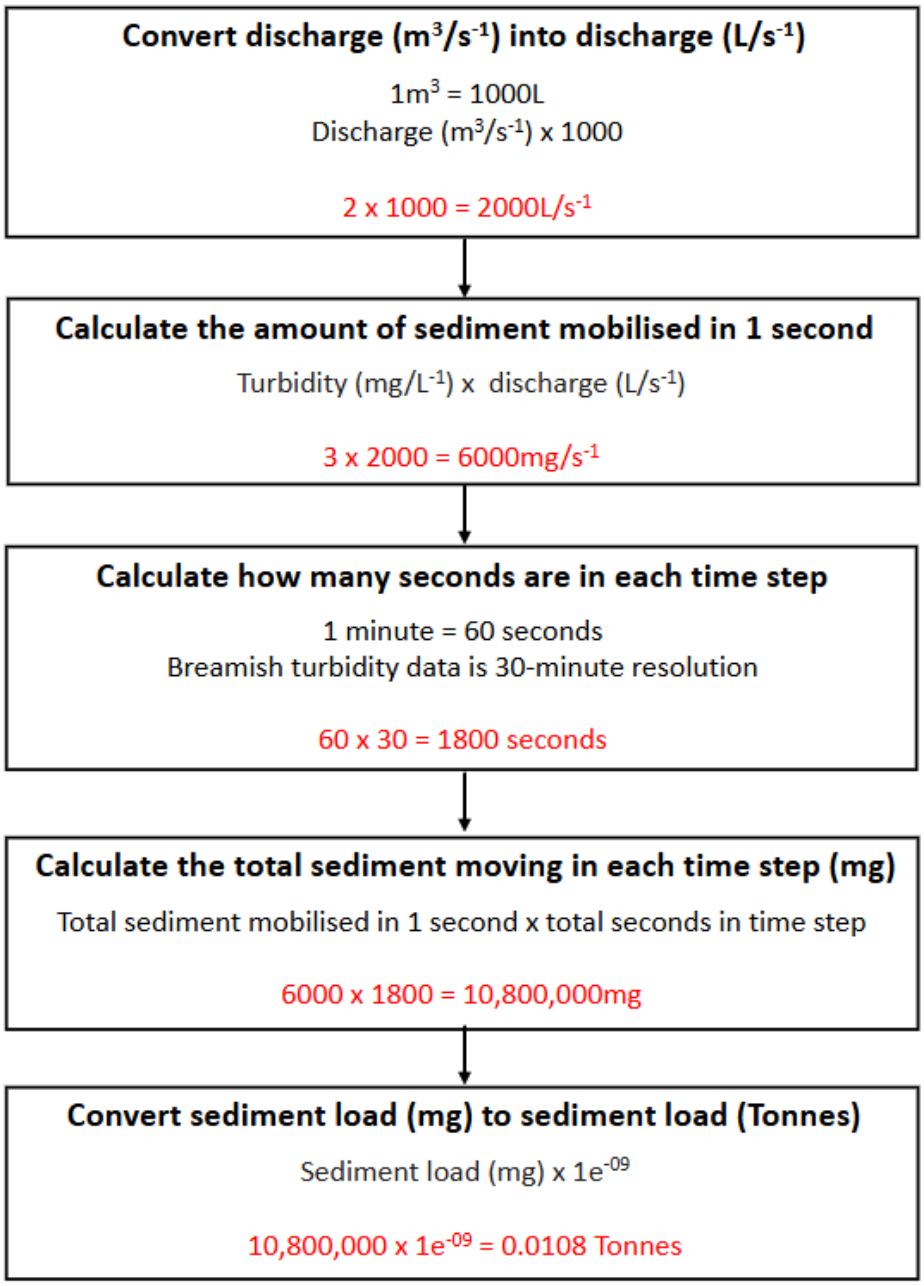


Figure 12. Flow chart detailing the methodology for calculating sediment loads (tonnes). Example calculations are shown in red.

Then the calculated sediment loads were input into Equation 8 to estimate specific sediment yield (SSY) by dividing the loads (L_{est}) by the Breamish catchment area (90.009km^2). Integrating this into analysis allows comparisons to be made between catchments with different drainage basin areas (Charlton, 2008).

$$SSY = \frac{L_{est}}{catcment\ a}$$ *Equation 8. Where L_{est} is sediment load.*

3.6.3 Particle Size Analysis:

Laboratory particle size analysis was conducted by the UKCEH on behalf of the Environment Agency and Tweed Forum. Samples were first rinsed using Thermo Scientific GenPure Pro UV ultrapure water (Olszewska *et al*., 2025), then added to sterile falcon tubes where excess water was removed via room temperature evaporation. The remaining material was mixed and representative sub-samples were taken for analysis. 5ml of 5% sodium hexametaphosphate was added to each sample which were then topped up to with 60ml with ultrapure water and vortex mixed for 30 seconds before being left for 24 hours (Olszewska *et al*., 2025). To remove large particles, the samples were vigorously shaken and filtered through a 3mm sieve before being added to the dispersion reservoir (Olszewska *et al*., 2025). A Malvern Panalytical Mastersizer 3000 hydro G dispersion unit was utilised, employing laser granulometry to measure sediment particle size (Olszewska *et al*., 2025). Statistical analysis was undertaken for this study using the GRADISTAT software in Microsoft EXCEL, utilising the laser granulometry setting to input samples into the multiple sample data input sheet. This enabled rapid analysis of grain size statistics, providing a robust basis for analysis and is appropriate for the spatial and temporal scales of this study (Blott and Pye, 2001).

3.6.4 Macronutrient Composition Analysis:

Total P (ICP-OES) (De Morais *et al.,* 2021) and total N (DUMAS method) (Ostrom *et al.,* 1998) analysis was also conducted by UKCEH. P forms and N species were determined through sequential P extraction (Hupfer *et al.*, 2009) and the potassium chloride (KCl) method (Stanford University, 2025). Further details on UKCEH laboratory procedures are available in the Harehope Sediment Monitoring report (Olszewska *et al*., 2025).

Sediment load calculations were utilised to calculate the mass of nutrients bound to sediment particles in flux, converting total P from mg/kg into metric tonnes (t), total N from % into tonnes, and P forms (Table 4) from µg/g to tonnes. Determining the organic macronutrient composition provides detailed insight into the sedimentary source areas, whereby organic components

indicate agricultural inputs and inorganic components indicate additional inputs such as artificial fertiliser (Mainstone and Parr, 2002).

Table 4. Overview of common operational sediment phosphorus fractions based on sequential sediment extraction procedures.

Classification	Species	Release driver	Natural condition release likelihood
Labile	Directly available, pore water and loosely bound / absorbed P.	Desorption and diffusion.	High
Reductant Soluble	Bound to Fe- hydroxides and Mn-compounds.	Anoxia and redox potential.	High
Metal Absorbed	Absorbed to metal oxides.	Mineralisation.	Medium / High
Organic	Organic material.	High pH.	Medium / High
Apatite	Bound to carbonates and apatite P.	Low pH.	Medium
Residual	Refractory compounds.	n/a	Low

3.7 Modelling Critical Source Areas + Pollutant Delivery:

The SCIMAP model was used to analyse fine sediment and macronutrient delivery within the River Breamish catchment. The spatial distribution of sediment and nutrient transfer was determined, thus enabling critical identification of spatially variable sources (Perks *et al.,* 2017). While the aforementioned hysteresis analysis indicates the source proximity, it does not pinpoint the location, thus SCIMAP modelling is essential to quantitatively corroborate the identification of diffuse sources (Reaney *et al.,* 2019). Key model inputs include the land cover, topography, elevation and rainfall within a specified catchment area (Wells and Reaney, 2014). The model incorporates a risk based approach to generate a 5m resolution erosion risk map output (Reaney *et al.,* 2019); from this critical source areas where mobilisation and land connectivity (calculated as the connectivity index (Perks *et al.,* 2017)) contribute to excess sediment delivery and associated water quality issues can be determined (Figure 13). Generation of the risk map output requires multiple processing steps (Figure 13).

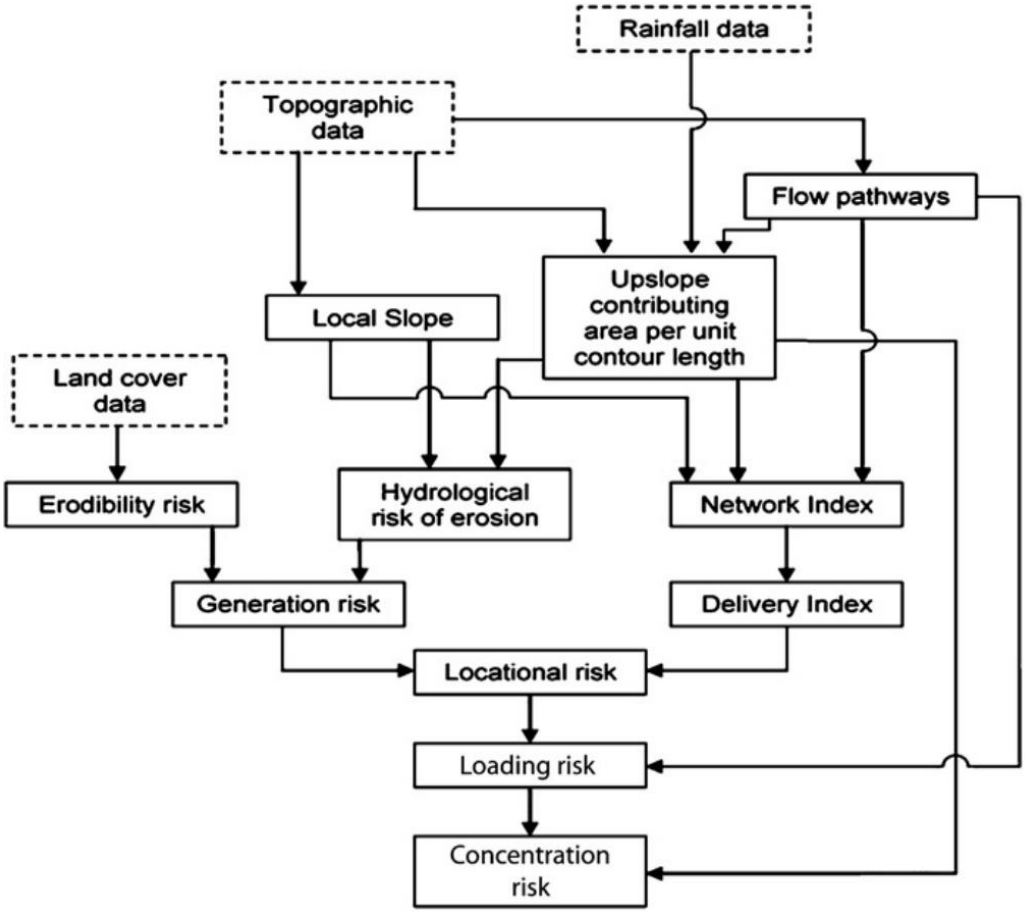


Figure 13. Overview of the landscape elements incorporated within the SCIMAP model of fine sediment diffuse pollution risk.
(Source: Reaney et al., 2011, p.1019, Fig.3)

Land cover and local erosion potential are critical inputs of the SCIMAP model. The Breamish catchment was categorised into different land use types utilising data from the CEH Land cover Map (Morton *et al.,* 2011). Different land use types are assigned a risk weighting (R_w) to represent the land surface erodibility and the mobilisation potential. This study encompasses modelling of fine sediment, NO_3 and PO_4, therefore, the risk weightings were altered for each parameter (Table 5) to optimise the application of model outputs for accurately identifying spatial patterns of water quality risk areas within the River Breamish catchment. The risk weightings employed are derived from inverse modelling across 11 UK catchments including the Till catchment performed by Milledge *et al.,* (2012) to identify optimised risk weightings for SCIMAP land cover classes that best fit in-stream observations (Milledge *et al.*, 2012). The River Breamish is situated within the wider Till catchment, therefore the risk weightings applied are appropriate for the scope of this research (Table 5).

Table 5. Optimised land cover risk weightings (R_w) used to assess the spatial pattern of erodibility in the SCIMAP model of the River Breamish catchment. Parameter values include fine sediment (Perks et al., 2017), phosphate (PO_4) and nitrate (NO_3) (Milledge et al., 2012).

Land Cover	Risk Weighting Value (Rw)		
	Fine Sediment	Phosphate	Nitrate
Arable	1.00	0.53	0.87
Improved Grassland	0.30	0.71	0.72
Natural Grassland	0.15	0.07	0.10
Moorland	0.05	0.41	0.09
Woodland	0.05	0.51	0.69
Urban	0.01	0.77	0.58
Bog	0.05	0.59	0.49
Water	0.00	0.00	0.00

The model then calculates the spatially distributed erosion potential (*E*) (Equation 9) using a stream power calculation as a function of slope inclination and water depth within a gridded area (Milledge *et al.,* 2012). The SCIMAP model combines the upper slope contributing area (*A*) as a proxy of potential surface flow volume and the local slope gradient (β) as a proxy for flow speed (Equation 9).

$$E = R_w \cdot A \cdot tan\beta$$

Equation 9. Where R_w is land cover risk weights, A is upslope contributing area and β is gradient.

Calculation of spatially distributed erosion potential (*E*) enables formulation of the connectivity index (C_i) based upon the Network Index algorithm (Lane *et al.,* 2004). This index considers the wetness required to generate surface runoff and subsequently connect source areas to the watercourse. The SCIMAP model incorporates a time integrated approach, characterising spatial and temporal constraints and addressing the frequency and length of established connections between source areas and the watercourse (Lane, Reaney and Heathwaite, 2009). Taking this into account, the predicted fine sediment erosion risk (L_j) (Equation 10) is calculated considering the upslope contributing area (*j*) monotonically increasing with distance throughout the drainage network, downstream of the diffuse pollutant source (Perks *et al.,* 2017).

$$L_j = \sum_{i=1}^{j} E_i \cdot C_i$$ *Equation 10. Where j is the upslope contributing area, E_i is the calculated spatially distributed erosion potential index and C_i is the connectivity index.*

Critical source areas can then be identified where a significant erodible source, overland flow and a connected pathway combine to transfer material into the channel (Heathwaite, Sharpley and Gburek, 2000). The accumulation of risk through the landscape and into the channel network is expressed by the in-channel risk (C_j) (Equation 11); this is a summation of the contributing upstream catchment area for an individual point (a_i) and the rainfall weighting factor (r_i) which normalises upstream area precipitation across the catchment.

$$C_j = \frac{\sum_{i=1}^{j} E_i \cdot C_i}{\sum_{i=1}^{j} a_i \cdot r_i}$$ *Equation 11. Where E_i is the calculated spatially distributed erosion potential index, C_i is connectivity index, a_i is the upstream catchment area for a given point and r_i is the rainfall weighting factor.*

Overall, the SCIMAP risk based modelling approach produces two output maps: erosion risk and hydrological connectivity risk. Subsequent analysis within ArcGIS Pro combined the two to enable identification of potential source areas following the critical source area approach and highlighted the spatial variability of fine sediment risk routing across the channel network (Perks *et al.*, 2017). Catchment observations were noted during fieldwork to identify transport pathways where land connectivity facilitates sediment delivery and ground truth SCIMAP outputs.

3.7.1 Sources of Model Uncertainty:

There are implicit assumptions which contribute to SCIMAP output uncertainties. Application of land use risk weightings assumes that there is a correlation between soil type and land cover and that seasonal variations in erodibility values are implicitly accounted for (Reaney *et al.*, 2011). Such uncertainty is reduced within this study by applying optimised land cover risk weightings identified by scientific research and outlined by Perks *et al.,* (2017) and Milledge *et al.,* (2012) (Table 5). Additionally, SCIMAP cannot confirm specific source location and is a predictive

approach, therefore the identification of risk areas are indicative predictions rather than certain (Walling, 1999), reflecting a key limitation of the conceptual model approach.

3.8 Integrating Methodologies + Objectives Achieved:

Overall, hysteresis analysis indicates the proximity of sediment sources to the channel and SCIMAP quantitatively corroborate the identification of diffuse sources pinpointing the location of critical source areas (Reaney *et al.,* 2019). The TIMS sampling framework and SCIMAP modelling approach both provide data on sediment pressures as normalised units by the unit area, indicating spatial patterns of sediment transport and enabling understanding of catchment sediment / macronutrient dynamics (Perks *et al.*, 2017). The results from each individual component of the methodology (hysteresis, TIMS, SCIMAP) are presented in an integrated manor through development of a schematic diagram illustrating the controls, potential pathways and processes operating within the Breamish catchment. Overall, this methodology supports the achievement of this study's objectives (Table 6).

Table 6. List of objectives achieved through application of the study methodology.

Method used:	Objective achieved:
Obtained high frequency monitoring data (~30 minute intervals) from two Environment Agency EXO3 Multiparameter Sondes (YSI, 2024) located upstream [55.4775°, -1.8810°] and downstream [55.4955°, -1.9102°] of the River Breamish restoration area.	To acquire and quality control high-resolution hydrometeorological (river level) and water quality (turbidity) datasets between February 2024 to May 2025 from two Environment Agency monitoring stations located upstream and downstream of the River Breamish Restoration area.
Characterised and quantified river level / turbidity relations through hysteresis loops and calculation of the HI_{mid} in MATLAB (R2024a).	To assess sediment source proximity through river level-turbidity hysteresis analysis and implement a hysteresis index (Lawler et al., 2006) to determine the hysteric strength.
Analysed material from three time integrated sediment samplers at approximately two month intervals between January 2024-October 2024. Obtained channel cross sectional geometry to enable calculation of sediment load and macronutrient composition of sediment flux.	To determine the particle size and nutrient composition of sediment flux at regular intervals through analysis of material captured by time integrated sediment samplers during pre-restoration sampling (February 2024-October 2024)(Phillips, Russell and Walling, 2000).
Applied optimised land cover risk weightings to the SCIMAP model to generate multiple risk maps, enabling analysis of fine sediment, NO_3 and PO_4 delivery in the River Breamish catchment.	To identify the critical source areas delivering fine sediment and macronutrient pollutants within the River Breamish catchment using the SCIMAP model (Reaney *et al.,* 2011).
Made qualitative observations and took photographs of identified potential sources and transport pathways relative to SCIMAP outputs during fieldwork.	To ground truth SCIMAP outputs during fieldwork, identifying potential pollutant sources and transport pathways where land connectivity facilitates delivery into the channel.
Integrated the results from each individual component of the methodology (hysteresis, TIMS, SCIMAP) through a schematic diagram illustrating the controls, potential pathways and processes operating within the Breamish catchment pre-restoration.	To integrate data from objectives 1-5, to provide a pre-restoration baseline of fine sediment and macronutrient dynamics, against which post-restoration conditions can be evaluated.

Chapter 4 – Results:

4.1 Catchment Context:

4.1.1 Hydrograph Nature During the Monitoring Period:

Notably, both the upstream and downstream sites follow similar patterns of river level during the pre-restoration monitoring period, however there are more gaps in the data due to increased sensor malfunctions at the upstream site. The downstream site sustains enhanced baseflow ~0.5m above the upstream site throughout the hydrological year (Figure 14). Despite this, similar peak values are recorded upstream and downstream of the restoration area during severe storm events. Daily mean river levels display seasonal variation throughout the study period (Figure 14). Larger magnitude storm events are observed during autumn 2024 at the upstream site and during winter 2024 at the downstream site with peak river levels of 2.14m and 2.46m respectively (Figure 14). While peak river levels are higher during autumn 2024 at the upstream site, comparison of daily mean river levels indicates that the average river levels are higher during winter 2024 for both upstream and downstream sites (0.25m and 0.73m respectively) (Figure 14). Comparatively, the lowest river levels are recorded during spring 2025 at both upstream (0.002m) and downstream (0.49m) sites; this dry period also exhibits the lowest mean river levels within the monitoring period (0.02m and 0.56m respectively) alongside low variation with ranging values of 0.14m and 0.23m respectively (Figure 14). River levels below 0.8m are sustained throughout the summer season before peak river levels exceeding 2m are reached from mid-autumn into winter at both the upstream and downstream sites. The October 2024-December 2024 sampling period generally exhibits higher river levels alongside greater variation at both upstream and downstream sites with an upstream range of 2.13m and a downstream range of 1.97m (Figure 14). Annual mean river levels during the pre-restoration period at the downstream site are more than double the upstream values: 0.73m compared to 0.27m in 2024 and 0.59m compared to 0.05m in 2025. There is also variation in bankfull river levels between the upstream (3.01m) and downstream (2.32m) sondes monitoring sites; between these, the mid-reach TIMS site exhibits a lower bankfull river level of 1.52m. Comparing these to the observed hydrographs indicates that under the channelised conditions during the pre-restoration monitoring period the water rarely enters the adjacent floodplains (Figure 14).

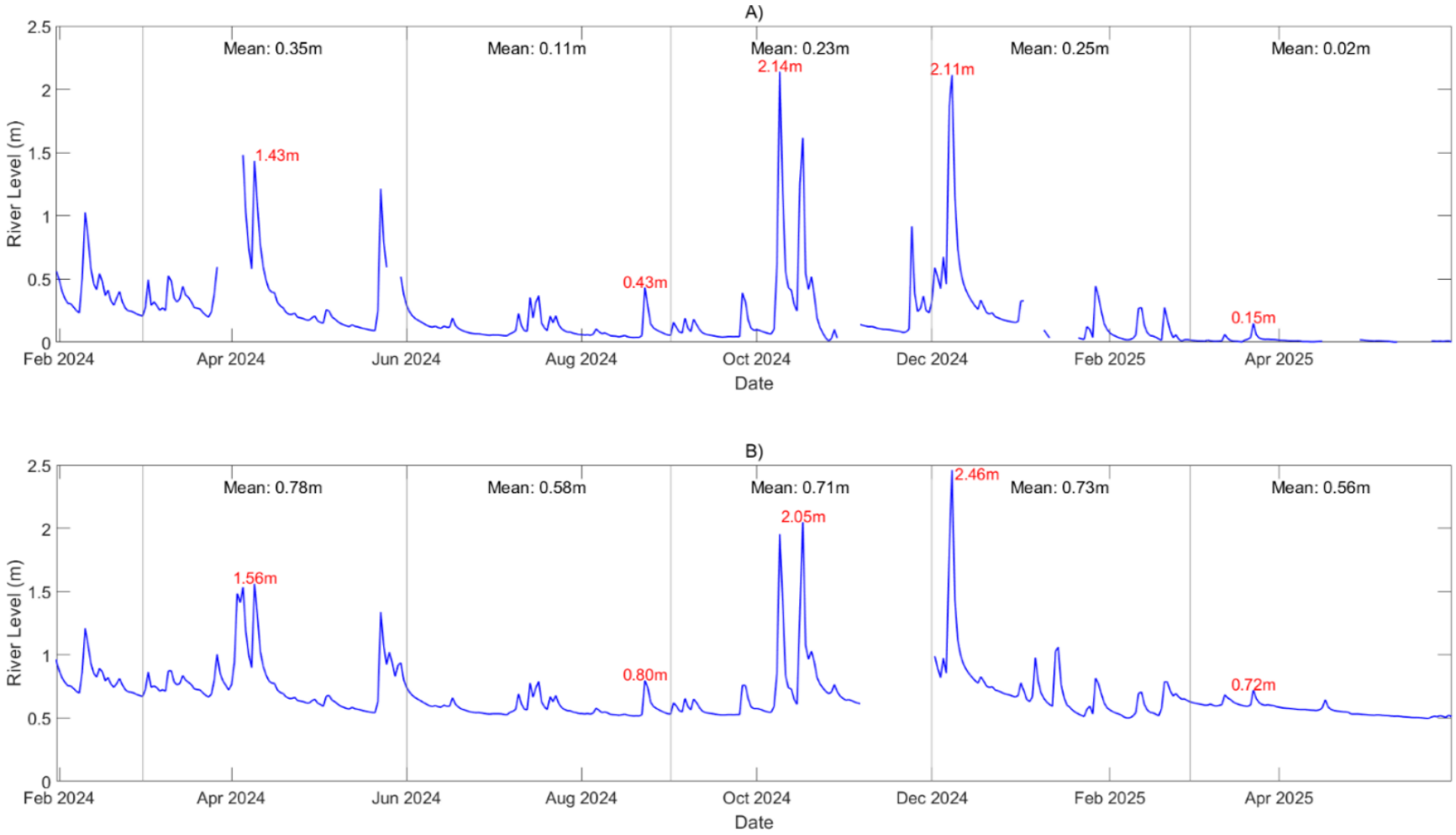


Figure 14. Hydrographs showing daily mean river level A) Upstream and B) Downstream of the River Breamish restoration area during the pre-restoration monitoring period (February 2024-May 2025). Solid lines indicate hydrological season divisions. The overall seasonal means are labelled in black and peaks are labelled in red.

4.2 Relationship Between River Level and Turbidity:

4.2.1 Qualitative Hysteresis Classification:

All four main hysteresis patterns are observed within this study (Figure 15). Both upstream and downstream hysteric behaviour is dominated by clockwise hysteresis, accounting for 41% of upstream and 54% of downstream storm events. At the upstream site the remaining events consist of a singular anticlockwise loop, clockwise F8 loops (16%), anticlockwise F8 loops (9%) and those with non-discernible hysteresis patterns (32%) (Table 7). Turbidity dynamics at the downstream monitoring site also exhibit clockwise F8 loops (11%), anticlockwise F8 loops (9%) and non-discernible patterns (26%) (Table 7).

Furthermore, on an intra-annual scale clockwise loops are dominant across both sites, except during autumn at the upstream site where events are dominated by non-discernible patterns (Table 8) and summer at the downstream site where clockwise F8 and non-discernible patterns are dominant (Table 8). Storm events displaying clockwise loops mainly occur during winter at the upstream site but during spring at the downstream site. An anticlockwise loop is only present at the upstream site, occurring as a singular unique event during winter 2024, reflecting an extended period of elevated turbidity levels during the falling limb (Table 8A).

Within this research, the hysteric behaviour observed is not always well-defined, with complex and non-clean patterns often exhibited. Many loops demonstrate initial micro-pulses of turbidity at the start of the rising limb, alongside sudden jumps in turbidity levels due to the noise associated with real time measurements (Figure 15). Additionally, some events occur successively without returning to baseflow, here the temporal overlap causes irregular, scattered plots rather than distinct, closed loops. Due to the complexity of fluvial sediment dynamics, secondary peaks and multiple turbidity pulses are sometimes observed during hydrograph recession, resulting in a multi-loop complex F8 pattern.

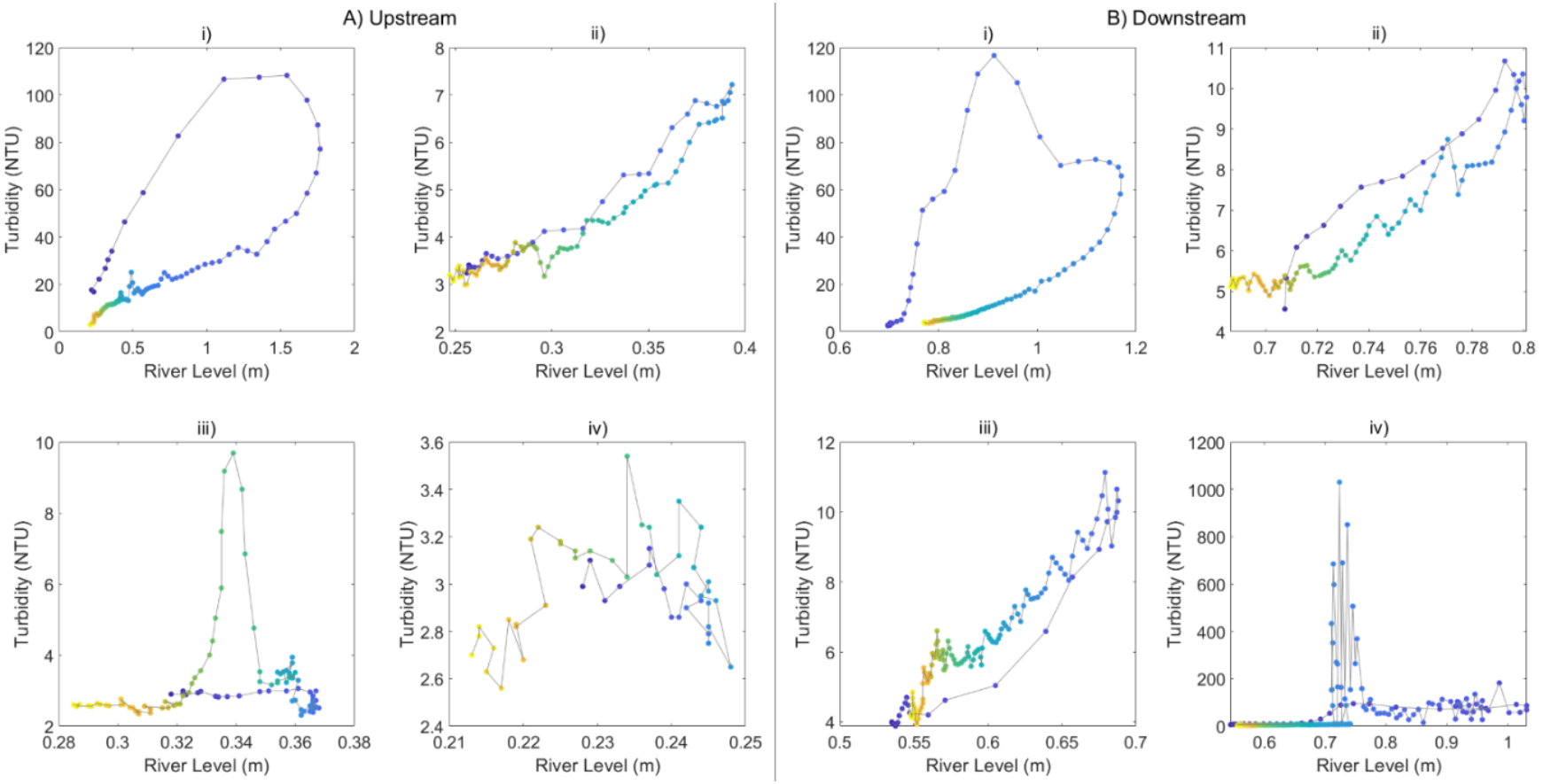


Figure 15. Examples of turbidity-river level hysteresis observed A) Upstream and B) Downstream of the River Breamish restoration area. The marker colour changes from dark blue, to green, to yellow during the flow event. Hysteresis classes include clockwise loop (Ai and Bi), clockwise figure of 8 (Aii and Bii), anticlockwise figure of 8 (Aiii and Biii), anticlockwise loop (Aiv) and non-discernible hysteresis pattern (Biv).

Table 7. Classification of river level-turbidity hysteresis loop patterns at each site.

Hysteresis class	Upstream	Downstream
Clockwise	18	29
Anticlockwise	1	0
Clockwise 8	7	6
Anticlockwise 8	4	5
Non-discernible	14	14

Table 8. Classification of river level-turbidity hysteresis loop patterns seasonally A) upstream and B) downstream of the River Breamish restoration area.

A) Upstream

Hysteresis class	Spring	Summer	Autumn	Winter
Clockwise	5	3	1	9
Anticlockwise	0	0	0	1
Clockwise 8	2	2	1	2
Anticlockwise 8	3	1	0	0
Non-discernible	2	2	8	2

B) Downstream

Hysteresis class	Spring	Summer	Autumn	Winter
Clockwise	13	2	3	11
Anticlockwise	0	0	0	0
Clockwise 8	1	3	1	1
Anticlockwise 8	1	0	2	2
Non-discernible	6	3	1	4

4.2.2 Hysteresis Sediment Load Transfer:

At the upstream site, although clockwise hysteresis is dominant in terms of frequency, events displaying non-discernible hysteresis patterns are dominant in terms of average sediment load transported (312.04t) (Table 9). Conversely, anticlockwise F8 loops are both the least frequently observed pattern and reflect low magnitude transfer events (average load of 1.33t) (Table 9). Downstream of the River Breamish restoration area, clockwise hysteresis patterns are dominant in both frequency and mass of sediment transport (244.85t) (Table 9). Clockwise F8 hystesis occurs more frequently downstream than anticlockwise F8 hysteresis, however clockwise F8 loops represent a smaller average load, transporting less than half the amount of sediment load per event than anticlockwise F8 loops (7.09t compared to 15.05t) (Table 9). During the pre-restoration monitoring period, the total sediment load transported by events displaying clockwise F8 is ~10t lower than anticlockwise F8 events, thus although events displaying clockwise F8 are more frequent, anticlockwise F8 events contribute more total sediment load into the catchment. There great variability in the sediment load of clockwise F8 loops at the downstream site, represented by a large interquartile range (IQR) >300m (Figure 16B), such variability is not present at the upstream site with more consistent loads and a small IQR (Figure 16). Non-discernible patterns also exhibit large variability at both sites, displaying a large IQR (Figure 16). The largest

magnitude events (90th percentile) all exhibit non-discernible hysteresis except the second largest event recorded downstream of the restoration (Table 10). Thus, for the highest magnitude transfer events, there is a change in hysteric behaviour represented by additional system complexity leading to loops characterised as non-discernible. It is important to note that these values only reflect storm events that occurred within the TIMS sediment sampling period (January 2024-October 2024), not the full pre-restoration monitoring period.

Table 9. Average sediment load (t) per event for hysteresis loop patterns observed A) upstream and B) downstream of the River Breamish restoration area during the sampling period January 2024-October 2024.

Hysteresis class	Upstream Average Load (Tonnes)	Downstream Average Load (Tonnes)
Clockwise	16.65	244.85
Anticlockwise	n/a	n/a
Clockwise 8	2.23	7.09
Anticlockwise 8	1.33	15.05
Non-discernible	312.04	62.68

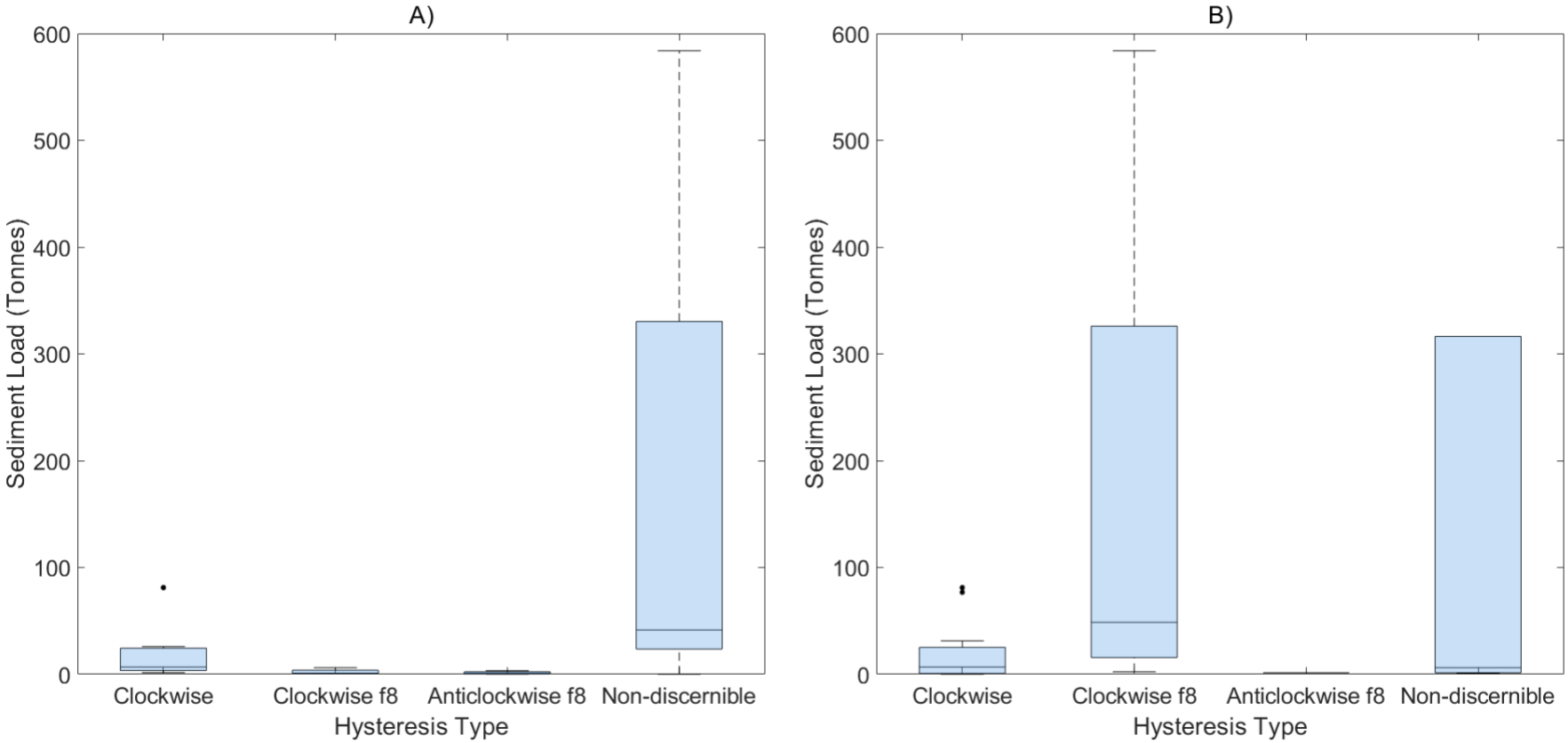


Figure 16. Box plots showing sediment load (tonnes) for hysteresis loop patterns observed A) upstream and B) downstream of the River Breamish restoration area during the sampling period January 2024-October 2024.

Table 10. Sediment load (tonnes) for high magnitude 90th percentile events A) upstream and B) downstream of the River Breamish restoration area.

Sediment Load (Tonnes)	Upstream Hysteresis Class	Downstream Hysteresis Class
1744.69	Non-discernible	Non-discernible
583.84	Non-discernible	Clockwise f8
1110.84	Non-discernible	Non-discernible

4.2.3 Quantitative Analysis of Hysteresis Magnitude:

HI_{mid} values are calculated within Table 11 and Table 12, whereby hysteresis dynamics are constrained by hydrological conditions for all single loop turbidity events (19 upstream, 25 downstream) occurring within the River Breamish restoration area during the pre-restoration monitoring period. Almost all events included in analysis have a positive $HI_{mid,}$ indicating clockwise hysteresis. There is one exception from this, occurring upstream during December 2024 where a negative HI_{mid} indicates anticlockwise hysteresis, thus supporting the manual classifications demonstrated within Table 8. HI_{mid} values upstream of the restoration area lie in the range of –0.04 to +8.82 with an overall mean of +2.55, whereas downstream HI_{mid} values show a smaller range from +0.05 to +7.97 alongside a lower mean of +1.82. Statistical analysis of HI_{mid} values indicated no statistically significant difference in hysteresis magnitude between upstream and downstream sites during the pre-restoration monitoring period. A non-parametric Mann-Whitney U test identifies no statistically significant difference between sites median central tendencies (p=0.58, Appendix 1). Similarly, although HI_{mid} variability appears larger at the upstream site, an F-test for equal variances shows this is also not statistically significant (p=0.18, Appendix 1).

Table 11. River level-turbidity HI_{mid} values from the monitoring site located upstream of the River Breamish restoration area. Storm events occur during the pre-restoration monitoring period (February 2024-May 2025). Hysteresis direction CW = Clockwise, ACW = Anticlockwise.

Event number	Event start	Event end	River level variable			Turbidity (NTU) at Q_{mid}		Hysteresis index	Hysteresis direction
			Q_{min}	Q_{max}	Q_{mid}	TU_{RL}	TU_{FL}		
1	08/02/2024	14/02/2024	0.23	1.22	0.72	55.64	9.29	4.99	CW
2	14/02/2024	18/02/2024	0.35	0.64	0.50	6.47	5.15	0.26	CW
3	18/02/2024	19/02/2024	0.35	0.46	0.40	4.93	3.84	0.28	CW
4	21/02/2024	24/02/2024	0.27	0.46	0.37	5.29	4.34	0.22	CW
5	02/03/2024	05/03/2024	0.25	0.74	0.49	29.49	8.39	2.52	CW
8	10/03/2024	14/03/2024	0.23	1.00	0.62	62.52	10.62	4.89	CW
13	04/05/2024	09/05/2024	0.15	0.36	0.26	46.97	5.25	7.95	CW
16	09/07/2024	13/07/2024	0.07	0.29	0.18	10.59	6.07	0.75	CW
17	13/07/2024	15/07/2024	0.08	0.57	0.33	15.71	7.32	1.15	CW
22	23/08/2024	30/08/2024	0.07	1.20	0.63	179.03	18.23	8.82	CW
31	24/11/2024	27/11/2024	0.21	1.77	0.99	150.05	28.39	4.29	CW
33	01/12/2024	04/12/2024	0.22	0.92	0.57	25.30	14.65	0.73	CW
35	05/12/2024	06/12/2024	0.46	0.83	0.65	13.77	11.17	0.23	CW
38	22/12/2024	23/12/2024	0.21	0.25	0.23	3.02	3.14	-0.04	ACW
38	27/01/2025	03/02/2025	0.00	0.82	0.41	55.77	10.88	4.13	CW
40	10/02/2025	16/02/2025	0.03	0.35	0.19	18.75	5.09	2.68	CW
41	20/02/2025	23/02/2025	0.00	0.66	0.33	17.83	8.69	1.05	CW
43	13/03/2025	16/03/2025	0.01	0.10	0.05	22.17	5.99	2.70	CW
44	23/03/2025	25/03/2025	0.04	0.22	0.13	5.53	3.08	0.80	CW

Table 12. River level-turbidity HI_{mid} values from the monitoring site located downstream of the River Breamish restoration area. Storm events occur during the pre-restoration monitoring period (February 2024-May 2025). Hysteresis direction CW = Clockwise, ACW = Anticlockwise.

Event number	Event start	Event end	River level variable			Turbidity (NTU) at Q_{mid}		Hysteresis index	Hysteresis direction
			Q_{min}	Q_{max}	Q_{mid}	TU_{RL}	TU_{FL}		
2	14/02/2024	18/02/2024	0.78	0.95	0.87	6.52	5.86	0.11	CW
4	21/02/2024	24/02/2024	0.73	0.85	0.79	5.15	4.23	0.22	CW
5	03/03/2024	05/03/2024	0.71	1.00	0.86	20.19	9.69	1.08	CW
8	10/03/2024	13/03/2024	0.70	1.17	0.93	112.26	12.51	7.97	CW
12	25/03/2024	01/04/2024	0.66	1.30	0.98	76.40	12.56	5.08	CW
13	01/04/2024	03/04/2024	0.71	1.11	0.91	17.20	7.42	1.32	CW
15	09/04/2024	13/04/2024	0.87	2.79	1.83	230.80	61.91	2.73	CW
17	04/05/2024	08/05/2024	0.60	0.75	0.67	5.51	3.42	0.61	CW
19	22/05/2024	26/05/2024	0.54	1.47	1.01	62.92	13.22	3.76	CW
20	26/05/2024	29/05/2024	0.80	1.33	1.07	25.73	12.55	1.05	CW
21	29/05/2024	30/05/2024	0.80	1.02	0.91	11.46	8.73	0.31	CW
22	30/05/2024	31/05/2024	0.84	1.03	0.94	13.33	8.84	0.51	CW
25	14/07/2024	16/07/2024	0.61	0.91	0.76	23.68	10.88	1.18	CW
30	23/08/2024	28/08/2024	0.56	1.27	0.92	101.31	16.91	4.99	CW
34	26/09/2024	01/10/2024	0.53	1.05	0.79	33.76	10.02	2.37	CW
37	28/10/2024	30/10/2024	0.69	0.80	0.74	7.60	6.44	0.18	CW
38	02/12/2024	04/12/2024	0.78	1.12	0.95	33.24	19.08	0.74	CW
40	05/12/2024	07/12/2024	0.83	1.07	0.95	13.91	13.22	0.05	CW
41	07/12/2024	15/12/2024	0.83	3.40	2.12	169.10	65.29	1.59	CW
45	05/01/2025	10/01/2025	0.62	1.06	0.84	43.79	19.01	1.30	CW
47	24/01/2025	27/01/2025	0.51	0.70	0.60	15.03	7.00	1.15	CW
49	10/02/2025	15/02/2025	0.52	0.75	0.63	30.07	5.82	4.17	CW
50	20/02/2025	23/02/2025	0.64	1.05	0.85	11.34	8.34	0.36	CW
52	12/03/2025	17/03/2025	0.60	0.71	0.65	7.11	2.18	2.26	CW
54	16/04/2025	19/04/2025	2.63	0.69	0.62	3.59	2.63	0.36	CW

4.2.4 Spatial Variation of Hysteresis Magnitude:

For storm events where HI_{mid} data is available at both the upstream and downstream sites, 50% show comparable values (<0.5 difference), 17% show a larger HI_{mid} at the downstream site for and 33% show a larger HI_{mid} at the upstream site (Table 13). The downstream HI_{mid} typically exceeds the upstream HI_{mid} during early spring (March 2024) and late winter (February 2025) (Table 13). Comparatively, the HI_{mid} is larger at the upstream site across spring (March 2024), summer (May and August 2024) and winter (February 2025) (Table 13). The upstream HI_{mid} is up to 13 times larger than the downstream equivalent, whereas the downstream HI_{mid} is only up to 1.6 times larger than the upstream equivalent (Table 13). Therefore, the difference between sites is more pronounced where the hysteresis magnitude is larger upstream of the River Breamish channelised reach.

Consistent hysteresis magnitude occurs between the upstream and downstream sites during storm events conditioned by moderate flows, transporting lower sediment loads (up to 8.53t) (Table 13). Conversely, events with contrasting hysteresis between sites coincide with larger storms during which sediment loads are higher (up to 25.97t) (Table 13). For example, the storm event occurring between 10th-14th March 2024 exhibits a sediment load of 24.42t, this is reflected by a large difference in hysteresis magnitude between upstream (4.89) and downstream (7.97) sites (Table 13). The contrasting magnitudes demonstrate the impact of additional tributary inputs between sites delivering additional sediment to the channel during larger storm events and consequently elevating the hysteresis magnitude downstream.

Additionally, results indicate that differential hysteresis magnitude between sites reflects mean turbidity values; where if mean turbidity is higher upstream, the HI_{mid} is also higher upstream and contrariwise (Table 13). For example, the storm event occurring between 2nd-5th March 2024 exhibits mean turbidity values of 11.29 NTU and 10.07 NTU at the upstream and downstream sites respectively; this is reflected by HI_{mid} values of 2.52 and 1.08 respectively (Table 13). This pattern is also present when investigating the impact of hydrological forcing, whereby the same storm event displays increasing river levels of 0.49m and 0.28m at the upstream and downstream sites (Table 13).

However, implementing the HI_{mid} as the response variable in multiple linear regression modelling shows that the relationship between mean turbidity values and HI_{mid} is not statistically significant both upstream (p=0.71; Appendix 2) and downstream (p=0.71; Appendix 3) of the River Breamish restoration area. The relationship between river level rise and HI_{mid} is also not statistically significant either upstream (p=0.22; Appendix 2) or downstream (p=0.76; Appendix 3). Such lack of statistical significance indicates additional controls on hysteresis behaviour such as seasonality and morphology. Both sites exhibit elevated HI_{mid} during late spring / early summer (Table 13), thus indicating the potential impact of seasonal land use modifying sediment availability.

Table 13. River level-turbidity HI_{mid} values and corresponding sediment load (tonnes), mean turbidity (NTU) and river level rise (m) for storm events occurring during the pre-restoration monitoring period (February 2024-May 2025).

A) Upstream

Event start	Event end	Hysteresis index	Sediment Load (Tonnes)	Mean Turbidity (NTU)	River level rise (m)
14/02/2024	18/02/2024	0.26	8.53	5.70	0.29
21/02/2024	24/02/2024	0.22	3.54	4.71	0.19
02/03/2024	05/03/2024	2.52	10.42	11.29	0.49
10/03/2024	14/03/2024	4.89	24.42	12.78	0.76
04/05/2024	09/05/2024	7.95	3.58	5.51	0.20
13/07/2024	15/07/2024	1.15	5.00	10.80	0.49
23/08/2024	30/08/2024	8.82	25.97	7.39	1.13
01/12/2024	04/12/2024	0.73	n/a	12.60	0.70
05/12/2024	06/12/2024	0.23	n/a	2.98	0.37
10/02/2025	16/02/2025	2.68	n/a	7.09	0.32
20/02/2025	23/02/2025	1.05	n/a	4.47	0.65
13/03/2025	16/03/2025	2.70	n/a	3.30	0.09

B) Downstream

Event start	Event end	Hysteresis index	Sediment Load (Tonnes)	Mean Turbidity (NTU)	River level rise (m)
14/02/2024	18/02/2024	0.11	8.53	5.77	0.17
21/02/2024	24/02/2024	0.22	3.54	4.66	0.12
02/03/2024	05/03/2024	1.08	10.42	10.07	0.28
10/03/2024	14/03/2024	7.97	24.42	15.84	0.47
04/05/2024	09/05/2024	0.61	3.58	5.92	0.15
13/07/2024	15/07/2024	1.18	5.00	12.33	0.30
23/08/2024	30/08/2024	4.99	25.97	9.83	0.71
01/12/2024	04/12/2024	0.74	n/a	20.12	0.33
05/12/2024	06/12/2024	0.05	n/a	48.73	2.56
10/02/2025	16/02/2025	4.17	n/a	7.44	0.41
20/02/2025	23/02/2025	0.36	n/a	4.26	0.12
13/03/2025	16/03/2025	2.26	n/a	2.61	0.14

4.3 Specific Sediment Yield:

Specific sediment yield is a measure of export per unit area, accounting for the catchment drainage area at the point of measurement / sampling. The specific sediment yield within the Breamish restoration area is variable with low specific sediment yield (<5t/km^2) during February, March and August 2024, intermediate specific sediment yield (20-30t/km^2) during October 2024 and high specific sediment yield (>30t/km^2) during June 2024 (Figure 17).

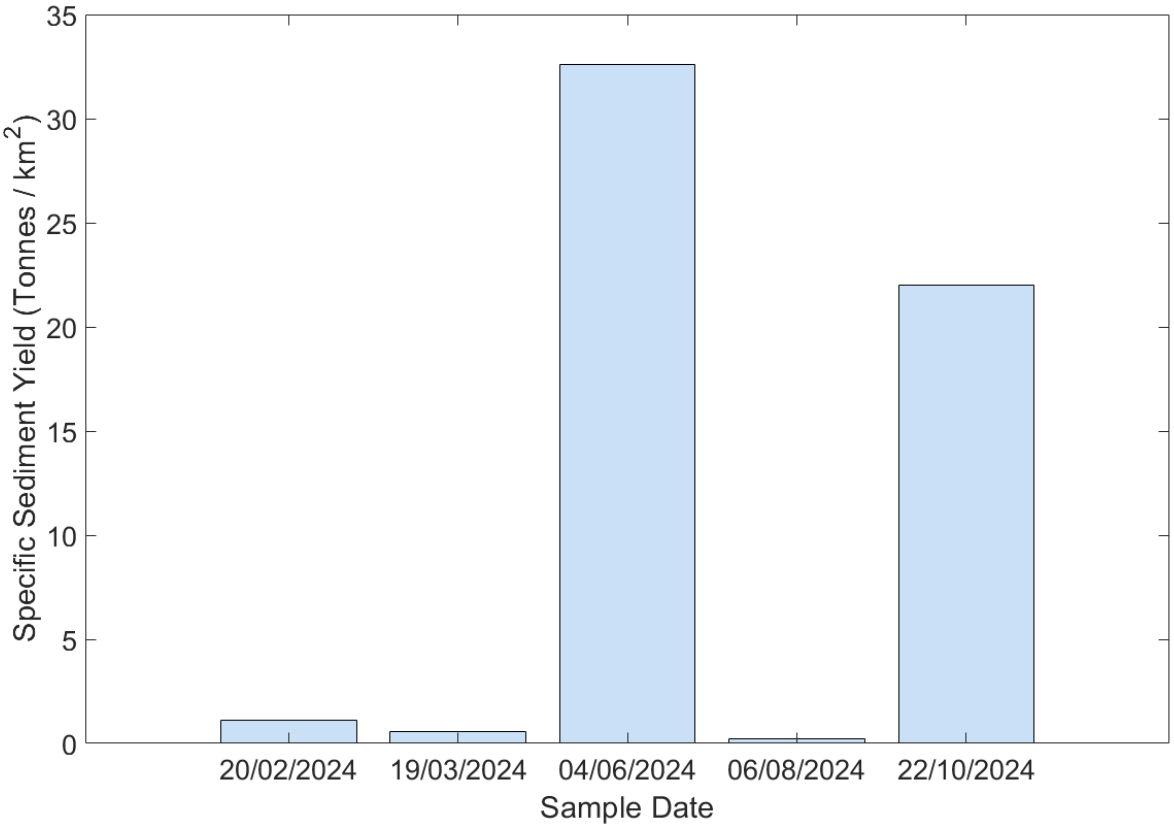


Figure 17. Specific sediment Yield (tonnes/km^2) in the River Breamish restoration area during the pre-restoration sampling period January 2024-October 2024.

4.4 Sediment Flux Particle Size Analysis:

Annual average median grain sizes upstream and downstream are 55µm and 18µm respectively, classified as silt (4µm-63µm). The average grain size composition of all samples collected at the upstream site is 56% silt, 38% sand and 6% clay (Table 14), whereas at the downstream site it is 76% silt, 14% sand and 10% clay (Table 14). Coarser sediment was sampled at the mid-reach site with an annual average median grain size of 112µm, classified as sand (63µm-1mm). The average grain size composition of all samples collected at the mid-reach site is 43% silt, 52% sand and 5%

clay (Table 14). At the mid-reach site gravel accounts for 0.01% of total sediment flux and was captured during the summer sampling period (Table 14).

Table 14. Annual mean percentage of gravel, sand, silt and clay volume in sediment samples collected across the River Breamish restoration area.

	Upstream	Mid-reach	Downstream
Gravel:	0	0.01	0
Sand:	38	52	14
Silt:	56	43	76
Clay:	6	5	10

Furthermore, there is some seasonal variation between sample grain size composition within the River Breamish restoration area. The samples are predominantly composed of mud across all seasons (Table 15), with mainly sandy silt texture. However, there are some grains of silty sand and sand textural groups present during spring, summer and autumn samples (Figure 18). Winter samples have the largest proportion of mud (78.46%) (Table 15) and no textural variation as all material falls within the sandy silt textural group (Figure 18). Conversely, summer samples have the lowest proportion of mud (59.19%) (Table 15) but the largest textural variation with material within sandy silt, silty sand and sand textural groups (Figure 18). During summer, autumn and winter the greatest proportion of grains are classified as coarse silt (14.86%, 15.57% and 21.54% respectively), however during spring they are predominantly fine sand (16.09%) (Table 15).

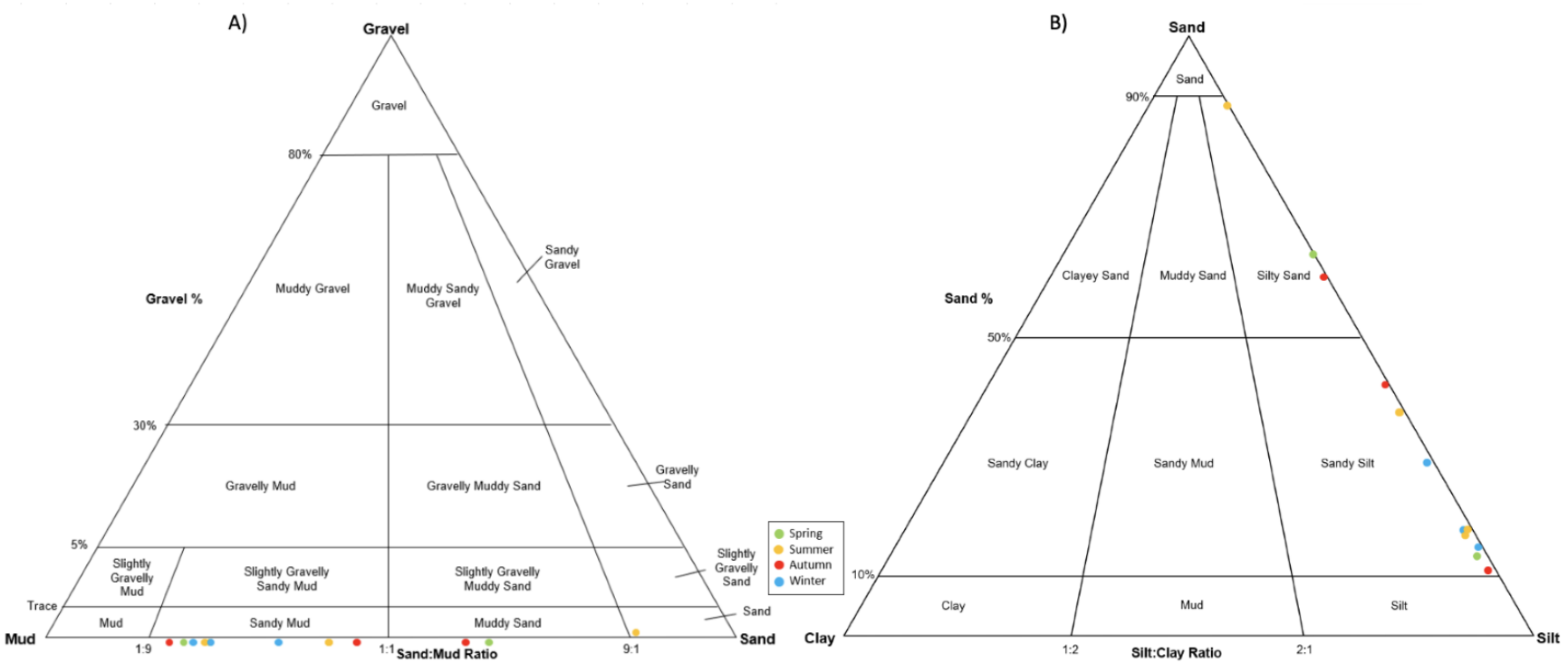


Figure 18. Grain size composition of time integrated sediment samples collected across the River Breamish restoration area A) Gravel-Sand-Mud and B) Sand-Silt-Clay.

Table 15. Seasonal average particle size composition of time integrated sediment samples collected across the River Breamish restoration area. Dominant composition is indicated in bold.

	Spring	Summer	Autumn	Winter
Gravel:	0.00%	0.03%	0.00%	0.00%
Sand:	39.79%	40.78%	38.84%	21.54%
Mud:	**60.21%**	**59.19%**	**61.16%**	**78.46%**
Average Composition (%):				
Very Fine Gravel:	0.00%	0.03%	0.00%	0.00%
Very Coarse Sand:	0.00%	0.90%	0.15%	0.00%
Coarse Sand:	0.72%	4.82%	4.06%	0.01%
Medium Sand:	10.33%	10.38%	10.27%	1.44%
Fine Sand:	**16.09%**	11.64%	11.50%	6.23%
Very Fine Sand:	12.65%	13.04%	12.87%	13.86%
Very Coarse Silt:	13.58%	14.28%	14.29%	20.50%
Coarse Silt:	15.34%	**14.86%**	**15.57%**	**21.52%**
Medium Silt:	13.42%	13.21%	13.93%	16.53%
Fine Silt:	11.14%	10.58%	11.09%	12.35%
Very Fine Silt:	5.90%	5.44%	5.56%	6.55%
Clay:	0.84%	0.82%	0.72%	1.02%

Moreover, grain size distributions are similar between seasons with the majority of grains distributed between 10-100µm (Figure 19). The distribution during winter is visually different than other seasons due to a lower proportion of grain class weight being recorded at 10µm (3.75%) compared to spring (4%), summer (4.4%) and autumn (4%) (Figure 19). Similarly, a small but substantial proportion of grains (>2%) exceed 100µm during spring, summer and winter (2%, 2.6% and 2.25% respectively), whereas during autumn only 1.8% are present the 100µm boundary (Figure 19). The grain size corresponding with the peak class weight varies from spring to winter ~20µm, ~15µm, ~15µm and ~30µm respectively (Figure 19. Overall, larger particles are more prominent during winter than other seasons, corresponding with higher magnitude flood events. It is important to note that the number of samples collected across each season varied (3 in winter, 2 in spring, 5 in summer, 3 in autumn) due to varied sampling intervals and insufficient mass available for analysis at the upstream site; consequently, this limits the accuracy of comparisons between sample periods.

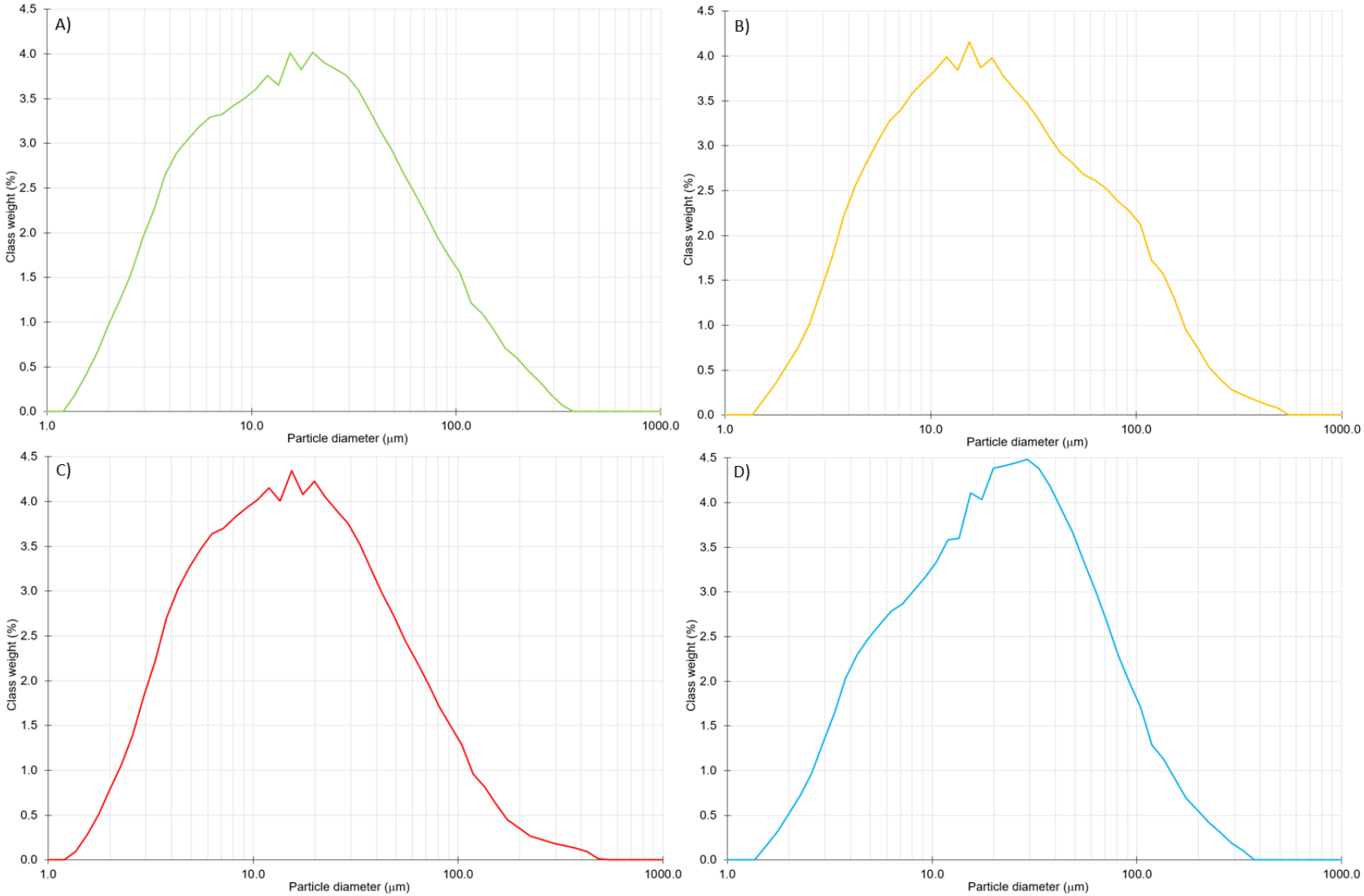


Figure 19. Seasonal grain size distributions of time integrated sediment samples collected across the River Breamish restoration area during A) Spring, B) Summer, C) Autumn and D) Winter.

4.4.2 Relationship Between River Level and Particle Size:

There is clear variation in the relationship between river level and grain size at each side of the River Breamish restoration area. Different d50 grain sizes are recorded across the reach at similar river levels; for example, for river levels of 1.2m, the upstream median grain size is 22µm (Figure 20A), whereas the downstream median grain size is 16µm (Figure 20B). Generally, higher d50 grain sizes are observed at the upstream site, with grain sizes more than five times larger than the downstream equivalent. Moreover, Pearson's product moment correlation coefficients indicate varied strength in the relationship between particle size and river levels across the River Breamish restoration area. Upstream of the channelised reach there is moderate positive linear relationship (R^2=0.41), suggesting that particle size increases / decreases in proportion to river levels (Figure 20A). Conversely, a weak negative linear relationship (R^2=0.04) is present downstream of the channelised reach, indicating that when river level increases, the particle size of sediment in flux decreases (Figure 20B). Regression modeling confirms that there is no statistically significant relationship between river level and particle size either upstream or downstream of the River Breamish restoration area (p=0.56 and p=0.74 respectively; Appendix 4).

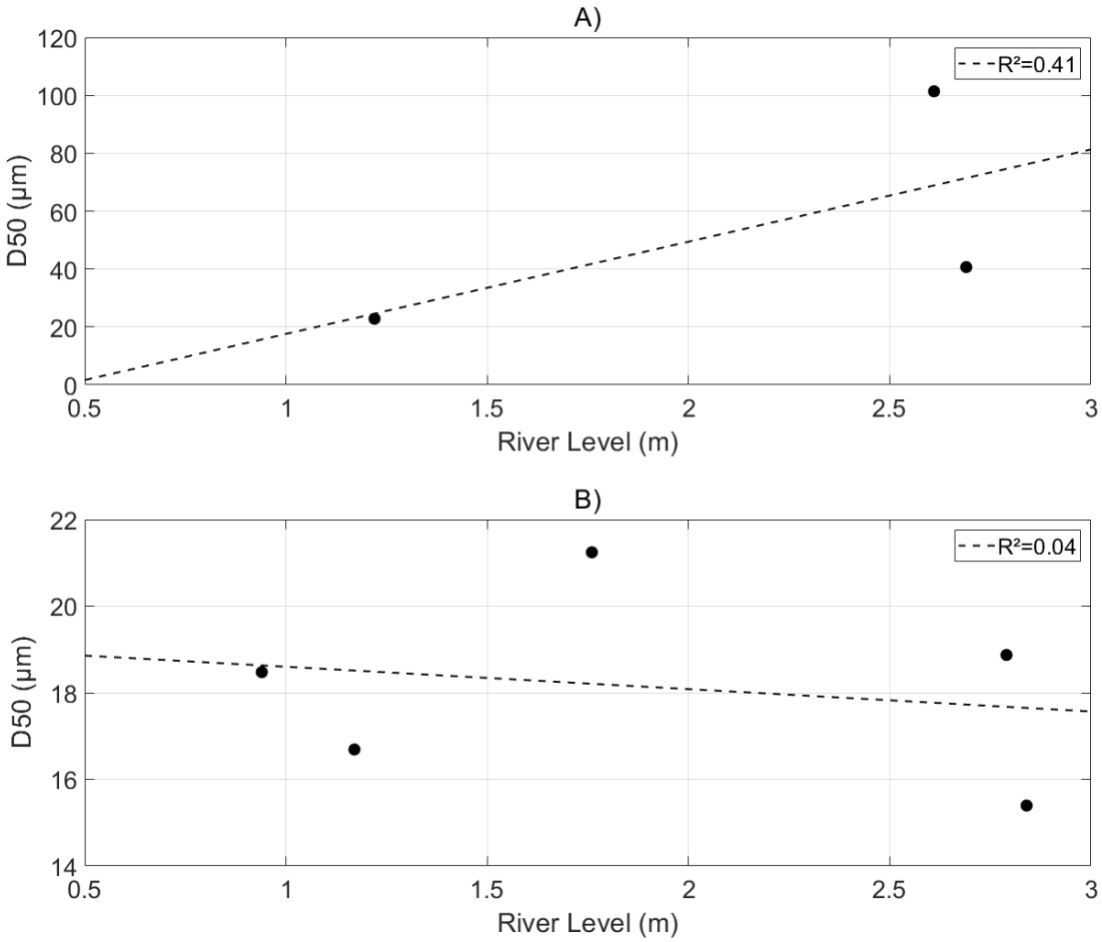


Figure 20. Relationship between river level (m) and median grain size (D50) from time integrated sediment samples collected A) Upstream and B) Downstream of the River Breamish restoration area. Note that Y-axis scales differ between subplots.

4.5 Sediment Flux Macronutrient Composition:

Increased variability in total N content between the upstream, mid-reach and downstream sites is indicated by a progressive increase in range (Table 16); this corresponds with an increase in peak total N content (6.16t, 6.75t and 9.09t respectively). Peak total N content is observed across the River Breamish restoration area within the sample collected in June 2024. Statistical testing is limited by the low sample number, however t-Tests indicate no statistically significant difference in total N content at the upstream and downstream sides of the restoration area (p=0.95; Appendix 5) or at the mid-reach and downstream sites (p=0.84; Appendix 6).

Similarly to total N content, peak total P concentrations are observed during early summer sampling (June 2024) with values of 2.79t, 2.89t and 2.39t at the upstream, mid-reach and downstream sites respectively (Table 16). The upstream site displays less variability than the mid-

reach and downstream sites with total P ranges of 0.79t, 2.87t and 2.36t respectively (Table 16). Again, statistical testing is limited by low sample numbers and data availability so comparison between the upstream and downstream sides of the restoration area could not be conducted. However, t-tests indicate that there is no statistically significant difference in total P values within mid-reach and downstream samples (p= 0.91; Appendix 7).

Table 16. Total N and total P content of time integrated sediment samples collected from upstream, mid-reach and downstream of the River Breamish restoration area.

Date	Site	Total N (Tonnes)	Total P (Tonnes)
20/02/2024	Upstream	0.13	n/a
	Mid-reach	0.16	n/a
	Downstream	0.27	n/a
19/03/2024	Mid-reach	0.18	n/a
	Downstream	0.19	n/a
04/06/2024	Upstream	6.16	2.79
	Mid-reach	6.75	2.89
	Downstream	9.09	2.39
06/08/2024	Mid-reach	0.10	0.02
	Downstream	0.12	0.03
22/10/2024	Upstream	4.75	2.00
	Mid-reach	5.94	2.25
	Downstream	0.99	n/a

4.5.1 Phosphorus Composition:

Apatite bound P is the dominant P form across the River Breamish channelised reach during all TIMS sampling periods (Figure 2). On average apatite bound P accounts for ~63%, ~70% and ~45% at the upstream, mid-reach and downstream sites respectively. Redundant soluble P represents the second largest contribution to P composition at the downstream site (Figure 21), accounting for ~21% of the total P. At the mid-reach and upstream sites redundant soluble P concentrations are lower, accounting for on average 11% and 13% of total P at each site respectively. Organic P measurements also exhibit the highest mean concentrations at the downstream site (15%) compared to the mid-reach (7%) and upstream (9%) sites. Labile and residual P form small amounts (<1% and <2%) of the total P across all sites (Figure 21).

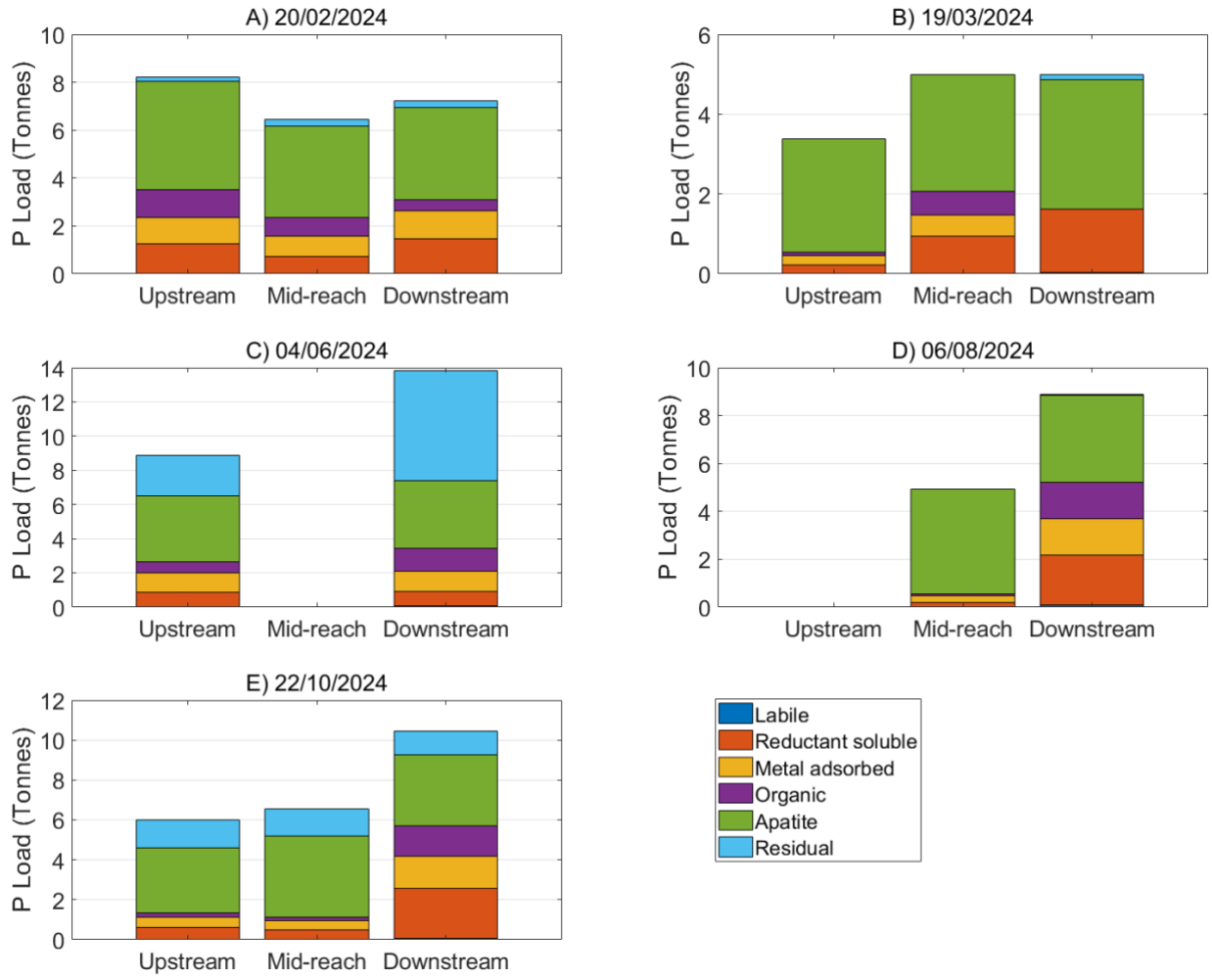


Figure 21. Phosphorus (P) composition of time integrated sediment samples collected from upstream, mid-reach and downstream of the River Breamish restoration area on A) 20/02/2024, B) 19/03/2024, C) 04/06/2024, D) 06/08/2024 and E) 22/10/2024.

4.5.2 Bioavailable Nitrogen Components:

The concentrations of bioavailable N components show temporal variation across the pre-restoration sampling period at the mid-reach and upstream sites. The mid-reach site is the only site with a complete set of results, here NH_4 concentrations range from 0.501t in June 2024 to below the limit of detection in February 2024 and August 2024 (Table 17); NO_3 + NO_2 concentrations range from 0.119t in winter to 0.007t in autumn (Table 17). Single factor ANOVA testing indicates that NO_3 + NO_2 concentrations at the mid-reach site do not reflect a statistically significant difference between seasons (p=0.051; Appendix 8). Mean concentrations of NH_4 are overall higher at the mid-reach site (0.309t) than upstream (0.189t). Conversely, mean NO_3 + NO_2 concentrations are higher at the upstream site (0.063t) than mid-reach (0.049t). The only

measurable sample from the downstream site was retrieved during winter, however NH_4 was below the limit of detection. The downstream sample exhibits the highest concentration of NO_3 + NO_2 across all sites (0.186t) (Table 17).

Table 17. Concentrations of available N (nitrate (NO_3) + nitrite (NO_2) and ammonium (NH_4)) in time integrated sediment samples collected from upstream, mid-reach and downstream of the River Breamish restoration area. <LOD indicates values below the limit of detection.

Date	Site	$NO_3 + NO_2$ (Tonnes)	NH_4 (Tonnes)
20/02/2024	Upstream	0.119	0.297
	Mid-reach	0.134	<LOD
	Downstream	0.186	<LOD
19/03/2024	Mid-reach	0.072	0.088
04/06/2024	Mid-reach	0.014	0.501
06/08/2024	Mid-reach	0.008	<LOD
22/10/2024	Upstream	0.007	0.081
	Mid-reach	0.017	0.338

4.6 SCIMAP Delivery Modelling:

4.6.1 Land Connectivity + Potential Storage Areas:

The land connectivity risk is presented across the River Breamish catchment; this is the same for all pollutants based on the scientific assumption that lateral surface flow is the dominant mechanism of delivery for diffuse pollutants (Figure 22). Areas of high connectivity risk (1) facilitating pollutant delivery into the channel are predominantly located on the land adjacent to the channel rather than within headwater areas which exhibit consistently low risk values (<0.5) (Figure 22). Variable connectivity risk is exhibited on the land adjacent to the channel within the middle section of the catchment, with landscape features such as roads and tractor tracks within agricultural fields contributing to the delivery of fine sediment into the channel (Figure 22). High connectivity is prominent around the channelised reach and restoration area where the western adjacent floodplain displays maximum risk values (1) (Figure 22). Floodplain landscape features are assigned a delivery risk value of 1 as the topographic index indicates that flat ground is both most likely to be saturated and most frequently saturated, consequently saturation excess

overland flow is generated which rapidly delivers diffuse pollutants into the channel. The proposed restoration will likely not reduce the delivery of pollutants from the catchment into the channel, instead it may alter the in-stream processing that impacts delivery out of the river onto the floodplain.

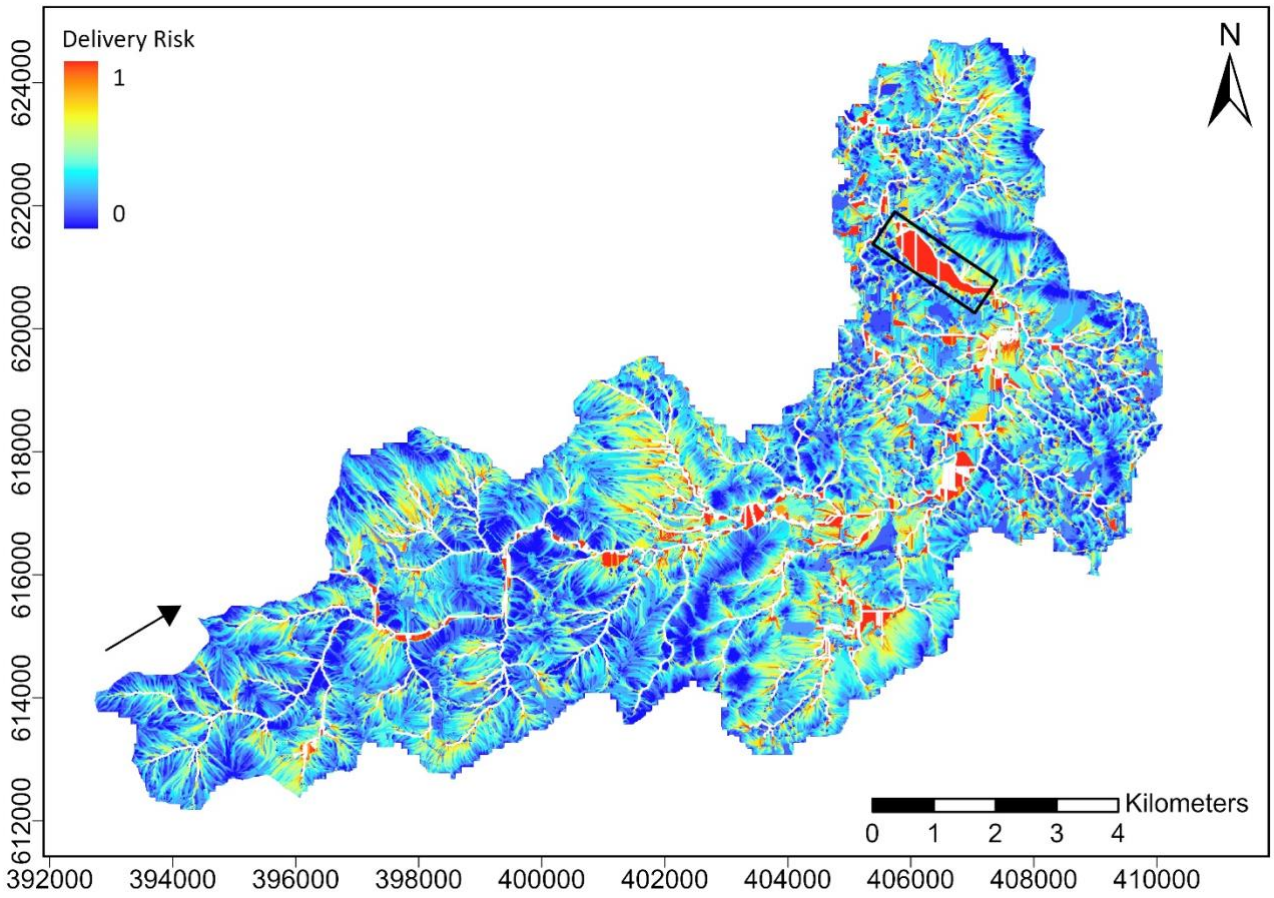


Figure 22. Connectivity / delivery risk map within the River Breamish catchment. The black rectangle indicates the restoration area and the black arrow indicates the direction of flow from south-west to north-east. Delivery risk values 1 is high and 0 is low.

4.6.2 Potential Critical Source Areas:

SCIMAP modelling displays variable fine sediment (Figure 23), NO3 (Figure 24) and PO4 (Figure 25) erosion risk across the Breamish catchment. The main channel sustains in-channel risk values <0.5 across all pollutant models. The lowest risk values are observed within the headwater areas, averaging to ±0.25σ within the main channel at the River Breamish restoration area (Figure 23; Figure 24; Figure 25). Potential critical source areas of fine sediment are identified within isolated low order tributaries / streams located upstream of the restoration area where both high

erosional risk (source) and high connectivity risk (pathway) combine to produce high in-channel risk >0.75 (Figure 23). While the south western headwaters display high erosion risk >0.75 (Figure 23), this is not translated to in-channel risk due to the aforementioned low delivery risk <0.5 (Figure 22). Therefore, while headwaters represent prominent diffuse sediment source areas, there is no pathway of sediment transport into the channel.

Similarly, macronutrient (NO_3 and PO_4) pollutant modelling indicates the highest erosional risk (>0.75) within the headwater areas characterised by high, steep sided slopes. In channel risk levels are locally variable with areas of elevated risk distributed throughout the north and central parts of the River Breamish catchment. NO_3 and PO_4 in-channel risks follow a broadly similar pattern, however PO_4 risk is more concentrated, with the highest risk values >+2.2σ occurring within a small area of the mid-catchment (Figure 25). There is a notably large patch of high in-channel risk located immediately downstream of the restoration area (Figure 24; Figure 25). While this does not contribute to the nutrient loadings within the restoration area, it will have critical impacts downstream, most prominently regarding NO_3 inputs (Figure 24). The River Breamish catchment overall displays a range of sedimentary and macronutrient source areas that are well connected to the channel, this translates to locally variable in-channel risk within the River Breamish catchment.

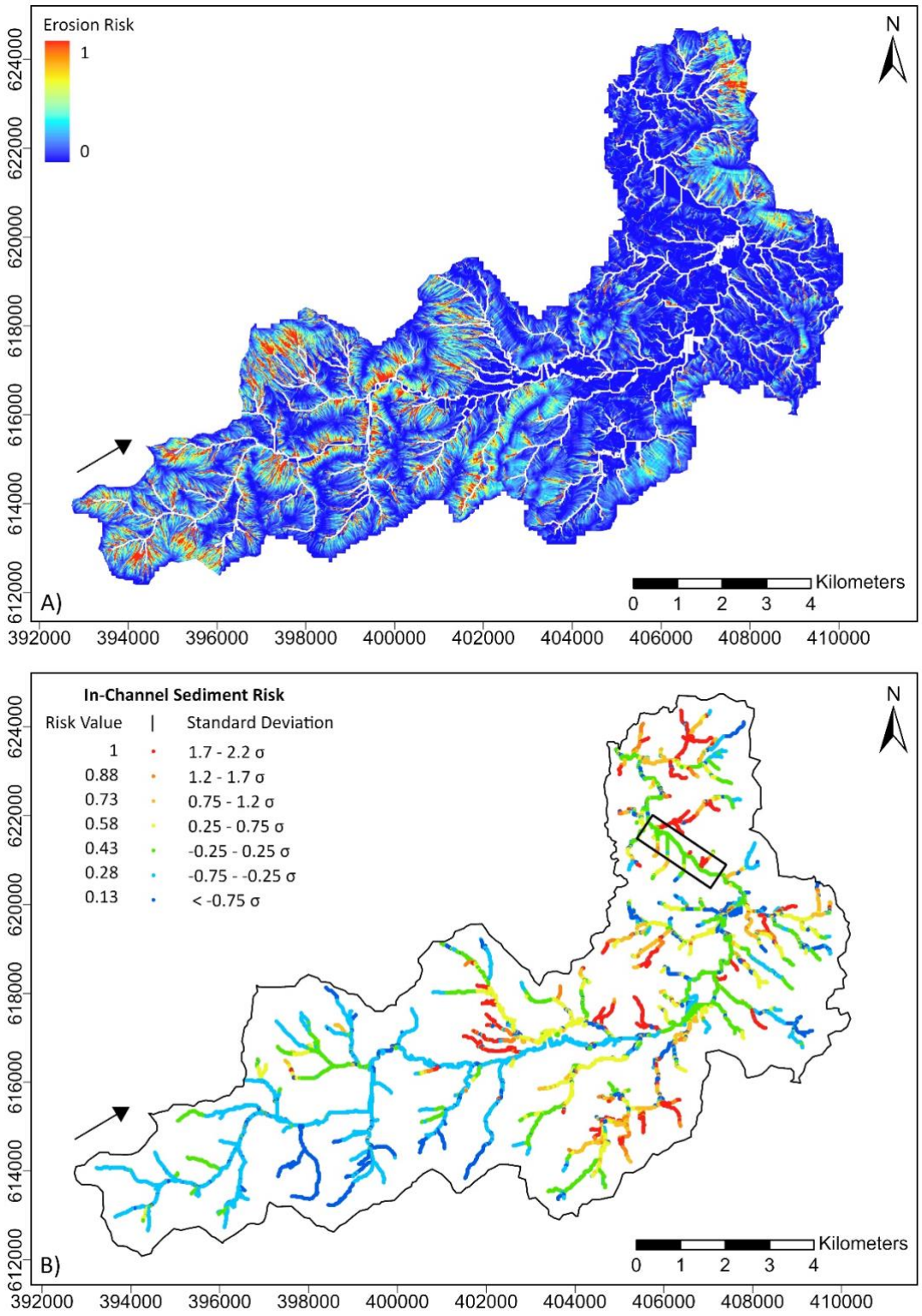


Figure 23. Distributed erosion and in-channel fine sediment risk within the River Breamish catchment produced by SCIMAP modelling. In-channel risk concentrations are expressed as risk values and as standard deviations from the mean. The black rectangle indicates the restoration area and the black arrows indicate the direction of flow from south-west to north-east. Risk values 1 is high and 0 is low.

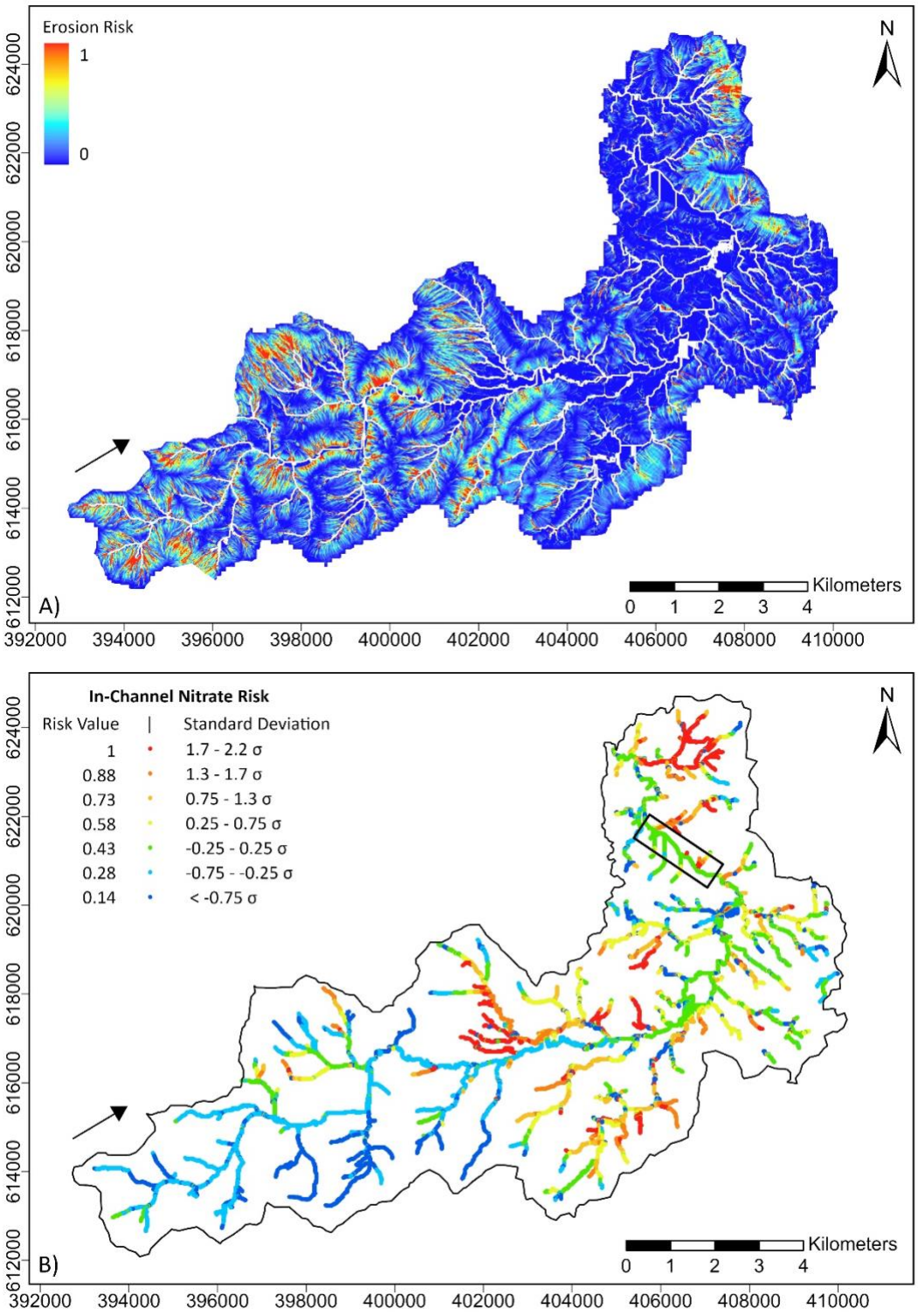


Figure 24. Distributed erosion and in-channel nitrate (NO_3) risk within the River Breamish catchment produced by SCIMAP modelling. In-channel risk concentrations are expressed as risk values and as standard deviations from the mean. The black rectangle indicates the restoration area and the black arrows indicate the direction of flow from south-west to north-east. Risk values 1 is high and 0 is low.

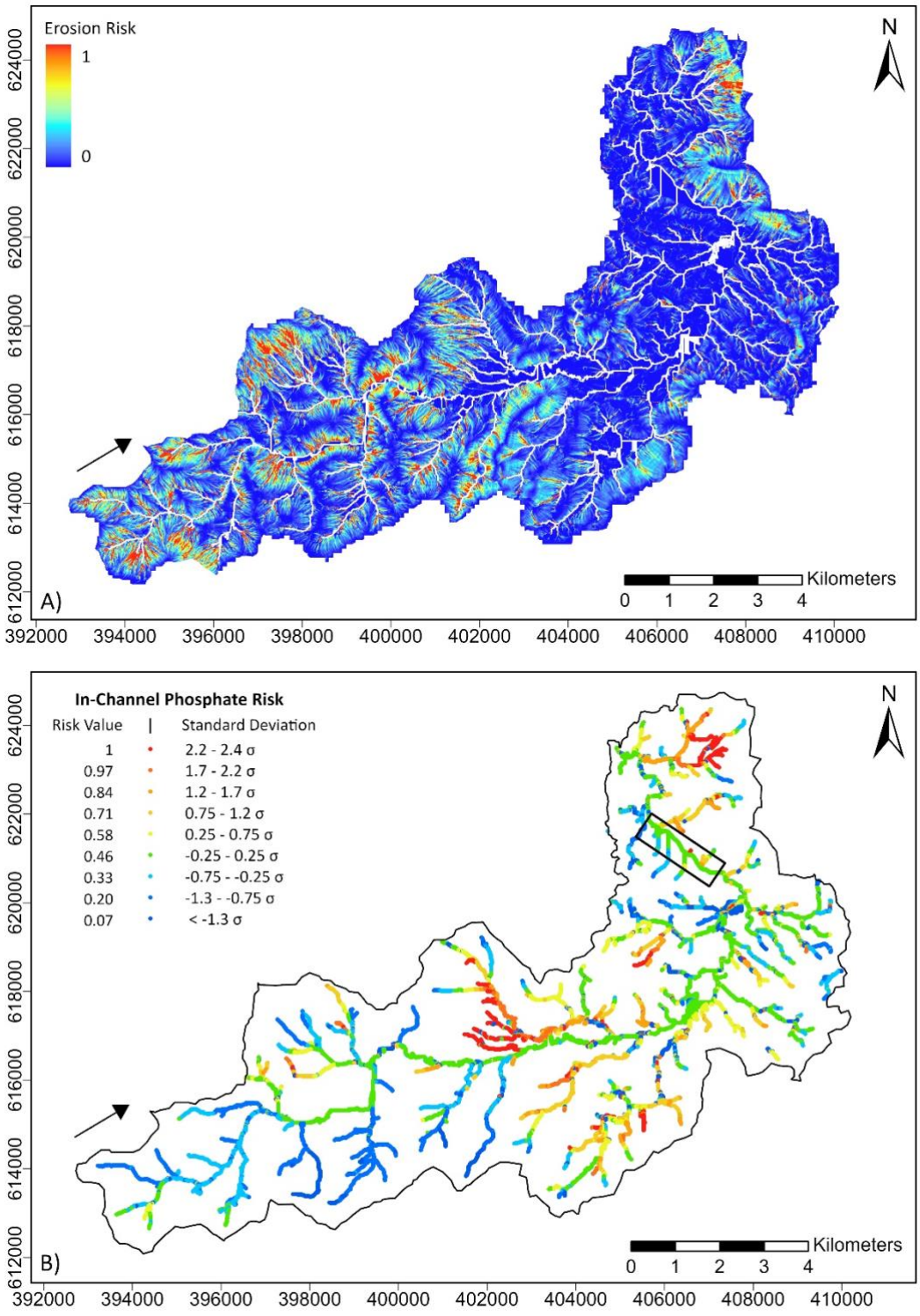


Figure 25. Distributed erosion and in-channel PO_4 risk within the River Breamish catchment produced by SCIMAP modelling. In-channel risk concentrations are expressed as risk values and as standard deviations from the mean. The black rectangle indicates the restoration area and the black arrows indicate the direction of flow from south-west to north-east. Risk values 1 is high and 0 is low.

Chapter 5 – Discussion:

This chapter is an interpretation of the study's key findings utilising relevant literature and observations made during fieldwork.

5.1 Relationship between river level and turbidity:

This section relates to research question 1, discussing the results of hysteresis analysis to determine the proximity of fine sediment source areas contributing to poor pre-restoration water quality.

5.1.1 Hysteresis Classification:

There is a clear dominance of clockwise hysteresis both upstream (41%) and downstream (52%) of the River Breamish restoration area (Table 7); this is reaffirmed by the prominence of positive HI_{mid} values (Table 11; Table 12). Clockwise hysteresis loops indicate rapid intra-event sediment exhaustion, whereby readily available sediment sources located proximal to the channel are mobilised via surface runoff processes during the rising limb of a storm hydrograph (Lawler *et al.*, 2006). The rapid sedimentary response to hydrological forcing (river level rise) means that while flow remains adequate during the later falling limb stages of a storm, not much fine sediment transport occurs due to lack of availability (Lawler *et al.*, 2006). The dominance of clockwise hysteresis within this study is concurrent with ongoing research suggesting that agricultural intensification is a major source of surplus fine sediment (Wilkinson and McElroy, 2007; Wharton, Mohajeri and Righetti, 2017; Jones *et al.,* 2011). The River Breamish restoration area is surrounded by multiple farms, with active use of tractors and machinery within the fields and on the slopes adjacent to the channel. Such land management leads to enhanced soil degradation / retrogression and enhances surface runoff processes, thus impacting sediment delivery (Perks *et al.,* 2015). Additionally, the tracks left in the soil from tractor wheels create channels of concentrated surface runoff (Figure 26) which enhance sediment erosion and rapidly deliver material into the channel during the onset of a rainfall event (Williams-Mounsey *et al.,* 2024). Moreover, the prevalence of clockwise hysteresis within the River Breamish restoration area (Table 7) demonstrates the lasting impacts of historic channelisation fundamentally altering fluvial environmental processes. The reduced floodplain buffering and storage potential within the channelised reach exacerbates the rapid delivery of material from the adjacent landscape via

surface runoff processes during the rising limb (Noe and Hupp, 2005). This is consistent with evidence of intra-event sediment exhaustion leading to clockwise hysteresis patterns (Lawler *et al.*, 2006).



Figure 26. Image showing tractor wheel tracks leading towards the River Breamish within an adjacent field.

Moreover, during the pre-restoration monitoring period there is a single isolated event displaying anticlockwise hysteresis at the upstream site (Figure 1; Table 7). This type of hysteresis demonstrates a falling limb turbidity extension (Lawler *et al.*, 2006), indicating diffuse sedimentary sourcing from the upper slope headwaters and delayed transport / lag time (Klein, 1984). It is deduced that this pattern occurred as a result of delayed transport via overland flow from the agricultural areas located further upstream in the middle section of the wider River Breamish catchment. The event occurred during winter 2024, displaying a low river level peak (0.25m) relative to the preceding events. This is consistent with the longstanding theory that anticlockwise hysteresis occurs during the winter season during relatively dry conditions (Klein, 1984), where increased soil absorption leads to very slow suspended sediment response, consequently leading to anticlockwise hysteresis as sediment peaks lag behind river level (Klein,

1984). This event transpires a negative HI_{mid} (Table 11), further supporting visual classification, showing methodological robustness of this research combining both manual and quantitative classifications.

Furthermore, this study observes more complex clockwise hysteresis patterns through complex F8 loops (Figure 1), suggesting multiple sources and pulses of sediment delivery. Clockwise F8 hysteresis accounts for 16% and 11% of events at the upstream and downstream sites respectively, indicating high turbidity levels following the hydrograph peak (Smith and Dragovich, 2009). It is deduced that eroding banks at either end of the channelised reach (Figure 27) continue to act as source of fine sediment after rainfall has ceased and flow has declined (Smith and Dragovich, 2009). This is facilitated by the increased water surface slope and velocity within the River Breamish channelised reach accelerating bed erosion during high flow events and consequently generating additional sedimentary input through bank erosion / incision (Todd-Burley *et al.,* 2021). Due to the cohesive properties, fine-grained particles are easily mobilised during peak flows (Zhang, 2014), consequently downstream sedimentation occurs within areas of inefficient filtration, thus demonstrating how the historic chanelisation of the River Breamish inhibits natural environmental functioning (Hohensinner, Hauer and Muhar, 2018). At the downstream site, the delayed contribution of sedimentary input during the falling limb is catalysed by rapid delivery of fine sediment from tributary sub-catchments (Smith and Dragovich, 2009). This site is located downstream of additional local sediment sources (Kirk Burn, Sheer Burn and Randy Burn) that are not hydrologically connected to the upstream site (Figure 6).

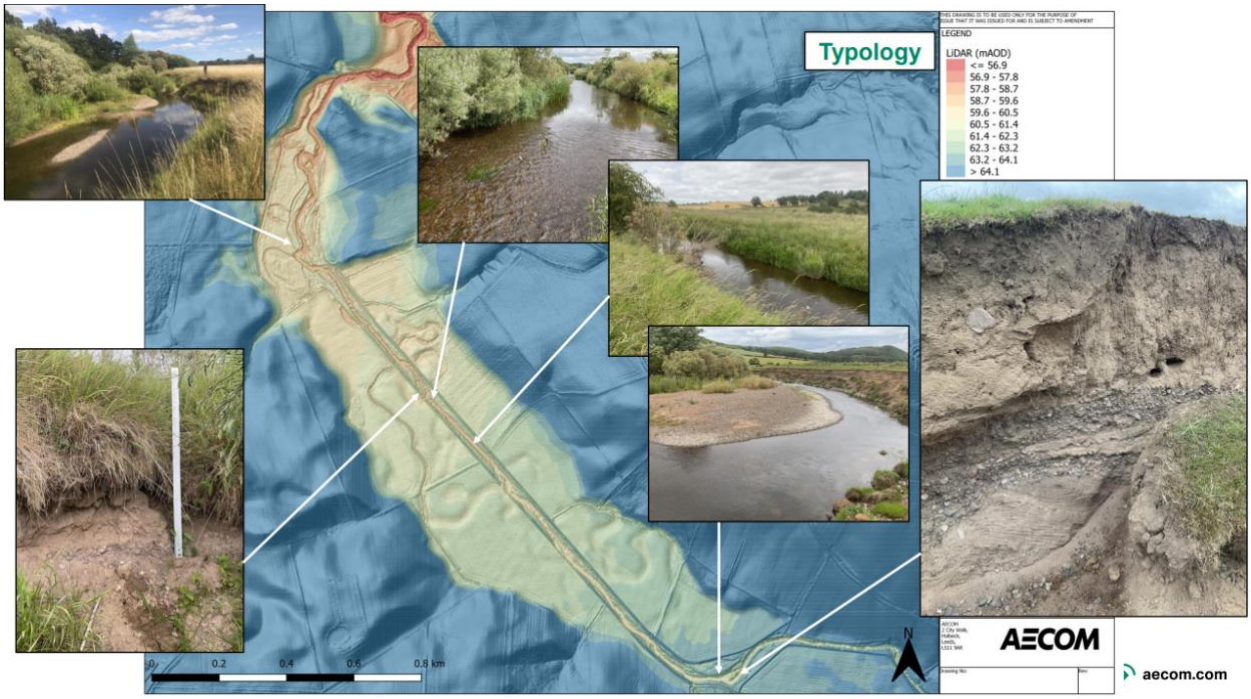


Figure 27. LIDAR Imagery from the River Breamish restoration area. Overlayed photos show differential channel conditions upstream, mid-reach and downstream of the channelised reach. (Source: Williams et al., 2023, p.6)

Anticlockwise F8 loops are less frequently observed, accounting for 9% of events at both the upstream and downstream sites, whereby multiple pulses of sediment are mobilised during a storm event (Table 7). Events displaying this pattern often followed large storm events, during which widespread rainfall across the Breamish basin activates distant sediment sources (Jing *et al.*, 2025). Sedimentary processes are enhanced during extreme events, such that sediment availability depletes, consequently reducing the transport capacity within the River Breamish and leading to temporary storage within the channel (White, 2008; Wilkes *et al.,* 2019). Woody debris was observed within the Breamish channelised reach, this will act as a natural dam, trapping and storing sediment under such conditions (Figure 28) (Mott, 2006). Subsequent flows disrupt sedimentary stores (Charlton, 2007) and entrainable fine grains are resuspended (Malutta *et al.*, 2020; Preziosi-Ribero *et al.,* 2020) and transported downstream through the channelised reach. Such remobilisation leads to the elevated sedimentary response during the rising limb for events with large preceding storms. This is combined with a further enhanced sediment signal during the falling limb resulting from distal inputs from the upper catchment headwaters, thus fine sediment continues to be mobilised through the catchment once rainfall has ceased (Smith and Dragovich,

2009; Klein, 1984). Overall, anticlockwise F8 loops are least frequently observed and represent the lowest magnitude transfer events (~1.33t at the upstream site) (Table 9), showing the activation of distant sources that only contribute after peak flow.



Figure 28. Image showing large woody debris within the River Breamish.

Additionally, hysteresis loops often display multiple pulses of turbidity, reflecting multiple phases of sedimentary input from the complex interconnected hydro-metrological, geomorphological and anthropogenic factors that combine to influence sediment dynamics within drainage basins (Reid *et al.,* 1997). These patterns are poorly understood due to the variability and complexity, are often observed during consecutive storms immediately following one another (Upadhayay, Granger and Collins, 2021). Events displaying non-discernible hysteresis patterns dominate the highest magnitude sediment transfer events across the reach (90th percentile) (Table 10), this implies that extreme flow events overwhelm typical sediment transport mechanisms, resulting in unpredictable and highly variable material movement (Vercruysse, Grabowski and Rickson, 2017). This is particularly prominent in channelised reaches like the Breamish as the limited lateral hydrological connectivity leads to complex sedimentary routing through multiple flow pathways and source mixing during large transfer events, thus increasing the systems unpredictability (Arnaud *et al*., 2019).

5.1.2 Frequency and Magnitude of Hysteresis Patterns:

A higher proportion of events exhibit clockwise hysteresis downstream than upstream of the channelised reach (Table 7). This is expected as previous research suggests that channelised reaches similar to the River Breamish exhibit accelerated downstream transport and accumulation due to reduced lateral hydrological connectivity impeding sediment exchange between the channel and floodplain (Opperman *et al*., 2010; Thoms, 2003), thus reducing opportunity for material deposition (McMillan and Noe, 2017).

Additionally, clockwise F8 loops occur more frequently than anticlockwise F8 loops downstream of the restoration area (Table 7), however less sediment is transported on average (7.09t compared to 15.05t respectively) (Table 9). This combined with high variability in sediment loadings during clockwise F8 events (IQR > 300t) (Figure 16) indicates inconsistent source activation and that such variability may occur due to interactions with additional tributary hydrological inputs (Wang *et al.,* 2025). An overall larger range of hysteresis magnitude (HI_{mid}) is present at the upstream site (8.86) alongside a higher mean (2.55), further indicating that the variable sediment mobilisation mechanisms are influenced by localised land management and farming activities such as use of tractors and machinery within the fields located immediately adjacent to the channel (Figure 26) (Williams-Mounsey *et al.,* 2024). Additionally, a more natural system morphology is present upstream, including meandering and sediment bars (Figure 27), thus allowing temporary storage and delaying sediment peaks and leading to greater magnitude hysteresis and a higher mean HI_{mid} (Corenblit *et al.*, 2024). Further sedimentary input is present at the upstream site due to exacerbated erosion and runoff from the road located immediately upstream of the sondes sensor (Lawler *et al.,* 2006). Comparatively, the downstream site's more consistent hysteresis magnitude (range of 7.92 and mean of 1.82) reaffirms the ideology that sediment is rapidly transported through the reach with limited opportunity for deposition / accumulation (Opperman *et al*., 2010; Thoms, 2003). There is no statistically significant difference in HI_{mid} at either end of the River Breamish restoration area (Appendix 1), as the high natural variability of sediment dynamics masks statistical signals due to the small number of events in the available sample size (19 upstream and 25 downstream). This demonstrates that the factors combining to influence sediment delivery are highly event specific and not consistently different between sites.

On an event scale, HI_{mid} values are generally more consistent across the River Breamish restoration area during moderate storm events transporting lower average sediment loads (<8.53t), whereas contrasting HI_{mid} values occur during larger storm events (<25.97t) (Table 13). This suggests an increase in spatial heterogeneity of sediment inputs under higher energy conditions due to activation of additional sediment sources and inputs such as the tributaries located between the upstream and downstream sites (Figure 6) (Smith and Dragovich, 2009). However, neither mean sediment loads or river level rise are statistically significant predictors of HI_{mid} ($p>0.05$, Appendix 2; Appendix 3), indicating additional drivers of sediment transport such as channel morphology and the lasting impacts of channelisation (Corenblit *et al.*, 2024). Overall, while analysis of HI_{mid} values indicates spatial variation, statistical testing reflects the underlying stochastic nature of sediment transport dynamics (Tsai, Hung and Wu, 2020).

Another key point is that the higher magnitude hysteresis patterns are often recorded during or immediately following more intense, UK Met Office recognised named storms (Table 18). During such events extreme rainfall leads to enhanced sediment erosion and delivery via the mechanisms of rainsplash and sheet wash (Kinnell, 2005; Stella *et al.,* 2013). Higher river levels during these storms correspond with increased sediment flux within the River Breamish restoration area (Reid *et al.,* 1997; Aich, Zimmermann and Elsenbeer, 2014) as exposed, unstable banks collapse, producing a large sediment influx (White, 2008; Wilkes *et al.,* 2019).

Table 18. HI_{mid} values recorded <7 days post-initial storm impact for UK Met Office recognised storms.

Name	Date Of Impact	Upstream HI_{mid}	Downstream HI_{mid}
Kathleen	04/04/2024		2.73
Lilian	22/08/2024	8.82	4.99
Bert	22/11/2024	4.29	
Éowyn	24/01/2025	4.13	1.15

This study's results align with the prevailing theory that agricultural intensification is a major source of fine sediment supply within watercourses, increasing sediment availability and delivery (Wilkinson and McElroy, 2007; Wharton, Mohajeri and Righetti, 2017; Jones *et al.,* 2011). Events exhibiting the highest sediment loads occur during the summer season (Table 13), coinciding with peak farming activity. This supports the assertion that the land disturbance associated with

tractor use on adjacent fields creates concentrated channels of surface runoff (Figure 26), enhancing fine sediment erosion and mobilisation processes within the Breamish restoration area (Williams-Mounsey *et al.,* 2024). Conversely, the reduction in agricultural activities during autumn and winter corresponds with lower HI_{mid} values (Table 11; Table 12) further reinforcing the influence of seasonal land management on sediment dynamics within the River Breamish catchment.

Overall, the hysteresis behaviour observed within the River Breamish supports the theory that sediment transport processes within small sub-catchments are typically dominated by clockwise hysteresis (Seeger *et al.,* 2004; Smith and Dragovich, 2009). Fine sediment within the River Breamish is mainly sourced from proximal areas on the land adjacent to the channel and within the channel bank, with occasional inputs from diffuse agricultural areas towards the headwaters (Figure 29).

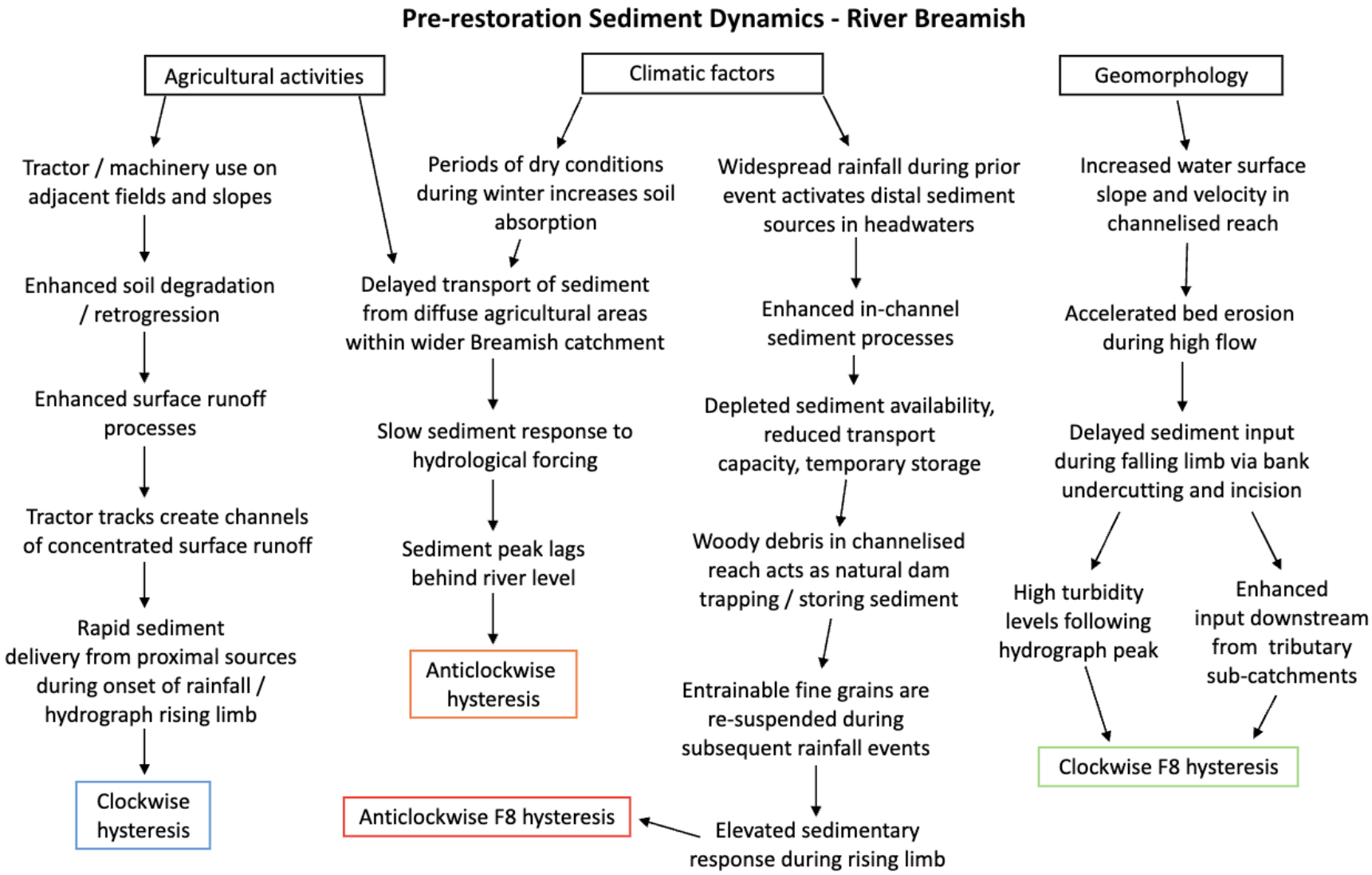


Figure 29. Summary of the main controls of observed river-level turbidity hysteresis dynamics observed within the River Breamish restoration area during the pre-restoration monitoring period. Includes sediment source, delivery, transport / storage and resultant hysteric response.

5.2 Composition of Sediment Flux in the River Breamish Restoration Area:

This section relates to research question 2, discussing analysis of time integrated sediment sampler material to determine the pre-restoration baseline composition of sediment in flux within the River Breamish.

This research shows clear variation in specific sediment yield between sampling periods with high specific sediment yield during June 2024, intermediate during October 2024 and low during February, March and August 2024 (Figure 17). There is uncertainty associated with these results due to the varied duration of TIMS sampling, for example while the highest sediment yield is associated with the sample collected in June 2024, this is also the longest sampling duration (46 days), so material had longer to accumulate.

5.2.1 Particle Size Composition:

Although suspended sediment load is deemed to be supply limited (Knighton, 1998), hydrological variables such as river level also impact fine sediment flux (Charlton, 2008). Within the Breamish catchment, there are opposing relationships between river level and D50 grain size at the upstream and downstream sites (Figure 20). The moderate positive linear correlation at the upstream site (Figure 2A) suggests that coarse sediment is transported and deposited under high flows; this reflects more natural catchment conditions as the site is situated upstream of the channelised section of the Breamish. Conversely, the weak negative linear correlation at the downstream site (Figure 21B) indicates a dominance of fine-grained sediment under high flow conditions and sediment trapping under lower energy conditions downstream. This shows decoupling between flow and sediment size in flux, demonstrating the lasting geomorphic and sedimentological impacts of historic channelisation. However, the relationship is not statistically significant at either end of the restoration area ($p>0.05$; Appendix 4), indicating that river level / flow alone does not explain grain size variation. Additional factors influencing the grain size of sediment flux within the Breamish include supply, differential land use and antecedent soil conditions (Reid *et al.,* 1997; Walling and Fang, 2003). It is therefore deduced that the difference in D50 grain size between sites reflects different external catchment controls on sediment dynamics and depositional environments within the River Breamish.

Moreover, the grain size composition of sediment flux exhibits intra-annual variability, whereby larger particles of uniform texture (silty sand) are prominent during winter (Figure 18; Table 15; Figure 19). Elevated river levels and increased stream power during winter mobilise larger particles and rapidly transport suspended sediment downstream (Ryan and Dixon, 2007), consequently reducing local variability. The constrained, narrow, uniform channel morphology within the Breamish pre-restoration leads to homogenised flow conditions during the high river levels exhibited during winter, thus limited sediment sorting leads to uniform textured sediment in flux (Carbonari, Reckling and Solari, 2020). Comparatively, summer exhibits greater textural variation than winter (Figure 18) alongside a stable, low baseflow and occasional short duration rainfall events (Figure 14); these intermittent flows lead to localised sediment sorting and the formation of depositional zones (Powell, 1998). Long term monitoring is necessary to determine the effects of restoration and floodplain reconnection on the grain size composition of sediment in flux as the surface area of particles represents a principle control of surface chemistry (Walling and Moorehead, 1989). The median grain size was smaller at the downstream side of the channelised reach (Table 14), this indicates more potential for downstream pollutant transport as finer grains offer greater surface area for macronutrient contaminant load attachment (Goharrokhi, Lobb and Owens, 2020). Therefore, understanding grain size in flux is a critical aspect to be considered when developing diffuse pollutant management strategies.

5.2.2 Macronutrient Load:

The pre-restoration baseline nutrient composition of sediment flux indicates that the River Breamish channelisation reduces flow heterogeneity, consequently limiting nutrient retention and promoting uniform nutrient transport downstream (Sear *et al.,* 2010). The observed increased variability in total N content from upstream to downstream and the progressive increase in peak values (Table 16) reflects the cumulative impact of upstream inputs and downstream deposition (Noe and Hupp, 2005) within the River Breamish pre-restoration. The increase in peak values downstream (Table 16) indicates increased susceptibility to leaching following artificial fertiliser application and agricultural runoff (Withers and Jarvie, 2008) within tributary sub-catchments that discharge into the Breamish channelised reach, consequently delivering additional nutrient rich sediment. The lack of statistically significant difference between sites ($p>0.05$; Appendix 5; Appendix 6) illustrates the impact of channelisation and

reduced potential for retention (Noe and Hupp, 2005) dampening the effect of spatially varying nutrient flux. However, statistical testing is limited by the small sample size (n=2-5), so caution is required when interpreting the results. Both total N content and total P concentrations are highest during early summer across the reach (Table 16), this reflects the increased agricultural activities during summer elevating macronutrient availability and delivery into the channel (Withers and Jarvie, 2008). Unlike N, P variability is more constrained at the upstream site (range of 0.79t) compared to the mid-reach and downstream sites (2.87tand 2.36t respectively) (Table 16), this indicates spatial differences in pollutant source and transport dynamics. Similarly to total N content, statistical testing indicates no significant difference between total P concentrations at the mid-reach and downstream sites ($p>0.05$; Appendix 7), thus reinforcing the ideology that channelisation reduces spatial heterogeneity in fluvial environments (Noe and Hupp, 2005).

5.2.3 Phosphorus Composition:

Under pre-restoration channelised conditions, apatite bound P (a particulate P form) is dominant at the upstream (63%) and mid-reach (70%) sites (Figure 21). This reflects limited biogeochemical cycling and reduced interaction with floodplain soils and vegetation, restricting the transformation of P into labile / bioavailable forms (Weigelhofer, Hein and Bondar-Kunze, 2018). Disconnection of the Breamish from its floodplains prevents sediment deposition in areas where microbial activity and plant uptake could facilitate the transformation of apatite P (Schönbrunner, Preiner and Hein, 2012). Therefore, P remains in its stable apatite form, illustrating the necessity for floodplain reconnection to reinstate natural processing and macronutrient cycling (Weigelhofer, Hein and Bondar-Kunze, 2018). There is a decreased proportion of apatite bound P at the downstream site, where samples exhibit a more diverse P composition with elevated redundant soluble P (~21%) and organic P (~15%) (Figure 21). This reflects additional inputs from tributaries that are not hydrologically connected to the upstream sampling site. Bioavailable organic matter from agricultural runoff is delivered into tributaries from agricultural areas, thus contributing to the heterogeneous sediment / nutrient samples downstream (Carpenter *et al.,* 1998). Redundant soluble and organic P are reactive, bioavailable P forms; as the pre-restoration channelised conditions remove the potential for floodplain retention and biological uptake, fine-grained sediment rich in bioavailable P is rapidly transported and accumulates downstream (Weigelhofer, Hein and Bondar-Kunze, 2018). This increases the risk of downstream

eutrophication and is a recognised pressure in channelised systems with limited capacity for nutrient retention (Carpenter *et al.,* 1998). It is therefore deduced that the tributaries flowing into the Breamish channelised reach represent a key pre-restoration pressure for macronutrient dynamics. Restoring the lateral connectivity within the River Breamish restoration area should reduce redundant soluble and organic P concentrations downstream by enabling biological nutrient uptake and increasing residence times (Weigelhofer, Hein and Bondar-Kunze, 2018), thus improving downstream water quality.

5.2.4 Bioavailable Nitrogen Components:

Due to limited data availability, analysis of bioavailable N is restricted. Mean NO_3 + NO_2 concentrations are higher at the upstream site (Table 17), indicating more efficient nitrification in the natural, meandering system upstream of the channelised reach. The downstream site exhibits the highest observed NO_3 + NO_2 concentrations (0.186t) (Table 17), similarly to P this indicates the influence of additional tributary inputs within the channelised reach. This also exemplifies the limited capacity for N transformation and retention within the channelised reach, consequently leading to downstream accumulation of oxidised N forms (Allan, Castillo and Capps, 2021). NH_4 concentrations are highest at the mid-reach site (Table 17), reflecting slower nitrification processes in the channelised reach. On an intra-annual scale, NH_4 concentrations at the mid-reach site peak in summer (0.501t) but fall below the limit of detection during winter and late summer (Table 17). This corresponds with the seasonal peak / decline in agricultural activities (Withers and Jarvie, 2008). While the intra-annual variation in bioavailable N forms is not statistically significant ($p=0.051$; Appendix 8), the results of this research indicate ongoing nutrient cycling within tributary sub-catchments, however the effectiveness within the River Breamish is constrained by the channelised conditions (Allan, Castillo and Capps, 2021).

5.3 Pollutant Delivery Modelling:

The following section relates to research question 1, discussing critical source area modelling results to pin-point the areas of excess fine sediment, nitrate and phosphate delivery into the River Breamish pre-restoration.

5.3.1 Critical Source Areas:

Critical source areas are identified where fine sediment sources are eroded via overland flow and transferred from the land into the channel by a connected pathway (Reaney et al., 2019). This translates to areas with erosional risk, connectivity risk and in-channel risk all >0.75. Co-located high erosion and connectivity risk values situate potential critical source areas of sediment predominantly in the riparian zone of low order tributaries located upstream of the restoration area in the middle section of the Breamish catchment (Figure 23). Agricultural farms and pastureland are located along many of these tributaries; thus it is deduced that the riparian zone is vulnerable to erosion due to livestock access. Livestock trampling / grazing on adjacent pastureland proximal to the channel exacerbates soil erosion and increases surface runoff due to soil compaction (Evans, 1997), consequently elevating fine sediment availability and delivery (Heathwaite, Burt and Trudgill, 1990). Macronutrient (NO_3 and PO_4) models also indicate low order tributaries as potential critical source areas of high risk (Figure 24; Figure 25). Artificial fertiliser application on agricultural land near flow pathways is rapidly delivered to the channel via overland flow under saturated soil conditions, consequently representing a critical source of elevated NO_3 and PO_4 within watercourses like the River Breamish (Withers and Jarvie, 2008; Sharpley and Smith, 1990). Additionally, macronutrient critical source areas are locally variable with significant in-channel risk indicated immediately downstream of the River Breamish restoration area (Figure 24; Figure 25). While this area does not contribute to the loadings within the channelised reach, it exerts crucial impact on downstream water quality and system functioning. This will not be addressed by the proposed restoration within the River Breamish, thus additional management may be required. Fewer very high-risk PO_4 zones with delivery risks >+2.2σ (Figure 25) indicates that PO_4 critical source areas are more spatially concentrated than NO_3 sources due to the spatial arrangement of high and low pollutant availability risk land covers. SCIMAP identifies the main channel of the River Breamish as low to intermediate sediment / macronutrient risk consistently below the mean (~0.1-1σ) (Figure 23; Figure 24; Figure 25). In-channel risk values average out downstream for all three models (sediment, NO_3 and PO_4), this reflects a universal property of fluvial catchments, whereby as catchment area increases, variability decreases (Birkinshaw and Bathurst, 2006). Thus, the risk variability reduces downstream and the modelled main channel exhibits low risk values towards the end of the

Breamish catchments drainage basin (Figure 23; Figure 24; Figure 25). The low in-channel sediment, NO_3 and PO_4 risk in the main channel also reflects SCIMAP's underlying assumptions that pollutants are primarily sourced from diffuse overland flow rather than in-channel processes. Additional potential source areas that are not fully captured by SCIMAP include the elevated delivery of sediment during storms due to unstable banks collapse (Figure 27), proximity to exposed areas of compacted soil and lack of vegetation.

In the wider context of the entire Tweed catchment, the critical source areas identified within the Breamish are moderately high risk, sitting above average but almost all within 2σ of the mean (Figure 30E). Overall SCIMAP does not predict severe fine sediment risk within the River Breamish restoration area relative to other locations within the wider Tweed catchment (Figure 30E). However, as aforementioned, the SCIMAP model overlooks critical sub-surface pathways and thus will likely give an incomplete representation of sediment transport dynamics. Additionally, SCIMAP risk values are relative so do not represent absolute concentrations; modelled pollutant concentrations could be relatively below average in the River Breamish while still being higher than desirable amounts for system functioning in absolute terms. Therefore, local observations and reach scale monitoring data are critical for developing a full understanding of pre-restoration sediment and macronutrient dynamics within the River Breamish.

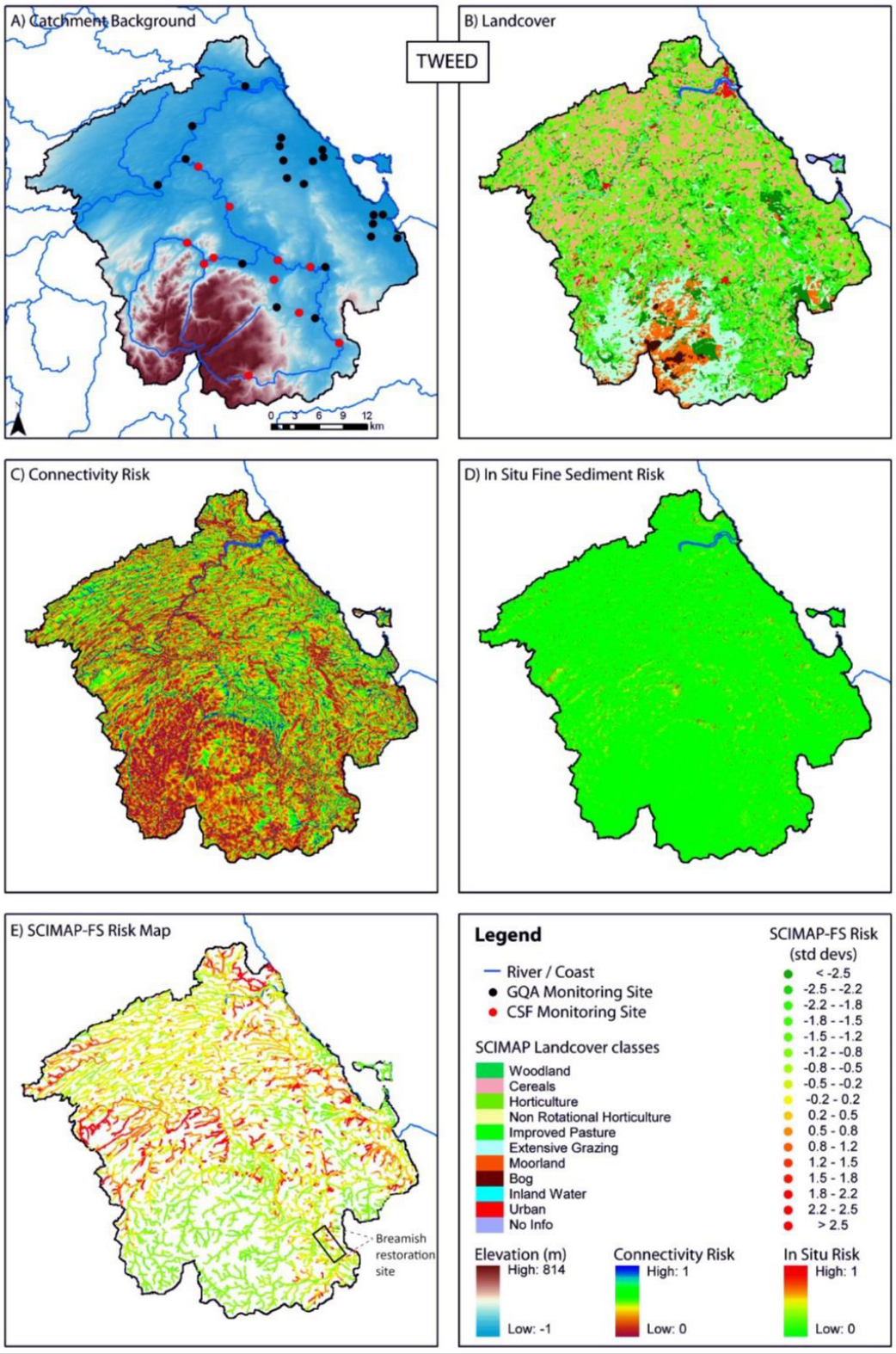


Figure 30. Breamish restoration site relative to full tweed catchment. The scale bar in map A is relevant to all maps.
(Source: Adapted from Milledge, 2012)

5.3.2 Land Connectivity Variability and Potential Storage Areas:

The results of SCIMAP modelling indicate that the land located adjacent to the channel nearby the channelised reach / restoration area has the highest connectivity risk (Figure 22). The western floodplain exhibits maximum connectivity risk (Figure 22) due to the flat topography alongside high likelihood and frequency of saturation excess overland flow facilitating rapid delivery of pollutants into the channel (Beven and Kirkby, 1979). This corresponds with hysteresis analysis, where the dominance of clockwise loops (Table 1) alongside high HI_{mid} values (Table 11) indicates rapid sediment delivery from connected adjacent land areas. Conversely, headwaters exhibit consistently low connectivity risk across sediment, NO_3 and PO_4 models (<0.5) (Figure 22), suggesting limited lateral transfer of pollutants from diffuse areas located further away from the channel. Between these areas, the mid-catchment exhibits variable connectivity risk (Figure 22), attributed to landscape features such as roads and tractor tracks creating preferential pathways of pollutant delivery (Figure 26) (Williams-Mounsey *et al.,* 2024). The risk patterns also indicate the influence of rill and gully networks within the headland areas, where established gullies provide well connected delivery pathways which mobilise sediment and nutrients into the channel (Possen *et al.*, 1996). These areas act as key pathways for the transfer of diffuse pollutants into the stream, supporting the conceptual model of lateral surface flow dominance in pollutant transport.

The variable concept of connectivity is represented where headwater areas have a high erosional risk (Figure 23; Figure 24; Figure 25) but are functionally disconnected from the channel (low delivery risk) (Figure 22), thus headlands exert a non-critical contribution to relative in-channel sediment and nutrient pollutant risk across the Breamish. These areas of disconnection act as potential temporary storage areas, where fine sediment accumulates on the ground until a high magnitude storm of sufficient duration enables connection with the channel (Lane, Reaney and Heathwaite, 2009), thus enabling these areas to act as diffuse critical source areas. There is evidence of such connection occurring within the Breamish as when large storm events activate these storage areas there is a large delayed sedimentary input into the channel (Table 10), leading to peaking sediment during the falling limb and clockwise F8 hysteresis within the restoration area (Figure 15; Table 1). These results corroborate previous research within UK catchments where the release of fine sediment from temporary storage areas usually disconnected from the

channel was observed during high intensity storms once a variable threshold was exceeded (Perks *et al.*, 2015). This represents a critical risk for the Breamish catchment as the magnitude and frequency of intense rainfall and extreme storm events are projected to increase under future climatic change (Dankers *et al.,* 2013); the connection of headland areas during such events would represent large additional critical source areas contributing excess pollutants into the channel upstream of the restoration area. Overall, while a high in-channel risk is not observed within headwater areas (Figure 23; Figure 24; Figure 25), the potential release of pollutants from temporary storage zones represents a large threat to fine sediment and macronutrient dynamics under projected future climatic conditions (Dankers *et al.,* 2013). This represents a key limitation of the SCIMAP model, as the results are temporally constrained, with no potential for developing results under projected climatic scenarios to explore magnitude frequency relations.

It is paramount to note that the SCIMAP model only shows surface flow connectivity, therefore the sub-surface pathways delivering sediment and macronutrients to the channel are overlooked (Deasy *et al.,* 2009). Research indicates that sediment and NO_3 can also be transferred through sub-surface flows (Bilotta *et al.,* 2008), therefore the patterns of NO_3 and fine sediment risk likely differ and may reflect a higher risk than indicated by SCIMAP. NO_3 is soluble and has the ability to travel via surface and subsurface pathways, whereas PO_4 have stronger particle binding properties so solely follow surface flow pathways (Ballantine, Walling and Leeks, 2009). Critical source areas are determined with consideration of the SCIMAP assumption of surface flows transfer and the omission of sub-surface flows which could also be delivering sediment into the channel.

5.4 Wider Implications for Future Catchment Management:

This section relates to research question 3, integrating analysis to discuss the practicality and effectiveness of floodplain reconnection for reducing downstream transport of sediment and macronutrients, overall improving water quality within agricultural catchments like the River Breamish.

This research demonstrates the combined application of multiple methodologies to understand sediment and macronutrient dynamics within fluvial environments. It is suggested that these methodologies are employed by the Environment Agency and Tweed Forum during post-

restoration monitoring and evaluation of success to improve the understanding of diffuse fine sediment and macronutrient pollutants within the Breamish catchment. Combining SCIMAP spatial outputs with event based hysteresis patterns gives a comprehensive picture of when and where sediment and nutrient delivery risks are highest. SCIMAP modelling is freely available and can be used following restoration to identify and target land use management to further sub-field critical source areas (SCIMAP, 2025). The results of this research enhance the understanding of key pollutant (sediment, NO_3 and PO_4) source, delivery and composition flux, however additional analysis is required to integrate the influence of sub-surface pathways transporting sediment and NO_3 into the channel.

The pre-restoration channelised conditions offer limited opportunity for pollutant retention and storage, consequently increasing downstream transport to areas with inefficient filtration (Noe and Hupp, 2005), accelerating eutrophication and water quality deterioration (Ekholm *et al.*, 2000). Reconnecting the channel with the western floodplain (which SCIMAP indicates has the highest connectivity risk) under the LIFE WADER restoration project should increase residence times, encourage sediment deposition and enhance sediment retention by creating a sediment sink, consequently reducing suspended sediment concentrations in the channel (Sear *et al.,* 2010). Floodplain vegetation and microbial communities should promote nutrient cycling and uptake (McMillan and Noe, 2017), however the impact will likely be delayed due to the time lag between completion of works and development of stable vegetation increasing soil carbon and accelerating microbial activity (Gordon, Dorothy and Lenhart, 2020). Empirical data and risk modelling within this research supports the concept of utilising floodplain reconnection for improving water quality. Foodplain reconnection is deemed a practical and evidence-based intervention to reduce in-channel pollutant flux within channelised reaches and restore natural hydro-ecological functioning within agricultural catchments.

Moreover, integrated hysteresis and SCIMAP analysis indicate key critical source areas of sediment, NO_3 and PO_4 situated within areas proximal to the channel in the mid and lower Breamish catchment. Therefore, while floodplain reconnection will likely reduce sediment and macronutrient pollutants via in stream processing, additional intervention is necessary to mitigate elevated pollutant delivery from proximal critical source areas located on the adjacent land.

Therefore, floodplain reconnection should be implemented as part of a broader integrated catchment management plan to increase the long term resilience and success of restoration. Development of further restoration schemes targeting riparian zones and near channel erosional areas is recommended as these are the dominant sediment contributors within the Breamish catchment. Evergreen riparian vegetation should be planted along the riverbanks to create additional buffer zones, stabilise the soil and reduce land-channel connectivity within the SCIMAP identified high risk areas (Wood and Armitage, 1997; Lowrance *et al.,* 1984). Research suggests that woodland buffer zones intercept and temporarily store fine sediment, thus reducing the rate of sediment bound pollutant delivery via prominent surface runoff pathways (Perks *et al.*, 2015). Additionally, community engagement with farmers and landowners would educate them on the negative impacts of artificial fertiliser application and promote sustainable land management practices.

It is paramount that actions are taken to effectively manage fine sediment and macronutrient dynamics within fluvial environments as projected climate change is expected to alter the hydrological regime, further increasing pollutant availability and delivery (Dankers *et al.,* 2014; Leitner *et al.,* 2015). The results of this research indicate that headwater areas within the Breamish catchment have the potential to develop from storage areas into critical source areas under enhanced hydrological conditions, therefore it is critical that these areas are targeted in broader catchment management. There is a time lag of up to 30 years between the implementation of restoration strategies and observed impact (Bryant, 1995), thus emphasising the necessity for rapid action within the Breamish as downstream water quality faces critical deterioration. Employment of the recommended strategies would reduce the delivery of fine sediment into the Breamish channel, thus benefitting channel water quality and downstream ecosystem health.

The complex interactions between the natural and anthropogenic controls of pre-restoration fine sediment and macronutrient dynamics within the River Breamish are displayed within Figure 31. These must be considered when developing and implementing future restoration strategies within the River Breamish catchment and can be used as a reference for post-restoration evaluation (Figure 31).

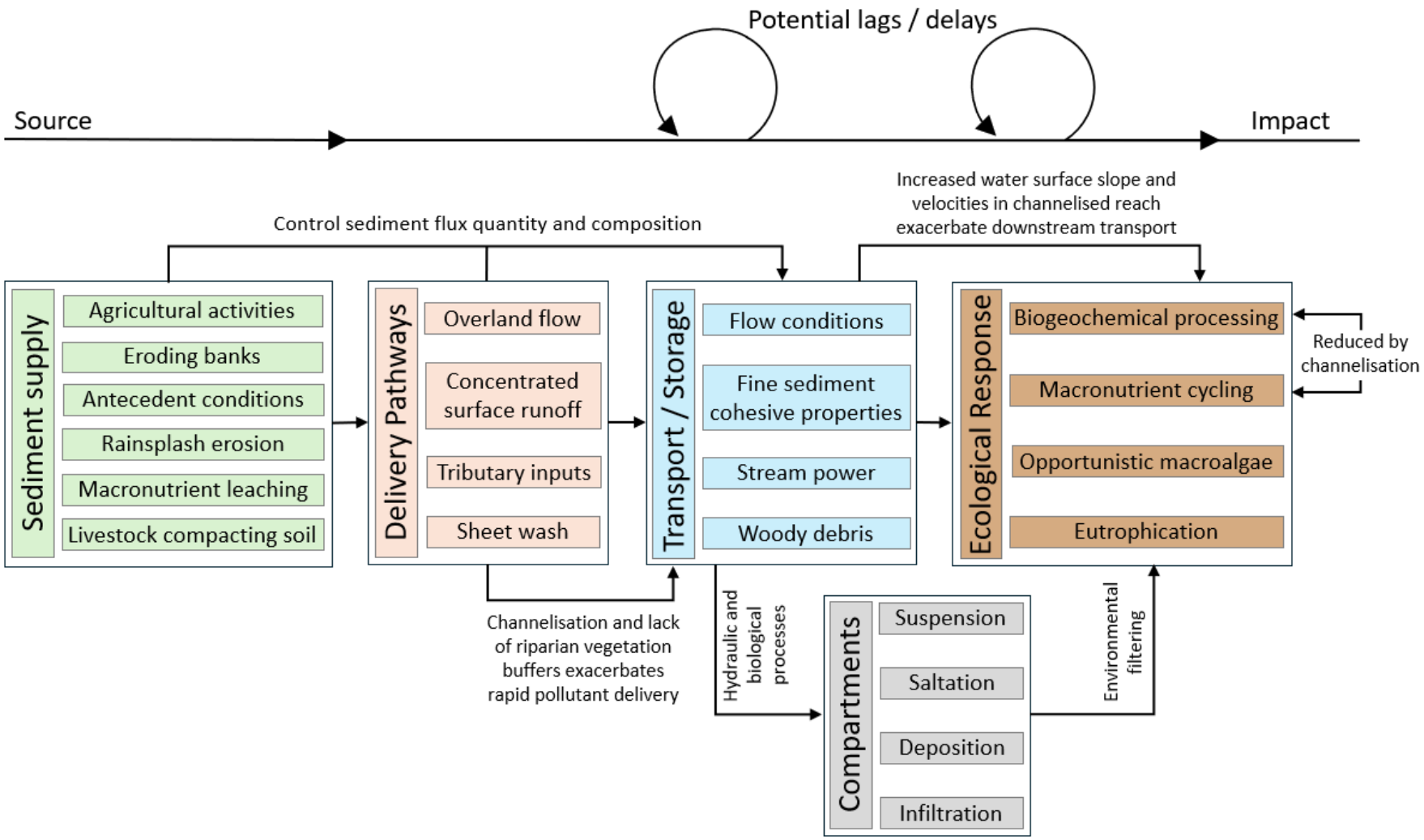


Figure 31. Schematic diagram summarising the natural and anthropogenic controls of fine sediment and macronutrient dynamics within the River Breamish, Northumberland prior to restoration including potential lags / delays and ecological response.

5.5 Study Limitations and Evaluation of Methods Applied:

Although the findings of this study provide valuable insights into pre-restoration sediment and macronutrient dynamics in the River Breamish catchment, several limitations must be acknowledged.

Hysteresis analysis was influenced by external noise especially where the system's sedimentary response was fast. Data interpretation was occasionally challenged by irregular and noisy hysteresis due to sensor failure, storm overlaps and compound events, reflecting the real time nature of monitoring data. Additionally, hysteresis analysis was limited by the constraints of the applied HI_{mid}, which is limited to quantifying the magnitude of singular loop hysteresis (Lawler *et al.,* 2006). Due to the intricate nature of fine sediment dynamics, many observed hysteric patterns were complex, displaying multiple pulses of turbidity, subsequently these were excluded from HI_{mid} analysis. The HI_{mid} applied within this study was deemed the most suitable for the pre-restoration monitoring data available due to its ability to produce reliable results from temporally constrained datasets (Jing *et al.,* 2025). Analysis should be extended during further research by applying Lloyd *et al's.,* (2016) differential hysteresis index as this method is tailored to analyse complex patterns including multi-peak and F8 loops (Lloyd *et al.,* 2016). Despite such limitations, the combination of manual loop classification and quantitative HI_{mid} calculation offers a robust assessment of sediment dynamics and identification of source proximity.

Sediment samples collected at consistent, high temporal resolution is necessary for understanding the composition of sediment flux within fluvial catchments (Knighton, 1998; Schindler Wildhaber *et al.*, 2012). While the initial sample design of this research planned monthly TIMS sample collections over a 12-month period, insufficient mass accumulated for analysis, so the period was extended to two months. Due to fieldwork and site access constraints, there were variable durations between sample collections. Although the variable sampling durations were integrated into load calculations, the inconsistent durations could impact the result precision and limit comparison between sample periods (Perks et al., 2017). The sampling intervals within this study are unable to capture short term variability during storm peaks. While monitoring data enabled analysis of sediment dynamics on an event scale, more frequent and detailed analysis of P and N speciation would support understanding of macronutrient bioavailability. Previous

research indicates TIMS have an inherent bias as non-isokinetic sampling preferentially retains coarse grained suspended sediment (Perks, Warburton and Bracken, 2014), thus the sampled material is not representative of ambient particle size (McDonald, Lamoureux and Warburton, 2010). This could introduce bias to the particle size and compositional analysis within this research. Additionally, the downstream sediment loads utilised in this study will likely be underestimated due to the remote nature of sediment load calculations. Typical practice would weigh TIMS samples before analysis, however this was an oversight during UKCEH laboratory processing, thus sediment loads for each TIMS sampling period were calculated using the upstream sondes monitoring station data. Due to the temporal constraints of masters research these sediment loads were also utilised for determining macronutrient composition at the mid-reach and downstream sites. Resultantly, the additional tributary inputs between upstream and downstream sites are not accounted for and the compositions will be underestimated. Utilising TIMS in this research demonstrates the analytical potential of TIMS samplers and supports future application during post-restoration monitoring to understanding the impact of restoration on sediment and macronutrient fluxes within the River Breamish.

A key limitation with SCIMAP modelling is that it does not account for the biogeochemical transformations that influence transfer and storage of nutrients due to the conceptual nature of the model. While SCIMAP indicates the likelihood of surface flow pathways and frequency of connection, it does not indicate the length of a flow pathway and how quickly a pollutant is routed to the channel. Although SCIMAP is a conceptual model, previous research utilises complex physically based distributed hydrological modelling to exemplify the effectiveness of SCIMAP's approach to modelling hydrological connectivity through application of the Network Index algorithm (Lane, Reaney and Heathwaite, 2009). This research critically indicates the explanatory power of SCIMAP outputs (Lane, Reaney and Heathwaite, 2009, therefore while it does not account for biogeochemical processing, it does indicate the frequency of surface flow which is inherently mobilised faster than sub-surface flow. Consequently, there is less opportunity for biogeochemical processing of pollutant materials. Future research should investigate further modelling methodologies for representing biogeochemical transformations. Moreover, SCIMAP outputs do not account for the in-channel processes specific to channelised reaches such as bank incision. As SCIMAP is based on static inputs, the outputs do not account for seasonal land use

change, spatially varying precipitation patterns and dynamic / episodic pollutant mobilisation through bank collapse (Perks *et al.,* 2017). The model provides spatial context but represents relative risk rather than absolute risk, indicating potential critical source areas where restoration / management should be targeted. However, the model is limited to identifying the location of source areas, it does not quantify the volume of pollutants being delivered into the channel and the extent of influence / impact. Future research could integrate SCIMAP with field studies of bank stability and sediment fingerprinting methods to verify the sources of fine sediment.

This research was overall limited by the inherent time and resource limitations of a master's research project. The one year temporal duration limits interpretation of inter-annual variability impacting sediment and macronutrient dynamics within the system; long term sondes monitoring and higher frequency (monthly) TIMS sediment sampling would enhance future research. Additionally, the statistical power of analysis was reduced by the limited incidence of storm events during the pre-restoration monitoring period.

Despite the outlined limitations, the methods applied within this research were appropriate and complementary, providing a pre-restoration baseline to use as a reference for evaluating the impacts of floodplain reconnection and informing implementation of further catchment scale management strategies.

Chapter 6 – Conclusion:

This chapter summarises the key outcomes of this research with respect to the research questions listed in chapter one. Overall, this research has developed a critical understanding of the fundamental behaviour of sediment and macronutrient dynamics prior to restoration within the River Breamish catchment.

Addressing research question 1, hysteresis analysis indicates that the source areas of fine sediment contributing to poor pre-restoration water quality within the River Breamish are predominantly located within areas proximal to the channel. Clockwise hysteresis patterns are dominant within the River Breamish prior to restoration, indicating rapid sediment mobilisation from proximal sources via surface runoff processes during the rising limb of a storm hydrograph. The rapid delivery of material is exacerbated by agricultural activities and the use of tractors / machinery on the adjacent fields creating channels of concentrated surface runoff and degrading the soil. Additionally, limited floodplain buffering alongside the increased water surface slope and velocity within the River Breamish channelised reach accelerate bed erosion during high flow events and consequently generate additional sedimentary inputs through bank erosion / incision. While the majority of sediment is sourced from localised inputs, occasional complex F8 and anticlockwise loops demonstrate the importance of diffuse headwater sources and additional tributary sub-catchment contributions following large magnitude storm events. These sources are less frequently activated and represent lower magnitude sediment transfer events. It is therefore deduced that proximal agricultural land and eroding banks within the Breamish restoration area are the dominant sources of fine sediment driving water quality deterioration, with occasional input from distal areas during more extreme hydrological conditions.

Furthermore, SCIMAP modelling identifies critical source areas of fine sediment, NO_3 and PO_4 within the near channel riparian zone of low order tributaries in the mid-catchment further upstream of the River Breamish restoration area. These results corroborate hysteresis analysis, indicating rapid transfer of pollutants from near channel sources associated with agricultural activities degrading the soil, consequently exacerbating erosion and pollutant delivery via surface runoff processes. Fertiliser use on the adjacent farmland further contributes to elevated NO_3 and PO_4 concentrations with localised risk zones identified mid-catchment and immediately

downstream of the River Breamish restoration area. While the main channel exhibits relatively low to intermediate pollutant risk, the surrounding landscape connectivity is a critical driver of fine sediment and macronutrient delivery. Distal headwater areas exhibit high erosional risk, however are functionally disconnected from the channel except during high magnitude storms. This further corroborates hysteresis analysis indicating the occasional input from distal areas during more extreme hydrological conditions. Such areas represent storage zones that are at risk of becoming connected and shifting to large critical source areas under future projected climate change, where more frequent extreme rainfall could mobilise large pollutant loads with detrimental impacts on the River Breamish water quality.

Next, addressing research question 2, the baseline composition of sediment flux within the River Breamish restoration area demonstrates the combined pressures of current agricultural land management and historic channelisation fundamentally altering channel processing. Particle size analysis indicates accumulation of fine-grained sediment downstream of the channelised reach, where the more natural morphology promotes deposition. Fine sediments represent a critical water quality pressure due to the larger surface area for nutrient contaminant load attachment, thus representing enhanced risk of pollutant transport downstream. The impact of agricultural activities is demonstrated through analysis of total N and total P fluxes which peak during summer, alongside peak agricultural activity and fertiliser application. The lack of statistically significant difference between upstream and downstream sites illustrates the impact of channelisation and reduced potential for retention dampening the effect of spatially varying fluxes. Analysis of P composition highlights a dominance of particulate forms at the upstream and mid-reach sites, thus reflecting the limited biogeochemical cycling under pre-restoration channelised conditions. The downstream site exhibits more immediately bioavailable and reactive P forms, representing the key downstream eutrophication risk exacerbated by reduced floodplain buffering and biological uptake. While analysis of bioavailable N forms was limited, higher concentrations of bioavailable $NO_3 + NO_2$ were present downstream, also demonstrating the lack of biogeochemical transformations and macronutrient cycling occurring within the pre-restoration channelised reach.

Finally, addressing research question 3, this research indicates the potential of floodplain reconnection as a practical and effective method of reducing downstream sediment transport and improving water quality within agricultural catchments like the River Breamish. Re-establishing lateral hydrological connectivity through floodplain reconnection increases resistance time, promotes deposition and enhances biogeochemical processing of sediment bound nutrients. Although floodplain reconnection has the potential to reduce sediment and macronutrient pollutants via in stream processing, additional intervention is necessary to mitigate elevated pollutant delivery from the proximal critical source areas on the adjacent land. Therefore, floodplain reconnection should be implemented as part of a broader integrated catchment management scheme to increase the long term resilience and success of the River Breamish restoration. Further restoration should target the critical source areas identified by SCIMAP modelling. It is recommended that evergreen riparian vegetation is planted along the riverbanks to create additional buffer zones, stabilise soil and reduce land-channel connectivity within these areas. Catchment management incorporating these additional approaches alongside floodplain reconnection has the capacity to reduce pollutant flux, restore hydro-ecological functioning and improve downstream water quality within the River Breamish. Overall, additional post-restoration monitoring is critical for establishing the impact of floodplain reconnection on fine sediment and macronutrient dynamics within the River Breamish.

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Appendices:

Data Availability:

The raw data files and MATLAB scripts utilised in this research are available for download upon request.

Appendix 1: Statistical analysis of upstream and downstream HI_{mid} values.

Test	p-value	Test Statistic	Reject Null
"Mann-Whitney U"	0.58578	451	0
"F-test"	0.17547	1.806	0

Appendix 2: Multiple linear regression using HI_{mid} as the response variable at the upstream site.

```
--- Upstream Regression ---

Linear regression model:
    HImid ~ 1 + MeanTurbidity + RiverLevelRise

Estimated Coefficients:
                       Estimate      SE        tStat      pValue
                       ________    _______    ________    _______

    (Intercept)          1.3694     2.1241     0.64472    0.53519
    MeanTurbidity      -0.11262    0.29235    -0.38524    0.70901
    RiverLevelRise        4.691     3.5331      1.3277    0.21696


Number of observations: 12, Error degrees of freedom: 9
Root Mean Squared Error: 2.99
R-squared: 0.173,  Adjusted R-Squared: -0.011
F-statistic vs. constant model: 0.94, p-value = 0.426
```

Appendix 3: Multiple linear regression using HI_{mid} as the response variable at the downstream site.

```
--- Downstream Regression ---

Linear regression model:
    HImid ~ 1 + MeanTurbidity + RiverLevelRise

Estimated Coefficients:
```

	Estimate	SE	tStat	pValue
(Intercept)	2.3506	1.2276	1.9148	0.087786
MeanTurbidity	-0.073453	0.18855	-0.38956	0.70592
RiverLevelRise	1.1056	3.4876	0.317	0.75847

```
Number of observations: 12, Error degrees of freedom: 9
Root Mean Squared Error: 2.72
R-squared: 0.0187,  Adjusted R-Squared: -0.199
F-statistic vs. constant model: 0.0857, p-value = 0.919
```

Appendix 4: Linear regression between peak river level and d50 grain size.

Upstream:

Regression Statistics	
Multiple R	0.64
R Square	0.41
Adjusted R Sc	-0.18
Standard Erro	44.80
Observations	3.00

ANOVA

	df	*SS*	*MS*	*F*	*Significance F*
Regression	1	1386.46	1386.46	0.69	0.56
Residual	1	2006.72	2006.72		
Total	2	3393.18			

	Coefficients	*Standard Error*	*t Stat*	*P-value*	*Lower 95%*	*Upper 95%*	*Lower 95.0%*	*Upper 95.0%*
Intercept	-14.28	87.21	-0.16	0.90	-1122.37	1093.81	-1122.37	1093.81
Peak Flow	31.85	38.32	0.83	0.56	-455.07	518.78	-455.07	518.78

Downstream:

Regression Statistics	
Multiple R	0.21
R Square	0.04
Adjusted R Sc	-0.28
Standard Erro	2.52
Observations	5.00

ANOVA

	df	*SS*	*MS*	*F*	*Significance F*
Regression	1	0.84	0.84	0.13	0.74
Residual	3	19.07	6.36		
Total	4	19.91			

	Coefficients	*Standard Error*	*t Stat*	*P-value*	*Lower 95%*	*Upper 95%*	*Lower 95.0%*	*Upper 95.0%*
Intercept	19.12	2.93	6.53	0.01	9.81	28.43	9.81	28.43
Peak Flow	-0.52	1.42	-0.36	0.74	-5.04	4.01	-5.04	4.01

Appendix 5: t-Test of total N between upstream and downstream sites

t-Test: Two-Sample Assuming Unequal Variances

	Upstream	Downstream
Mean	3.6808	3.4518
Variance	9.9478	24.0021
Observations	3	3
Hypothesized Mean Difference	0	
df	3	
t Stat	0.0681	
P(T<=t) one-tail	0.4750	
t Critical one-tail	2.3534	
P(T<=t) two-tail	0.9500	
t Critical two-tail	3.1824	

Appendix 6: t-Test of total N between mid-reach and downstream sites

t-Test: Two-Sample Assuming Unequal Variances

	Mid-reach	Downstream
Mean	2.6263	2.1337
Variance	11.5946	15.2593
Observations	5	5
Hypothesized Mean Difference	0	
df	8	
t Stat	0.2126	
P(T<=t) one-tail	0.4185	
t Critical one-tail	1.8595	
P(T<=t) two-tail	0.8370	
t Critical two-tail	2.3060	

Appendix 7: t-Test of total P between mid-reach and downstream sites.

t-Test: Two-Sample Assuming Unequal Variances

	Mid-reach	Downstream
Mean	1.4535	1.2101
Variance	4.1129	2.7974
Observations	2	2
Hypothesized Mean Difference	0	
df	2	
t Stat	0.1309	
P(T<=t) one-tail	0.4539	
t Critical one-tail	2.9200	
P(T<=t) two-tail	0.9078	
t Critical two-tail	4.3027	

Appendix 8: Single factor ANOVA tests of bioavailable NO_3 / NO_2 between seasonal samples from the mid-reach site.

Anova: Single Factor

SUMMARY

Groups	Count	Sum	Average	Variance
Spring	1	0.0716	0.0716	0
Summer	2	0.0220	0.0110	1.88059E-05
Autumn	1	0.0172	0.0172	0
Winter	1	0.1340	0.1340	0

ANOVA

Source of Variation	SS	df	MS	F	P-value	F crit
Between Groups	0.0116	3	0.0039	206.4631	0.0511	215.7073
Within Groups	1.88E-05	1	1.88059E-05			
Total	0.0117	4				